Contracting frictions, geography, and multinational firms: Evidence from Mexico*

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Abstract

I explore how contracting frictions and geography influence the trade costs faced by multinationals' affiliates in Mexico relative to those faced by domestic firms. I document two key facts. First, distance to firms' home countries influences their sourcing patterns. Second, sectors with a larger presence of foreign affiliates are more intensive in relationship-specific inputs. I develop a small open economy model with multiple sectors, imperfect contracting, input relationship specificity, global sourcing, and multinational production. I compute a set of counterfactual equilibria to gauge the relative importance of contracting frictions, trade costs, and productivity in the price advantage of multinationals over domestic firms. Contrary to priors, foreign firms seem disadvantaged relative to domestic firms in trade costs and contracting frictions. Eliminating all differences between foreign and domestic firms would reduce real GNP by 2.7 percent; eliminating only productivity differences would reduce it by 2.2 percent.

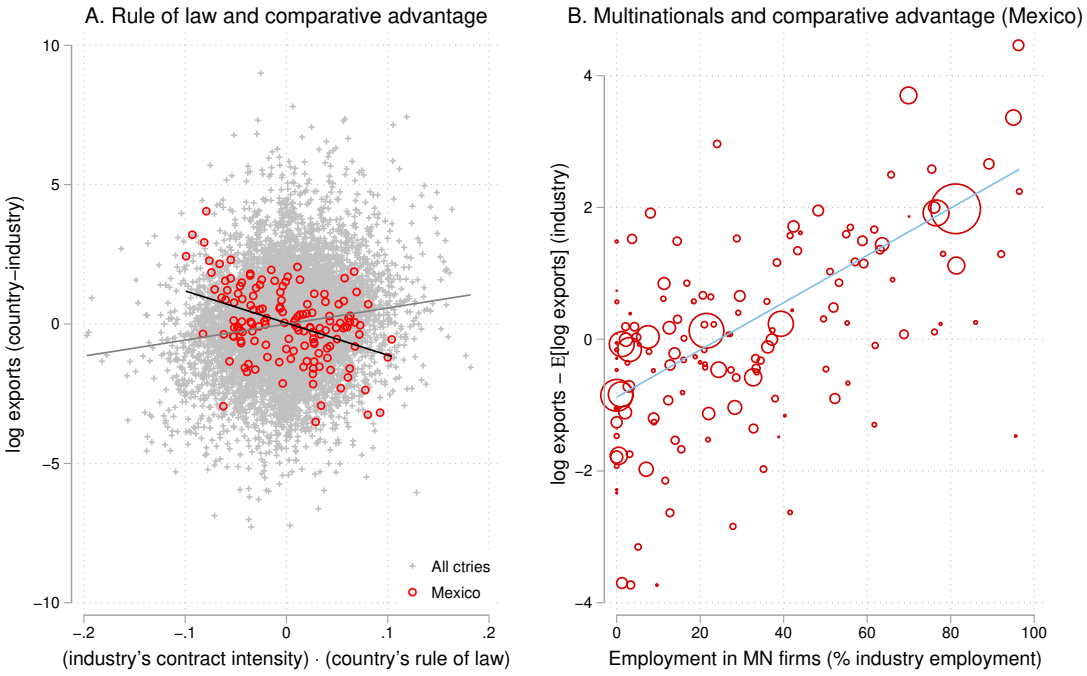
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1. Introduction

Weak contract enforcement institutions make firm-to-firm transactions riskier, which increases the cost of doing business. These distortions can reduce aggregate productivity (Boehm, 2022; Boehm and Oberfield, 2020) and be a determinant of countries' industry mix and comparative advantage (Levchenko, 2007; Nunn, 2007; Costinot, 2009). Panel A of Figure 1 shows evidence of the latter: Countries with better contract enforcement institutions export relatively more in industries that rely more on them. However, some countries seem to defy this pattern. For example, given the quality of its institutions relative to other countries', Mexico exports disproportionately more in industries that need strong contract enforcement and disproportionately less in those that do not.

Figure 1. Contract enforcement, comparative advantage, and multinationals



Panel A shows the partial relation between log exports and a measure of comparative advantage in contract-intensive industries, with industry and country fixed effects and controls for factor endowment interactions. "Rule of law" comes from Kaufmann et al. (2004) and "contract intensity" from Nunn (2007). Observations for Mexico appear in red. In comparisons of data points (industries) for the same country, a higher value on the y-axis means that the country's exports are higher relative to the world's average. The interpretation of a higher value on the x-axis depends on whether the country's rule of law index is above or below the world's average. If *above*, then a higher value implies a *more* contract-intensive industry. If *below*, as is the case for Mexico, a higher value implies a *less* contract-intensive industry. In either case, a positive fit is evidence that export patterns are consistent with revealed comparative advantage, and a negative fit implies the opposite. Panel B shows the relation between the share of foreign multinational (MN) firms in industry employment in Mexico and the residual from a regression of (log) country–industry exports on industry and country fixed effects, factor endowments interactions, and a contract enforcement interaction. Marker sizes are proportional to industry employment.

How is Mexico able to go against the current? Alviarez (2019) offers a possible explanation, documenting

that multinational (MN) firms tend to operate in industries in which domestic firms are less productive. The evidence from Mexico is consistent with this account: Deviations from the global pattern are positively correlated with the importance that MN firms have in the industry (panel B, Figure 1). This suggests that something unique to MN firms allows them to thrive in contract-intensive sectors in countries where contract enforcement institutions are weak. Understanding what this unique characteristic is (and whether local firms can learn or acquire it) is therefore important for welfare and policy-making in the host countries.

In this paper, I study the relative importance of four factors that could contribute to foreign MN firms' advantage over domestic firms: (1) lower costs of inbound international trade, (2) lower costs of outbound international trade, (3) weaker contracting frictions, and (4) higher efficiency. The first factor rests on the idea that some of the tasks required to source intermediate inputs need not be conducted from the physical location of the buyer's plant. For example, if a foreign firm is importing from its home country, it could save on costs by coordinating logistics from its headquarters (HQ) instead of from its plant abroad. The second factor reflects the same logic with respect to customers: It is less costly to coordinate distribution to a market in which a firm already has physical presence.[1] The third factor reflects the possibility that, conditional on the strength of contract enforcement at a source location, foreign MN firms may still face lower contracting frictions than their domestic peers. The fourth factor is the default explanation: that foreign MN firms simply have higher productivity in these sectors than domestic firms thanks to intangible assets such as proprietary technologies or organizational know-how. In this view, foreign firms face the same contract enforcement environment as domestic firms but are better able to compensate for it with lower unit costs.

Using plant-level data from Mexico's economic census and firm-level Mexican customs data, I document four empirical patterns. First, the economic activity of foreign and domestic firms is distributed differently across manufacturing industries—some of which are almost completely dominated by foreign firms. Second, I show that sectors with a large presence of foreign affiliates are more intensive in relationship-specific inputs. Third, I document that trade flows respond not only to the geographic distance between a foreign country and Mexico but also to the distance between a foreign country and the *country of origin* of a foreign firm in Mexico. I interpret this novel finding as reflecting the portion of trade costs dependent not on the location of *production* but on the firm's plant network, such as costs related to coordination of logistics. Fourth, I show evidence that differences in contract enforcement institutions across potential suppliers' countries are relevant for the patterns of trade in the data—and, most importantly, that foreign firms seem to have a lower sensitivity to these institutional differences.

Informed by the empirical evidence, I consider three potential factors that could give foreign affiliates in Mexico an advantage over their domestic peers in contract-intensive industries: (1) higher productivity, (2) lower international trade costs, and (3) lower contracting frictions. On the basis of Antràs et al. (2017), I develop a small open economy model with multiple sectors, imperfect contracting, input relationship

[1] A priori, lower costs of inbound and outbound trade do not have to benefit contract-intensive industries in particular. Thus, to explain the pattern in Figure 1, they would need to be lower specifically for countries specialized in customized or relationship-specific inputs.

specificity, and multinational production, with each channel represented by a different set of parameters. The first factor is represented by a Hicks-neutral productivity parameter that varies by home country and industry. The second factor is accounted for by a second set of bilateral trade costs, which vary with firms' home countries and supplier countries ("sourcing" countries). The third factor is modeled as an additional iceberg cost that also varies with firms' sourcing and home countries but affects the prices only of relationship-specific inputs. Building on Acemoglu, Antràs and Helpman (2007), this result is derived within the model. A contribution of this paper is that, despite the inclusion of contracting frictions, the model preserves the main results of the standard approach of Eaton and Kortum (2002).

In the quantitative section of this paper, I assess the relative importance of these three factors. Using the hat algebra approach of Dekle et al. (2008), I calculate three counterfactual scenarios in which I replace foreign firms' value of the corresponding parameter with that of Mexican firms to "turn off" any advantage that multinationals may have in each factor. The fact that the counterfactual "shocks" are defined in terms of the relative values between foreign and domestic firms implies that I cannot pick their values exogenously (e.g., a "10 percent increase in trade costs"), as is usual in the literature following Dekle et al.'s (2008) approach. Instead, I use the model to identify the shocks, following an approach similar in spirit to that in Head and Mayer (2014).

With these measures in hand, I compute the counterfactual scenarios. I first calculate the "overall" joint effect as a benchmark, where I set multinationals' productivity, communication costs, and contracting friction parameters to match Mexican firms'. I find a 2.7 percent reduction in welfare (measured by real expenditure) accompanied by a reallocation of value added and employment within the manufacturing sector away from multinational-dominated industries. Repeating the exercise for trade cost and contracting friction shocks, respectively, I find welfare reductions of 0.2 percent in both cases. In contrast, bringing foreign firms' productivity to the level of domestic firms' reduces welfare by 3.6 percent. I interpret these results as indicating that, consistent with priors in the field, "productivity" (net of any effect of contracting or trade frictions) is indeed the main factor behind the differences between domestic and foreign firms.

This paper speaks to three strands of literature. First, it contributes to the large literature on multinationals. On the theory side, a large body of work models firm decisions on production location, input sourcing, and sales (Antràs et al., 2023, 2022; Arkolakis et al., 2018; Antràs et al., 2017; Tintelnot, 2016). Most recent papers—for example, Antràs et al. (2023), Antràs et al. (2022), and Arkolakis et al. (2023)—have focused on the extensive-margin problem of multinationals' optimal plant location. The model in this paper is based to a large extent on the one in Antràs et al. (2017), but I abstract from extensive-margin considerations by turning their model of the global economy into one centered on a small open economy, following the criteria in Demidova et al. (2022). In exchange, I added communication costs and contracting frictions to their model, which requires me to generalize the model in Acemoglu, Antràs and Helpman (2007) to an asymmetric case in which suppliers are in different locations. On the empirical side, most of the work on foreign multinationals' effect on the host economy has focused on how domestic firms and workers are

affected via knowledge spillovers or economic linkages (Holmes et al., 2015; Lu et al., 2017; Jiang et al., 2018; Alfaro-Ureña et al., 2022; Sampson, 2022). This paper explores potential benefits to the host economy mediated not by knowledge transfers but by foreign affiliates' ability to thrive in contract-intensive sectors that would otherwise not be present in a host country such as Mexico.

Second, this paper contributes to the literature on contract enforcement institutions and trade (Levchenko, 2007; Nunn, 2007; Costinot, 2009; Nunn and Trefler, 2014). In line with this literature, I find that relationship-specific intermediate inputs tend to be purchased from countries with better contract enforcement institutions. However, I also document that this relationship is weaker for foreign affiliates. Therefore, conclusions based on country-level data would underestimate the impact that contract enforcement institutions have on the comparative advantage of domestic firms, especially in economies with a large presence of foreign multinationals.

Third, this paper contributes to the quantitative literature on contract enforcement institutions and development (Acemoglu et al., 2007; Boehm, 2022; Boehm and Oberfield, 2020). Most of this literature has focused on closed-economy models where the only relevant institutions are the local ones. This paper extends Acemoglu, Antràs and Helpman's (2007) model of contracting frictions to a multicountry, multiproduct international setting by embedding it into an Eaton and Kortum (2002) model of intermediate input sourcing by heterogeneous final good firms as in Antràs et al. (2017). Hence, the model allows firms to get around (weak) local contract enforcement institutions by importing. More importantly, the model shows that contracting frictions can be accounted for by an additional friction parameter that behaves like an iceberg trade cost, a familiar object in the trade field. Moreover, by separately accounting for differences in contracting frictions and trade costs, this paper contributes by opening the "black box" of differences in measured productivity between multinational and domestic firms.

Finally, Chor and Ma (2021) deserves special mention because it is theoretically the closest to this paper. Similarly to my paper, it also embeds contracting frictions into Antràs et al.'s (2017) global model of multinational firms and input sourcing and features a new contracting frictions term in import trade shares. However, there are differences. First, Chor and Ma's (2021) focus is on the global economy but mine on one small open economy. Second, they model whether sourcing is conducted within or outside firm boundaries, a distinction I do not make. Third, their model, in contrast to mine, allows for neither multinational production nor export platforms. Fourth, they model bargaining assuming simultaneous Nash bargaining, while I use asymptotic Shapley values. As discussed in Antràs (2016), the latter approach has the advantage of being consistent with the possibility of moral hazard, which the former lacks.

2. Empirical evidence

2.1 Data

The empirical exercise of this paper is based on two main datasets from Mexico's national statistical agency, INEGI: the economic census and firm-level customs data.

2.1.1 Economic census

I use data from the manufacturing section of the Mexican economic censuses of 2009, 2014, and 2019. The census is gathered every five years and contains information at the *plant* level, defined as the "economic unit permanently located in a single physical location and delimited by a building, under the management of a single owner" (own translation from Instituto Nacional de Estadística y Geografía, 2020). This means that the census covers all economic establishments, regardless of whether they are incorporated. In terms of geographic coverage, it covers all localities with at least 2,500 people, all district capitals (*cabeceras municipales*) regardless of size, all industrial parks and corridors, and all rural localities and establishments deemed of "economic importance." The remaining plants, those in rural areas with little economic activity but of large geographic extension, are sampled. The data always correspond to economic activity during the previous year—in this case, 2008, 2013, and 2018.

Each plant is assigned a unique identification code (ID), firm ID (for those part of a multiplant firm), and industry classification (SCIAN, Mexico's version of NAICS, at six digits) based on its main economic activity.[2] The censuses contain a wealth of information typical of this type of survey. For each plant, there is information on employment, labor costs, intermediate consumption, fiscal and financial costs, sales, other sources of revenue, output, inventories, and capital stock. Most importantly for my purposes, the survey also asks whether there is any foreign participation in the firm's equity, along with its percentage and country of origin. These three items allow me to distinguish foreign affiliates (defined as plants with 50 or more percent foreign equity) from domestic-owned plants and the country where the HQ is located.[3]

The second most important set of information from the censuses is the product-level disaggregation of intermediate input purchases.[4] This more detailed information is not available for most plants, but I have it for multiplant firms and large plants, a group known as EGE (*establecimientos grandes y empresas*).[5] While not representative of overall employment, the EGE group covers most of the Mexican manufacturing sector in terms of revenue, expenditures, and value added. For each intermediate input, there is information

[2] Since I work at the firm level, I assign to multiplant firms the industrial code for the plant with the largest value added.

[3] In the few cases in which plants belonging to the same firm are classified differently on multinational status or are assigned more than one HQ country, I pick the classification of the plant with the largest value added.

[4] This information unpacks the variable "intermediate consumption of raw materials and intermediate inputs owned by the establishment," available for each plant in the census main dataset. As the name suggests, it covers only intermediate inputs purchased by the plant for processing, not any materials handled by their contractors for contract manufacturing.

[5] An EGE plant is one that satisfies at least one of the following requirements: (1) has annual revenue of at least 50 million pesos or employs at least 50 people (either directly or through a contractor); (2) is part of a multiplant firm with plants in more than one state; (3) is part of the nonprobabilistic sample of the annual economic survey; (4) is part of the IMMEX (*maquiladora*) program; or (5) is classified in one of the 25 SCIAN six-digit industries that have full coverage (which are not listed).

on its unit of measure, quantity, and value; for the last two, the data also distinguish between domestic purchases and imports. However, the imports are not further disaggregated by source country, which is information that I need for this paper. To fill this gap, I link the census information with firm-level customs data, described in the following section. First, however, I note that the reason I use the census instead of the annual economic survey (with its higher-frequency data) is that the latter does not include variables to identify multinationals or information at the product level.

2.1.2 Customs data

I use data from the **Profile of Export Manufacturing Firms (PEME)**, a statistical project that links information from the annual economic surveys, economic censuses, and IMMEX program with administrative custom data. In contrast to the census, this dataset is at firm level.[6,7] Since the census covers firms of all sizes but PEME covers only those engaged in international trade, the set of linked plants should account for only a small fraction of all the census plants.[8] In fact, 99 percent of the plants in the 2019 census do *not* match with any firm in PEME. Nonetheless, in terms of economic aggregates, this 1 percent accounts for a significant proportion of manufacturing activity, as the following table shows.[9]

Table 1. Share of linked plants in manufacturing sector totals of selected census variables, 2018

							Employment		
Plant status	Sales	Exports	Expenses	Imports	Output	Value added	Direct	Indirect	Total
Only in census	0.34	0.09	0.32	0.13	0.34	0.38	0.65	0.37	0.62
In census and PEME	0.66	0.91	0.64	0.87	0.66	0.62	0.35	0.63	0.38

Note: The amounts for sales, expenses, exports, and imports correspond to "goods and services" aggregates, which exclude sources of income (e.g., subsidies, resale of unprocessed goods) or expenditure (e.g., taxes, interest payments) not directly related to production. "Direct" labor expenses correspond to workers hired directly by the plant; "indirect" corresponds to workers supplied by third-parties (e.g., outsourced cleaning services).

The information in PEME is at the year–firm–product–country level for both exports and imports. For example, for any firm f, I observe the amount (in current US$) sold to or purchased from country i of good g in year t. Combined, these two datasets give me detailed information on the complete vector of trade

[6] An important issue unresolved by INEGI is that its definitions of a "firm" used for the construction of the PEME database and the censuses do not coincide. The main reason seems to be that PEME uses a tax ID (RFC) to identify firms but the census uses the national registry of firms. Hence, in some cases, a census "firm" contains more than one PEME "firm," and in others, the opposite happens. INEGI staff explained to me that these inconsistencies could be caused by firms changing their RFC (so that both the old and new are linked to the same firm ID from a census) or by firms having more than one RFC (likelier for large firms). I followed their recommendation to always prioritize the PEME classification in these instances.

[7] Despite the availability of correspondence tables for the PEME and census firm IDs, there are several remaining issues beyond my capacity to fix. For instance, 100 percent of the output value and employment reported in PEME is accounted for by firms with a match in the 2019 economic census, but these observations account for only 39 percent and 36 percent of total exports and imports in PEME, respectively. I was not able to obtain an explanation from INEGI about this or figure out it myself.

[8] A well-known stylized fact in the literature is that firms that import or export tend to be more productive and larger than those that do not.

[9] In some cases, the export/import totals from PEME do not coincide with those in the census. In fact, some firms that the former classes as not engaging in international trade appear according to the data in the latter to do so, and vice versa. Given that the information in PEME comes from administrative databases but the census is self-reported, I defer to the former when discrepancies arise.

flows, by product and trade partner, for the largest manufacturing firms in Mexico.

2.1.3 Other datasets

Though the census and customs data constitute the main information source for this project, I also use a set of publicly available auxiliary datasets.

Quality of contract enforcement institutions. I proxy this variable with the *rule of law index* from the **Worldwide Governance Indicators (WGI)** database of Kaufmann and Kraay (2022). This is a "perception"-based indicator, derived, according to its documentation, from "several hundred variables obtained from 31 different data sources" and meant to reflect the opinions of survey respondents, nongovernmental organization (NGOs), commercial businesses, and the public sector.

Typology of intermediate goods. I classify the intermediate inputs in both the census (which has information on the domestic/imported split) and PEME (which disaggregates imports by country) as either *standardized* or *relationship specific* using the classification constructed for Rauch (1999).[10] Rauch classifies all internationally traded goods (defined by SITC rev. 2 three- or four-digit code) into three categories on the basis of whether the good (1) is traded on an organized exchange (such as the London Metal Exchange), (2) has "reference prices" (i.e., "prices can be quoted without mentioning the name of the manufacturer"), or (3) neither. This last category is meant to capture those goods for which brands are relevant. Rauch considers goods in the first two groups "homogeneous" and the latter "differentiated." When focusing on intermediate inputs, I rename these categories "standardized" and "relationship specific," respectively.[11]

Trade, gravity and concordances. All the variables commonly used in gravity regressions and the country-level international trade data used in this paper come from the *Gravity* and *BACI* datasets of Conte and Mayer (2022) and Gaulier and Zignago (2010), respectively. Both are available online from the CEPII website. Finally, at different stages of the data cleaning process, I used several correspondence tables to navigate the different product and industry classifications used by these datasets. For example, to typify goods in PEME (defined according to different *Harmonized System (HS)* revisions) using Rauch's categories (defined according to the SITC rev. 2), I used the correspondence tables published by the United Nations Statistics Division.[12]

[10] The author has a webpage from which the data can be downloaded and which also has detailed information on its construction and correct use.

[11] For more precise definitions and a longer explanation, see section 3.5.

[12] Available here.

2.2 Empirical evidence

2.2.1 Aggregate descriptive statistics

According to the 2019 economic census, foreign firms represented only around 0.4 percent of manufacturing plants in Mexico. Their weight in Mexican manufacturing is orders of magnitude higher than this small share, however: They represent 27 percent of employment, 44 percent of output and sales, and a staggering 73 and 75 percent of imports and exports, respectively.[13] In line with a stylized fact in the trade literature, Mexico's foreign firms are trade intensive: Only 1 percent of its domestic firms import or export, but 94 percent of its foreign-affiliated plants engage in international trade. Among all plants that trade, foreign ones represent 40 percent of the plants' number and 59 percent of their employment and value added.

Table 2. Foreign presence in Mexico's manufacturing sector, 2018

Group	Plants	Employment	Value added	Exports	Imports
Mexican, autarkic	559,861	1,974,258	286,893	0	0
Mexican, traders	3,243	657,582	329,090	271,110	138,645
Foreign plants	2,253	983,709	481,978	793,224	366,090
Total	565,357	3,615,549	1,097,962	1,064,334	504,736

Note: "Autarkic" groups plants that do not engage in international trade. Value added, exports, and imports are in millions of current (2018) pesos.

Among this group of plants that engage in international trade, several patterns exist. First, the plants seem much more diversified in terms of import sourcing than in terms of of sales markets: The United States accounts for 67 percent of their sales and Mexico for another 22 percent. In contrast, the combined share of these two countries falls to 50 percent with respect to these plants' materials purchases. Second, domestic plants' sales are divided almost evenly between Mexico and the United States, but American plants in Mexico generate 85 percent of their revenue in the latter. Other foreign firms also direct the predominant share of their sales, albeit smaller, to the US (66 percent). Third, even among the firms that engage in international trade, Mexico remains the most important source of inputs for the Mexican plants, representing 47 percent of their expenditures of this kind. In contrast, foreign plants source only 14 percent of their materials purchases from the Mexican market. Within foreign plants, American ones divide their input expenses almost evenly between the US and other foreign countries; other foreign-owned plants source their inputs mostly from outside Mexico and the US.

[13] A similar pattern is present for domestic firms that engage in international trade: Despite accounting for only 0.6 percent of plants in manufacturing, they represent 18 percent of employment, 30 percent of value added, and 27 and 25 percent of imports and exports.

Table 3. Sourcing and marketing patterns of Mexican and foreign traders, 2018

Panel A. Sales

	Markets			
Nationality	Mexico	US	Other	Total
Mexican	852,445	821,093	171,537	1,845,076
American	200,356	2,021,333	164,909	2,386,598
Other foreign	455,739	1,774,723	474,483	2,704,946
Total	1,508,540	4,617,150	810,929	6,936,619

Panel B. Expenditures

	Input sourcing countries			
Nationality	Mexico	US	Other	Total
Mexican	651,968	367,255	357,198	1,376,421
American	209,276	860,053	870,446	1,939,776
Other foreign	410,528	307,626	1,632,389	2,350,543
Total	1,271,773	1,534,934	2,860,033	5,666,740

Note: Sales and expenditures are in millions of current (2018) pesos.

Overall, the basic facts about foreign firms in Mexico are consistent with evidence from other countries. First, they are few in number but of disproportionate heft in terms of both employment and value added. Second, they are more trade intensive than their domestic counterparts, even in comparison only to Mexican traders. Both facts are consistent with multinationals being *more productive* than even the most productive domestic firms (those that trade internationally) in a world with internal increasing returns to scale due to fixed operating and trading costs, as in Krugman (1980) and Melitz (2003). Third, among the firms engaged in trade, the Mexican ones trade relatively more with Mexico than do foreign firms, and the American firms follow the same pattern but with the US. Since all of these plants operate in the same country, this difference cannot be attributable to the traditional bilateral trade costs that vary by country pair. The divergence suggests some additional component that depends on the country of origin of the parent firm (the HQ country).

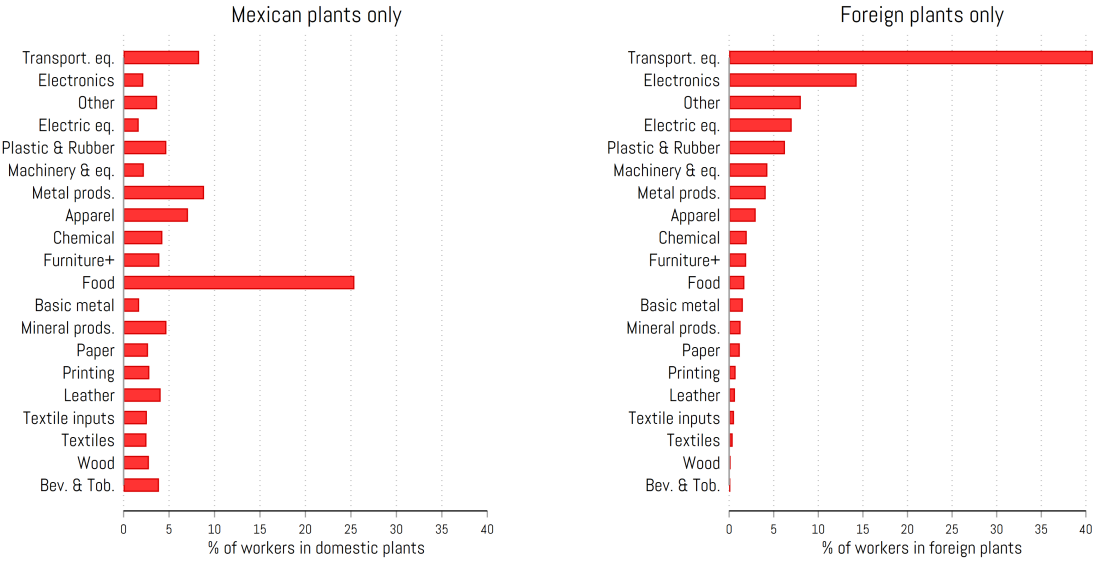
2.2.2 Stylized facts

1. Foreign plants are in different industries from domestic plants. Most of foreign firms' economic activity is concentrated in one industry, *transportation equipment*. This is true whether we measure the

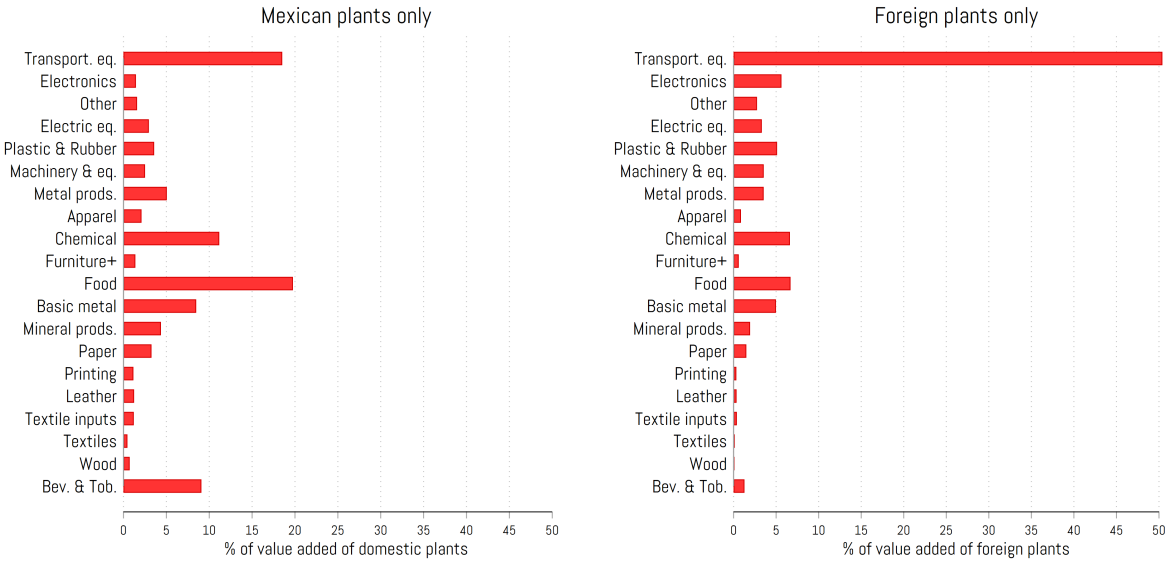
activity in terms of value added (50 percent) or employment (41 percent). In contrast, although the largest share of employment in Mexican plants is in the food industry (25 percent), when activity is measured in terms of value added, the food industry shares its predominance with transportation equipment, with 20 and 19 percent of value added, respectively.

Figure 2. Distribution of employment and value added across industries, 2018

(a) Employment



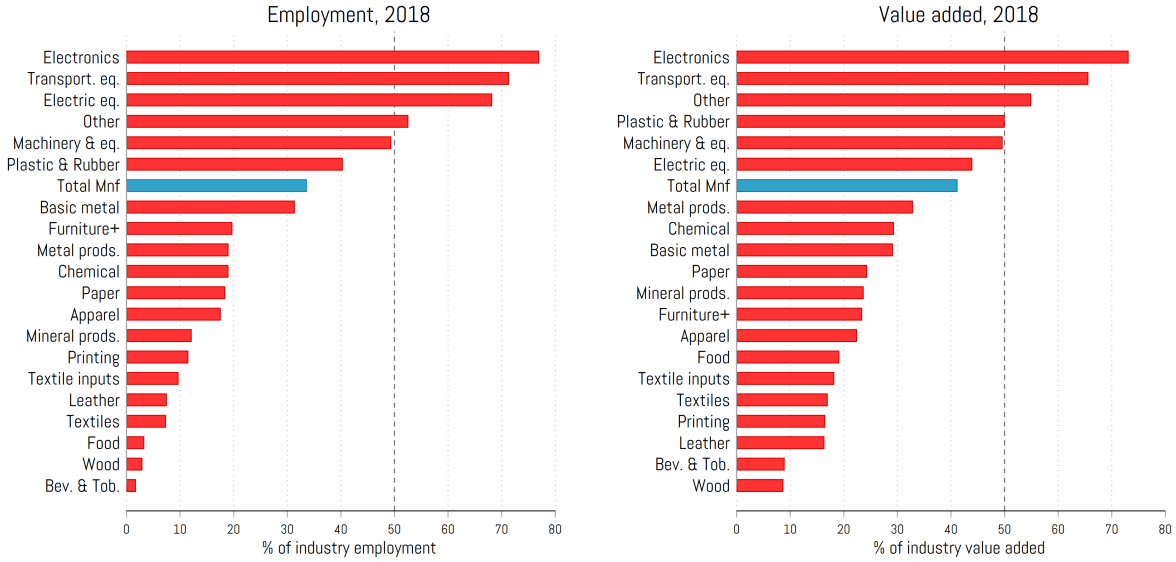
(b) Value added



Note: Industries are in descending order of foreign firms' employment in all graphs to facilitate comparison. "Bev. & Tob." stands for "beverages and tobacco."

2. Some industries are almost completely dominated by foreign plants. Not only do foreign plants have a different distribution of economic activity across industries than Mexican plants, but in many cases, they constitute most of the industry itself. For example, despite representing a small fraction of foreign plants' total employment (14 percent) and value added (6 percent), 77 percent of the workers and 73 of the value added in the *electronics* industry corresponds to foreign establishments. A similar pattern characterizes the transportation equipment industry.

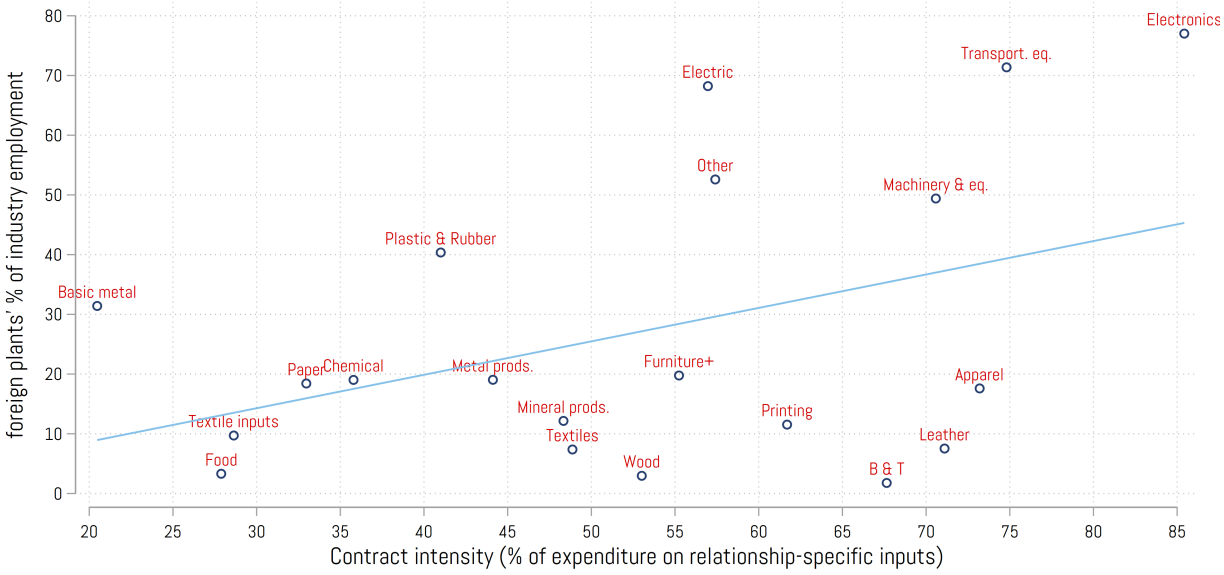
Figure 3. Share of foreign employment and value added by industry, 2018



Note: Industries 324 ("petroleum and coal products manufacturing") and 325110 ("manufacturing of basic petrochemicals from natural gas and refined petroleum") are excluded because, for the most part, foreign investment is restricted in them. "Bev. & Tob." stands for "beverages and tobacco."

3. Industries with a larger presence of foreign firms are more contract intensive. The industries in which multinationals are relatively more important in Mexico tend to be more *contract intensive*, as the following figure shows. Following Nunn (2007), the contract intensity measure is meant to capture "the importance of relationship-specific investments," and it is measured as the proportion of intermediate inputs that are relationship specific, where the proportions can be obtained from input–output tables (as in Nunn's (2007) original paper) and relationship specificity is measured with Rauch's (1999) measures. These patterns is robust to weighting of industries by their size and to the use of Nunn's original contract intensity measures constructed from US data or similar measures built from Mexican data.

Figure 4. Contract intensity and foreign share in employment, by industry, 2018



Note: Industries 324 ("petroleum and coal products manufacturing") and 325110 ("manufacturing of basic petrochemicals from natural gas and refined petroleum") are excluded because, for the most part, foreign investment is restricted in them. "B & T" stands for "beverages and tobacco."

Relationship-specific inputs are those that need to be customized to the buyer's specifications. Customizing an input requires the supplier to make a "relationship-specific" investment, in the sense that its return is higher in the context of the relationship. This implies that, once the investment is made, it cannot be recovered (or it can be only partially recovered) or repurposed for use by other buyers (or it can be repurposed only at an extra cost). Because of this, a self-interested buyer has an incentive to renegotiate the terms of the transaction in its favor after the supplier invests, taking advantage of the fact that the supplier's outside option is worse than it was when the original contract was negotiated. Preempting this, suppliers may look for ways to be compensated for the risk they are taking—be it asking for a higher price initially or underinvesting. Either way, the implication is that costs will be higher than in a scenario in which the original contract cannot be renegotiated.

In light of this, it is reasonable to expect that in countries with better *contract enforcement institutions*, the risk of a supplier's being taken advantage of by a buyer should be lower. Therefore, industries heavily reliant on relationship-specific inputs will have lower costs, ceteris paribus, the better their contract enforcement environment. This implies that in contract-intensive industries, the quality of contract enforcement institutions can be a source of comparative advantage just as physical or human capital endowments are. This is exactly what the foundational papers of Nunn (2007), Levchenko (2007), and Costinot (2009) show holds in international trade data: Countries with better contract enforcement institutions tend to export more goods from contract-intensive industries.

Implicitly, in these papers, it is assumed that relationship-specific inputs are always sourced domestically or, if not, that the local institutions are always the relevant ones. However, once we consider that intermediate inputs can also be internationally traded, this conclusion requires some qualification. First, by definition, an international transaction involves two countries and thus two sets of contract enforcement institutions, which may be of very different quality. Second, the consensus in the literature is that it is the *exporter*'s institutions that are expected to be more relevant, for reasons detailed in section 3.5.2. Therefore, a contract-intensive industry may still be competitive in a country with weak contract enforcement institutions as long as firms can source relationship-specific inputs from abroad at reasonable cost. A potential explanation for the pattern in Figure 4 is then that foreign firms in Mexico can access relationship-specific inputs at lower cost than their domestic peers.

I consider two potential features that could in practice lower the costs foreign firms face when procuring relationship-specific inputs. First, conditional on the quality of contract enforcement in Mexico, foreign firms may face effectively **lower contracting frictions** than domestic firms. This can be rationalized as multinationals having better *reputational capital*,[14] which implies that their suppliers have more confidence that they will not be cheated, which reduces underinvestment and thus costs. Second, because it is the most productive firms that select into multinational production, which allows them to reduce their marginal cost by exploiting internal economies of scale (see Helpman et al., 2004), the likelihood that they represent a large share of any given supplier's revenue is high. Given this, there is a higher probability that the suppliers, even if de jure separate entities from their buyers, agree to delegate major decisions to them—an arrangement known in the business literature as a *quasi-hierarchical structure* or *centralized control*.[15]

4. Geography and contract enforcement institutions are relevant correlates for firms' import patterns. A second factor that could also give foreign firms an edge when they source relationship-specific inputs is that they may simply face **lower trade costs** in importing (see Anderson and van Wincoop, 2004). To the extent that some trade costs are related to tasks that can be undertaken from anywhere in the world (e.g., search, logistics coordination, any tasks facilitated by shared social norms between suppliers and customers), by having plants in more than one country, multinationals can reduce their trade costs by conducting such tasks from their lowest-cost location. To back up this assertion, I document a novel fact: In addition to the distance between the exporting and importing countries, used as a proxy of traditional trade costs, another distance influences trade patterns—that from a supplier's or customer's location to the *plant network* of a multinational firm.[16] Imagine a Japanese firm wants to send South Korean inputs to its plant in Mexico. Why should coordination of logistics or search for a suitable supplier fall entirely to the management team in Mexico when the job could be arguably be done more easily from Japan, given its proximity to South

[14] On the relationship between reputation and contract enforcement, see MacLeod (2007).

[15] See Zhou and Xu (2012), Heide (2003), and Heide and John (1992). Areas in which this type of relationship are common include product design, production processes, and quality control procedures.

[16] A third bilateral friction called *multinational production (MP) costs* appears in the literature but is used to study not trade but foreign direct investment flows. In empirical exercises, MP costs are proxied by the distance between the HQ country and location of its plants. See Ramondo and Rodríguez-Clare (2013) and Ramondo (2014).

Korea? To reflect this idea, I call this second set of frictions *communication costs*. Given that I do not have information on the set of locations in which foreign firms operate in addition to Mexico, I use the restricted proxy of distance to a plant's HQ country.

Evidence of both factors is given in the following regression table. I run gravity regressions of firms' imports for each country and input type on explanatory variables representing contract enforcement and geography forces. In all cases, I use Poisson pseudo-maximum-likelihood (PPML) as suggested by Santos Silva and Tenreyro (2006). Row 1 accounts for communication costs.[17] I use six common proxies for trade costs, of which I report results on only for the log of the geographical distance between the HQ country (h) and sourcing country (j).[18] To test for Heckscher–Ohlin determinants of trade, including the role of contract enforcement institutions, I include an interaction between proxies for country-level endowments/fundamentals and input- or industry-level characteristics (rows 2–6). In particular, I include an input-level dummy for relationship specificity (RS_v, where v indexes the inputs) and a country-level index of rule of law (μ_j), which proxies for the quality of contract enforcement.[19] Finally, to check whether foreign firms differ from domestic firms in their sensitivity to contracting frictions, I add interactions of the previous two variables with a dummy for multinationals (MN_f). I run this regression over three subsets of firms: all firms, only domestic firms, and only foreign firms. For each subgroup, I run two specifications, one with exporter–year and importer–year fixed effects and another with an additional country pair dummy (such that only the time dimension is left as a source of variation) to test the robustness of the contracting frictions estimates.

[17] Given that I observe only firms in Mexico, traditional trade costs are no longer a dyadic variable and thus are not part of the table since they are absorbed by the exporter fixed effect.

[18] The other variables are (1) a dummy for shared border to account for any discontinuities at borders, dummies for (2) common official language and (3) a common former colonizer to account for nongeographical factors such as cultural or historical affinity that could also affect the cost of communication, (4) a dummy for a regional trade agreement, and (5) a dummy for a sourcing country that is the firm's own HQ country to control for any discontinuous "home bias."

[19] Similarly to Nunn (2007) and Levchenko (2007), I also control for physical and human capital endowments interacted with industries' intensities in these factors.

Table 4. Regression results

		PPML: firm f's imports of input v from country j				
		All firms		Domestic firms	Foreign firms	
		(1)	(2)	(3)	(4)	(5)
(1)	$\log dist_{hj}$	-0.452***			-0.464***	
		(0.0305)			(0.0368)	
(2)	RS_v	-1.744***	-1.639***	-1.403***	-0.509**	-0.503**
		(0.230)	(0.213)	(0.171)	(0.258)	(0.250)
(3)	$MN_f \times RS_v$	1.324***	1.201***			
		(0.114)	(0.179)			
(4)	$MN_f \times \mu_j$	-0.296**	0.366			
		(0.118)	(0.332)			
(5)	$RS_v \times \mu_j$	0.428***	0.390***	0.309***	0.209*	0.205*
		(0.101)	(0.101)	(0.113)	(0.126)	(0.122)
(6)	$MN_f \times RS_v \times \mu_j$	-0.242***	-0.202*			
		(0.0679)	(0.115)			
Observations		31,402,785	31,323,784	10,514,794	20,401,930	20,329,672
Exporter–Year FE		X	X	X	X	X
Importer–Year FE		X	X	X	X	X
Country Pair FE			X			X

Note: Clustered standard errors (firm's industry $k \times$ year t). *** p<0.01, ** p<0.05, * p<0.1. **Indices:** f denotes the firm, h denotes the country of the firm's parent company (for Mexican firms, $h = MEX$), j denotes the sourcing country of the good, v denotes a particular input (identified by eight-digit HS code). **Fixed effects:** An *importer* is defined as a home country–downstream industry ($h \times k$) pair; an *exporter* is defined as a source country–upstream industry pair ($j \times k'$). **Variables:** $\log dist_{hj}$ is the log of the population-weighted harmonic mean of the distances between all possible pairs of cities between countries h and j, RS_v is a binary variable equal to one if input v is relationship specific according to the Rauch (1999) typology, MN_f is a binary variable equal to one if firm f is foreign, and μ_j is the *rule of law index* of country j from Kaufmann and Kraay (2022). All regressions are estimated with the module for Poisson pseudo-maximum likelihood with multiple levels of fixed effects `ppmlhdfe` of Correia et al. (2020) in Stata SE, version 17.0. Before estimation, for each downstream industry–input pair, the dataset is rectangularized to contain all HQ country h, source country j, and year t combinations to account for zero trade flows.

There are three main takeaways from the table. First, the estimated coefficient associated with distance to HQ country is negative and statistically significant in both the full (column 1) and restricted sample (column 4) regressions.[20] The estimate is also economically significant: An elasticity of -0.452 implies that, ceteris

[20]It is also negative and significant when the sample is split between standardized and relationship-specific inputs and when the sample is restricted to either only US or non-US multinationals.

paribus, an American-owned plant operating in Mexico's electronics sector is predicted to import 54 percent less from Taiwan than does an otherwise identical Japanese-owned plant.[21] Second, row 5 echoes the the results in the literature on contract enforcement and comparative advantage: Relationship-specific inputs tend to be purchased from countries with better contract enforcement institutions.[22] Third, the significant estimates on the triple interaction in row 6 suggest that, when importing relationship-specific inputs, foreign firms may be less sensitive to contract enforcement institutions than their domestic counterparts.

5. Foreign firms have a geographical advantage when procuring relationship-specific inputs. While row 1 in the previous regression table shows that the distance between the HQ country and inputs markets is relevant for trade patterns, it is not enough to conclude that this fact benefits foreign firms. To explore whether this is so, I construct a measure of the "distance" between any country and global markets for each input using CEPII's gravity and BACI datasets. The measure is the weighted average distance to all countries (including itself), where the weights are the countries' share in global exports:

$$\text{dist}_h^k = \sum_j \text{dist}_{h,j} \times s_j^k$$

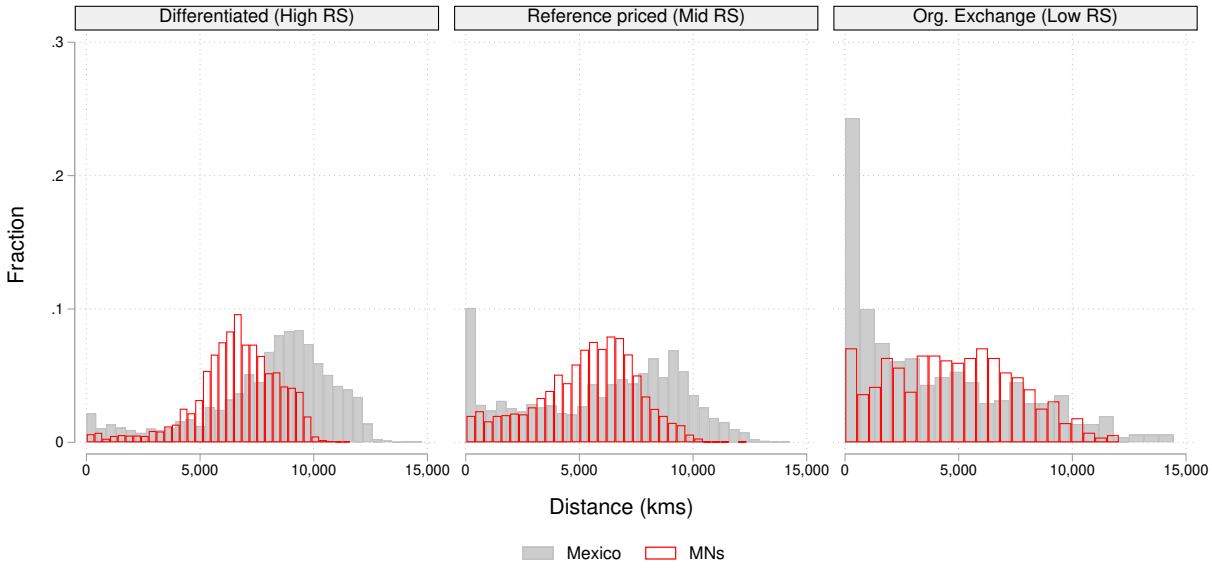
where $\text{dist}_{h,j}$ is the weighted harmonic average distance between countries h and j and s_j^k is country j's share in global exports of product k.[23] The following figure compares Mexico's distribution of distances with that of foreign countries with plants in Mexico, for the three groups of products in the Rauch (1999) classification.

[21] The distance between Taiwan and the US (12,241 km) is 5.66 times that between Taiwan and Japan (2,164 km), which implies a percentage change of $5.66^{-0.452} - 1 = -0.5431$.

[22] Despite the similarity, the result in row 5 does not capture the same pattern found in the earlier literature. Here the focus is on exporters of relationship-specific inputs, as in Berkowitz et al. (2006); Nunn (2007) and Levchenko (2007) focus on exporters of goods whose production is intensive in relationship-specific inputs.

[23] This measure is related to the *remoteness index*, which is also defined as a weighted average of distances but uses GDP shares as weights.

Figure 5. Distance to world suppliers: Mexico vs. foreign HQ



The main takeaway from this picture is that although Mexico is relatively closer to global suppliers of commodities (those traded in organized exchanges) than the HQ countries of foreign affiliates are, this relative proximity erodes and even reverses when we focus on relationship-specific inputs (the "differentiated" ones). Although not conclusive evidence on its own, this figure together with the regression results suggests that foreign firms may have advantages over domestic competitors beyond exogenous productivity differences. Informed by these patterns, in the next section, I develop a tractable theoretical model of international trade featuring contracting frictions and communication costs and mapping the three factors (productivity, contract frictions, and geography) to parameters identifiable with available data. These features will then allow me to implement counterfactual exercises that help answer the questions of this paper.

3. Model setup

3.1 Basic setup and notation

The model is centered on a small open economy (SOE), denoted by 0, which shares the world with other J countries, indexed by j, i, or h, depending on whether they represent an input source, a market, or an HQ country, respectively. The SOE is "small" in the sense that the domestic firms take foreign variables as given, but they still face a downward-sloping demand curve for their goods domestically and abroad, as in Bartelme et al. (2020). The economy is composed of K industries indexed by k. Each industry is composed of a continuum of goods $\omega \in \Omega^k$, each produced by a single firm. There is only one factor of production, labor.

3.2 Demand

Each country i has a representative agent who supplies an exogenous measure L_i of labor inelastically and has preferences represented by a two-tier utility function. The upper tier is Cobb–Douglas utility over all industries, where $\beta_k \geq 0$ is the expenditure share on industry k:

$$U_i = \prod_{k=1}^{K} \left(\frac{Q_i^k}{\beta_k} \right)^{\beta_k}, \quad \text{with} \quad \sum_{k=1}^{K} \beta_k = 1$$

The lower tier aggregates goods from the same industry via a constant elasticity of substitution (CES) utility function with elasticity $\sigma_k > 1$:

$$Q_i^k = \left(\int_{\omega \in \Omega_i^k} q_i^k(\omega)^{\frac{\sigma_k - 1}{\sigma_k}} d\omega \right)^{\frac{\sigma_k}{\sigma_k - 1}}$$

where $\Omega_i^k \subseteq \Omega^k$ is the subset of industry k varieties available to consumers in country i. The budget constraint is:

$$\sum_{k=1}^{K} P_i^k Q_i^k = E_i \equiv w_i L_i + \Pi_{ii} \tag{1}$$

where P_i^k is the consumer price index (CPI) of industry k and E_i is aggregate nominal spending in country i, defined as the sum of the aggregate wage income ($w_i L_i$) and aggregate profits by *domestic* firms, which are assumed to accrue to the representative consumer (Π_{ii}).[24]

3.3 Producers of final goods

There is an endogenous mass N_0^k of domestic firms in each industry k. Firms compete monopolistically in each market i they sell to. All domestic firms in industry k have a common productivity φ_0^k, and there are no domestically owned multinationals (MNs). There is an exogenous number of foreign affiliates operating in each industry k in the SOE, with productivities varying by industry and home country φ_h^k. Consumers in the SOE can import final goods at exogenous prices.

Final goods are assembled in plants that combine labor (L) with intermediate inputs using the following constant returns to scale (CRS) technology:

$$q_{h0}^k(\omega) = \varphi_h^k A_0^k \left(\frac{L^k}{\alpha_l^k} \right)^{\alpha_l^k} \left(\frac{M_s^k}{\alpha_s^k} \right)^{\alpha_s^k} \left(\frac{M_r^k}{\alpha_r^k} \right)^{\alpha_r^k} \quad \text{with } \alpha_l^k + \alpha_r^k + \alpha_s^k = 1$$

where A_0^k is an exogenous assembly productivity shifter for all firms in the SOE, L^k is the labor hired for assembly with a share in revenue equal to $\alpha_l^k \in [0, 1]$, M_s^k is a composite of *standardized* intermediate inputs with a share in revenue equal to $\alpha_s^k \in [0, 1]$, and M_r^k is a composite of *relationship-specific* intermediate inputs

[24] In other words, I assume that foreign firms operating in the SOE send their profits abroad.

with a share in revenue equal to $\alpha_r^k \in [0,1]$.[25] M_s^k and M_r^k are CES aggregates of a unit mass of intermediate inputs with elasticities $\zeta_s^k \geq 1$ and $\zeta_r^k \geq 1$, respectively:

$$M_x^k = \left(\int_0^1 m(\nu)^{\frac{\zeta_x^k - 1}{\zeta_x^k}} d\nu \right)^{\frac{\zeta_x^k}{\zeta_x^k - 1}}, \quad \text{for } x \in \{r, s\}$$

Both labor and intermediate input markets are perfectly competitive.

3.4 Producers of intermediate inputs

There is a competitive fringe of suppliers in each country j for both standardized and relationship-specific intermediate inputs, as in Antràs et al. (2017). Suppliers differ in two main aspects: the cost of labor in their location w_j and their productivity $a_j^{xk}(\nu)$, for $x \in \{r, s\}$. Each potential supplier's productivity is a random draw from a Fréchet distribution with shape parameter θ_x, and scale parameter T_j^{xk}:

$$\Pr\left(\frac{1}{a_j^x(\nu)} \geq a \right) = \exp\left\{ -T_j^{xk} a^{\theta_x} \right\}, \quad \text{for } x \in \{r, s\}$$

When a plant orders an input, it specifies not only quantitative features (e.g., size, weight, number of units) but also qualitative ones (e.g., materials, durability, color). I call these features *specifications*. For tractability, I assume that an intermediate input ν is characterized by a continuum of specifications, as in Acemoglu, Antràs and Helpman (2007) and Antràs and Helpman (2008). Then, $m(\nu)$ is a quality-adjusted measure of input ν:

$$m(\nu) = \exp\left\{ \int_0^1 \log m(\iota) d\iota \right\} \tag{2}$$

Each specification can be implemented with only labor:

$$m(\iota) = a_j^{xk}(\nu) l, \quad \text{for } x \in \{r, s\} \tag{3}$$

Note that the labor productivity $a_j^{xk}(\nu)$ is the same for all specifications $\iota \in [0,1]$.

3.5 Relationship specificity and contractibility

3.5.1 Relationship specificity

In section 3.3, I distinguished between standardized and relationship-specific inputs. An input is relationship specific if it has to be customized to the needs of the customer. This characteristic implies that relationship-specific inputs lack thick spot markets, with two consequences. On one hand, a firm can procure this type of input only by commissioning it from a specific supplier. On the other hand, once the

[25] The difference between a standardized and a relationship-specific input is explained in section 3.5.

input is produced (and the supplier has already incurred the production costs), its value is always higher inside the relationship than in the market. This fact gives the buyer an incentive to take advantage of the supplier by renegotiating a lower price ex post in what is known as the *hold-up problem*. In principle, higher customization requirements imply a higher hold-up risk, but I abstract from differences in the level of input customization by assuming *full* relationship specificity.[26]

Assumption 1. *Relationship-specific inputs are fully customized to the needs of their buyers and, thus, have no value for any other firm.*

By contrast, standardized inputs require no special customization and can be purchased on thick spot markets with no need for formal contracts. This distinction is captured by the next assumption.

Assumption 2. *Standardized intermediate inputs are not affected by contracting frictions.*

This assumption, together with perfect competition, imply that standardized inputs are priced at the factory door at marginal cost.

3.5.2 Contractibility

Hold-up risk also depends on how enforceable the initial agreement is, which I call *contractibility*. This concept captures the viability of (1) putting into writing the specifications required by the buyer and (2) inspecting and testing compliance with these requirements in a timely manner by the buyer (before money changes hands, for instance). It is reasonable to assume that the more intricate the specifications are, the more difficult it is to verify compliance. These two aspects are linked to the characteristics of the input being purchased.

In addition, if a breach of contract is identified, it becomes important that its content is understood, the breach verified, the dispute resolved, and compliance enforced by the third party imparting justice (whether courts and bailiffs or arbitration tribunals). This means that contractibility also hinges on the strength of contract enforcement institutions, which encompasses features such as accessibility (e.g., financial cost, geographical access), professionalism (e.g., availability of specialized courts), expediency (e.g., time required to enforce), and fairness (e.g., lack of corruption). This model focuses on this aspect of contractibility and abstracts from the first two.

When the transaction is international, it is not obvious which country's institutions are the relevant ones.[27] Berkowitz et al. (2004, 2006) show an asymmetry in the risks that buyers and sellers face in international

[26] See Antràs (2016) for a model with partial relationship specificity.

[27] It is not even clear whether *any* domestic institutions should matter at all since international transactions can leverage a set of international institutions created for them. First, contracts can specify a *choice of law* clause, where parties can opt for the law of any country (even a third one). Second, contracts can specify a *choice of forum* clause, where parties can agree on the jurisdiction that would resolve potential disputes (again, even a third country). Third, the parties can opt to solve their dispute via arbitration instead of in the courts. Finally, where courts are chosen for resolution, contract law has been standardized across countries via initiatives such as the New York Convention of 1958 and the Vienna Convention of 1980. These features should dampen the effect of country differences in institutional strength on cross-border transactions. However, they impact only the process of dispute *resolution*, not *compliance* with the resolution. If the losing party simply ignores a ruling, compliance can be enforced only by judicial execution organs in locations where that party has assets. Therefore, domestic institutions still influence international transactions (cf. Berkowitz et al., 2004, 2006).

transactions. On one hand, exporters face the risk of not being paid by the importer, but they have at their disposal tried-and-true tools to reduce this risk, such as requesting prepayment and using bills of exchange and letters of credit. On the other hand, importers face the risk of receiving defective goods. With this type of risk, mitigation implies hiring inspection and testing agents, who may not even be able to test every important specification or do so in a timely manner. The first implication of this asymmetry is that the *importer* is the party likeliest to need strong contract enforcement institutions. The second implication is that the relevant contract enforcement institutions are those of the *exporter*'s country: If the latter loses a dispute, unless it voluntarily abides by the ruling, compliance can be coerced only in a country where the exporter has assets, which is likeliest to be its home country.

Equation 2 expressed $m(\nu)$ as an aggregation of a continuum of specifications. I follow Acemoglu et al. (2007) and Antràs and Helpman (2008) in assuming that specifications in the range $[0, \mu_j]$ are perfectly contractible—that is, that they can be fully specified ex ante in a contract and this contract is costlessly enforceable ex post—while tasks in the range $(\mu_j, 1]$ are perfectly noncontractible. Hence, $\mu_j \in [0, 1]$ is the key parameter representing the degree of contractibility of any input sourced from country j.

$$m(\nu) = \exp\left\{ \underbrace{\int_0^{\mu_j} \log m(\iota) d\iota}_{\text{contractible}} + \underbrace{\int_{\mu_j}^{1} \log m(\iota) d\iota}_{\text{noncontractible}} \right\}$$

Given that contractible specifications are symmetric to each other, they share the same optimal level in equilibrium, denoted by $m_c(\nu)$. The same applies to noncontractible specifications, whose optimal level is denoted by $m_n(\nu)$. Hence, we can rewrite the production function of an intermediate input as:

$$m(\nu) = m_c(\nu)^{\mu_j} m_n(\nu)^{1-\mu_j} \tag{4}$$

3.5.3 Multinational firms and contracting frictions

I hypothesize that, ceteris paribus, multinationals are less likely to be affected by contracting frictions than domestic firms. This conjecture is based on the patterns in section 2.2.2, which I rationalize in two ways. First, when sourcing from abroad, foreign affiliates can do so from another affiliate of the same firm or from the firm HQ itself, in which case both parties' incentives are aligned and no hold-up problem arises. Second, even when the supplier is an unrelated party, foreign affiliates can use quasi-hierarchical structures to influence supplier behavior.[28] This second means of control is likelier to arise if the buyer's purchases represent a large share of the supplier's revenue.[29] Though I lack information on input suppliers in the data, multinationals are the likeliest candidates to satisfy this condition given the established empirical fact that they are among the largest and most productive firms. Thus, in the empirical exercise, I allow MN and

[28] See Zhou and Xu (2012) and Heide and John (1992).
[29] See Zhou and Xu (2012) and Heide (2003).

domestic firms to face different contracting frictions when they source relationship-specific inputs.

3.6 Fixed costs and variable frictions

3.6.1 Fixed costs

There is an infinitely large pool of potential entrants for each industry k and country h, and there are no constraints to entry except for a fixed cost of f_e^k units of home labor. The mass of domestic active firms N_0^k adjusts to guarantee that, in equilibrium, firms earn zero economic profits. In contrast, foreign affiliates can have positive profits. International trade does not entail extra fixed costs, so all firms will engage in international trade.

3.6.2 Variable frictions

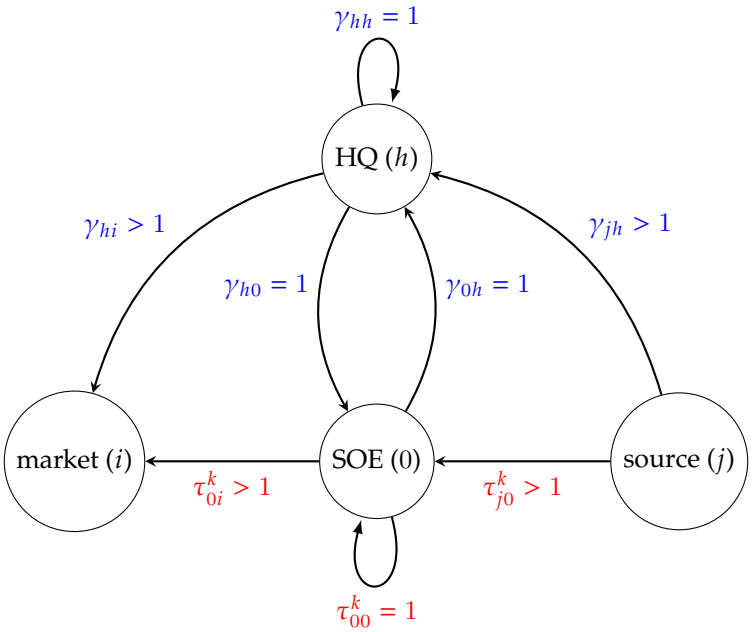
The model has two types of exogenous variable frictions: trade costs and communication costs. As is usual in the literature, I model both of them as (exogenous) iceberg costs, which means that for each unit sold in market i, a seller in country n must ship $t_{ni} > 1$ units to account for the fraction $t_{ni} - 1$ that is lost ("melts") in transit. These frictions also satisfy the triangle inequality, which means that there are no indirect routes (via a third country) cheaper than direct trade, and their value for within-country trade is normalized to one ($t_{ii} = 1$).

Trade costs. Anderson and van Wincoop (2004) broadly define trade costs as those incurred in shipping a good to final consumers on top of its marginal cost; such trade costs can include transportation costs, policy barriers (e.g., tariffs), information costs, currency exchange costs, regulatory costs, and *contract enforcement costs*. In this paper, trade costs are defined in the same way, except the contract enforcement component, which is modeled separately. Trade costs are denoted by τ_{j0}^k and τ_{0i}^k for imported inputs and exported goods, respectively.

Communication costs. Some trade costs, such a transportation costs, are intuitively related to the geographic distance between the exporter and importer locations, for domestic and foreign-owned firms alike. This is not necessarily the case for others, such as information or coordination costs, if a firm has plants in multiple locations. For example, it is reasonable to assume that, if a foreign affiliate sources locally, information gathering and coordination are conducted by its local agents but, if instead it is sourcing from abroad, such activities are conducted from the *closest* location where the firm has agents. The previous section showed evidence that the distance between the HQ and sourcing countries is negatively related to imports to their Mexican plants. To account for this relationship, I include a second set of iceberg variable trade costs, which I call *communication costs* and denote by γ_{jh} and γ_{hi}, for input sourcing and final product

sales, respectively.[30] Communication costs are normalized such that $\gamma_{hh} = 1$ and $\gamma_{0h} = \gamma_{h0} = 1$. The second normalization means that when sourcing or selling locally, foreign affiliates face the same trade and communication costs as domestic firms. The following figure summarizes the variable frictions in the model.

Figure 6. Variable frictions in the model: Trade and communication costs



3.7 Contracting and bargaining

Given that final good producers face a perfectly competitive fringe of potential suppliers in every country, when sourcing relationship-specific inputs, they will offer a take-it-or-leave-it contract in each country. However, once the contract is signed and the input is manufactured, the buyer is locked in and cannot change suppliers. Thus, the power balance between the parties changes to that of a bilateral monopoly. This accounts for the fact that buyers may have delivery deadlines that need to be met and that by the time they are able to inspect the inputs, there is not enough time to search for a new supplier. This creates the risk of suppliers trying to take advantage of the buyer and renegotiate the terms of the exchange in their favor.[31]

Sellers also face the risk of being taken advantage of by the buyer given that, by definition, relationship-specific inputs have zero value outside of the relationship, which incentivizes the buyer to also try to "renegotiate" the terms of exchange. I embed these features in the model by assuming that after contracts

[30] Antràs et al. (2023, page 21, Table 5) also show that foreign affiliates in the US import more from their HQ country and from countries in the HQ region. However, given that their paper focuses on the extensive-margin decision (which countries belong to firms' "sourcing set"), they do not model this finding. Like them, I find that foreign affiliates tend to import from countries closer to their HQ location. But in contrast to what they find, this effect does not seem driven by imports from the affiliates' HQ country.

[31] "Renegotiation" need not be taken literally. As illustrated by Antràs (2016), suppliers can de facto "renegotiate" the terms of exchange by substituting high-quality (but expensive) materials for low-quality (but cheap) ones. This risk is likelier to manifest the more difficult it is for the buyer—or a third party—infer which materials were used in the production of the input.

are signed but before trade happens, there is a renegotiation phase whose outcome determines the allocation of the rents from trade to all the parties involved.

3.7.1 Timing of contracting

1. Final good firms (buyers) post take-it-or-leave-it contracts to source each input v. These contracts stipulate two items:
   - the level of *contractible* specifications $m_c(v)$
   - a fee to win the contract $f(v)$
2. Suppliers bid for these contracts. Contracts are signed.
3. Simultaneously, buyers hire labor L, and suppliers choose how much to invest in *noncontractible* specifications $m_n(v)$ (contractible ones are fulfilled as stipulated in the contract).
4. Before handing their inputs to the buyer, the firm and its suppliers *bargain* over the distribution of the revenue $R(\varphi)$.
5. Trade happens, the final good is produced and sold, and rents are distributed according to the agreement reached in step 4.

3.7.2 Bargaining

Since buyers source more than one input, bargaining must be modeled multilaterally. In multilateral bargaining settings, a commonly used solution concept from cooperative game theory is the **Shapley value**, shown by Lloyd Shapley in 1953 (see Winter (2002)) to be the *unique* solution satisfying four axioms that, together, could be considered to characterize a "fair" distribution.[32] In addition, the allocation formula has an intuitive interpretation: It is the average marginal contribution of each player across all possible coalitions. For these reasons, I assume that the distribution of revenue between the buyer and its suppliers that arises in equilibrium from the bargaining process is characterized by their Shapley values.[33]

[32] The four axioms are: (1) efficiency (the players distribute among themselves the resources available to the grand coalition; i.e., no money is left on the table), (2) symmetry (if two players' contribution to any coalition is always the same, they should receive equal shares), (3) dummy (if a player doesn't contribute to any coalition, she should receive nothing), and (4) additivity (the rule that distributes the resources to the coalition members should be an additive operator on the space of all games).

[33] Multilateral Nash bargaining (NB) is an alternative solution concept, but there are good reasons not to use it. First, the Shapley value allows partial cooperation among subsets of agents, while NB does not. Second, and more importantly, the NB solution is incompatible with moral hazard. This conclusion is a consequence of two features of the model. On one hand, given that relationship-specific inputs have zero outside value, the NB solution allocates a *constant* share of revenue to each agent, regardless of its actions. On the other hand, given that each supplier is infinitesimal, it does not internalize the effect of its choices in total revenue. Together, these two features imply that under the NB sharing rule, all suppliers would try to free ride on the others. In equilibrium, this leads to zero supply of inputs (equation 4) and zero revenue.

4. Solving the model

4.1 Consumers' utility maximization

The preferences laid out in section 3.2 give rise to the following demand for good ω:

$$q_i^k(\omega) = p_i^k(\omega)^{-\sigma_k}(P_i^k)^{\sigma_k - 1}\beta_k E_i \tag{5}$$

where E_i is aggregate spending on country i and P_i^k is the CPI of industry k, defined as:

$$P_i^k \equiv \left(\int_{\omega\in\Omega_i^k} p_i^k(\omega)^{1-\sigma_k} d\omega\right)^{\frac{1}{1-\sigma_k}} \tag{6}$$

4.2 Plant's profit maximization

The marginal cost of any good ω is constant for all firms, with its value depending on the firm's home country, h:

$$c_h^k(\omega) = c_h^k \equiv \frac{(w_0)^{\alpha_l^k}(P_h^{sk})^{\alpha_s^k}(P_h^{rk})^{\alpha_r^k}}{\varphi_h^k A_0^k}$$

where P_h^{sk} and P_h^{rk} are the price indices for standardized and relationship-specific inputs, respectively. These indices are CES aggregators equal to:

$$P_h^{xk} \equiv \left(\int_0^1 p_h(v)^{1-\zeta_x^k} dv\right)^{\frac{1}{1-\zeta_x^k}}, \quad x \in \{r, s\} \tag{7}$$

Given that the marginal cost of production is constant, profit maximization can be solved for each destination market independently. The optimal price to sell to market i, inclusive of trade and communication costs, is also constant and the same for all goods assembled by firms from the same home country:

$$p_{hi}^k(\omega) = p_{hi}^k = \left(\frac{\sigma_k}{\sigma_k - 1}\right) c_h^k \tau_{0i}^k \gamma_{hi} \tag{8}$$

This implies that sales to any market i are also the same for all firms from the same country and assembly location. Given that there are no fixed costs for exporting, all active firms sell to all countries. Therefore, total revenue is also the same for all firms from a given home country h.[34]

$$R_h^k = \left(\frac{\sigma_k}{\sigma_k - 1}\right)^{1-\sigma_k} (c_h^k)^{1-\sigma_k} \left[\sum_{i=0}^{J} \beta_k E_i (P_i^k)^{\sigma_k - 1} (\tau_{0i}^k \gamma_{hi})^{1-\sigma_k}\right] \tag{9}$$

[34] When mapping the model to the data to conduct counterfactual exercises, I have to take a stand regarding which industries produce the intermediate inputs. I allocate the demand for standardized and relationship-specific inputs across all tradable sectors using information from firm-level input purchases and industry-level input–output tables. Strictly speaking, this means that equation 9 represents revenue only from *final good* sales, not total revenue. I postpone the inclusion of intermediate good sales until section 5.

4.3 Standardized inputs

Perfect competition in the standardized inputs market implies that they are priced at marginal cost plus trade and communication frictions. For example, if input ν is sourced from country j, its price would be:

$$p_{jh}^{sk}(\nu) = \frac{w_j \gamma_{jh} \tau_{j0}^k}{a_j^{sk}(\nu)} \tag{10}$$

Given that standardized inputs are homogeneous, firms always source from the cheapest location, so the *actual* price paid in equilibrium for input ν is:

$$p_h^{sk}(\nu) = \min_j \left\{ p_{jh}^{sk}(\nu) \right\}$$

The assumption that productivities are drawn from a Fréchet distribution implies that potential and actual prices are Weibull-distributed random variables. By taking advantage of the properties of the Weibull distribution, we can obtain a closed-form expression for the price index without having to determine the source country of each input. The closed form of equation 7 is:

$$P_h^{sk} = \left[\int_0^\infty p^{1-\zeta_s^k} d\Pr\left(p_h^{sk}(\nu) \leq p \right) \right]^{\frac{1}{1-\zeta_s^k}} = B_s^k \left[\sum_j T_j^{sk} (w_j \gamma_{jh} \tau_{j0}^k)^{-\theta_s} \right]^{-\frac{1}{\theta_s}} \tag{11}$$

where $B_s^k \equiv \Gamma\left(\frac{\theta_s + 1 - \zeta_s^k}{\theta_s} \right)^{\frac{1}{1-\zeta_s^k}}$. Similarly, the properties of the Weibull distribution imply that the share of expenditure in standardized inputs from country j is equal to the probability that an input is sourced from that country (χ_{jh}^{sk}):

$$\frac{X_{jh}^{sk}}{\sum_{j'} X_{j'h}^{sk}} = \chi_{jh}^{sk} \equiv \frac{T_j^{sk} (w_j \gamma_{jh} \tau_{j0}^k)^{-\theta_s}}{\sum_{j'} T_{j'}^{sk} (w_{j'} \gamma_{j'h} \tau_{j'0}^k)^{-\theta_s}} \tag{12}$$

4.4 Relationship-specific inputs

Unlike for standardized inputs, for relationship-specific inputs, the marginal-cost-plus-trade-costs formula is not the price paid. Instead, prices are an outcome of the sourcing game played by the buyer and its suppliers. First, I show how to calculate the players' revenue shares, which are modeled as their (asymptotic) Shapley values. Then, I use backward induction to determine the equilibrium level of any input $m_j(\nu)$. Third, I characterize the equilibrium quantities with contracting frictions relative to a frictionless benchmark. Finally, I obtain an expression for the price, inclusive of contracting frictions, that rationalizes trade flows and characterize it.

4.4.1 Bargaining solution

I assume that revenue is distributed among the agents according to their Shapley values. Thus, the first step is to obtain these functions. However, the Shapley value was originally defined for a *finite* number of players, but this model features a *continuum* of suppliers. I follow Acemoglu, Antràs and Helpman (2007) in adapting it to a game with a continuum of players in what is known as the *asymptotic* Shapley value.

Proposition 1 (Revenue shares)
Let input ν be produced by a supplier in country j.

1. *The supplier's share of rents is:*

$$s_j^k(\nu) = \left(\frac{\alpha_r^k \rho_k}{\alpha_r^k \rho_k + \rho_r^k}\right)\left(\frac{m_j^k(\nu)}{M_r^k}\right)^{\rho_r^k} \tag{13}$$

   where $\rho_r^k \equiv \frac{\zeta_r^k - 1}{\zeta_r^k}$ and $\rho_k \equiv \frac{\sigma_k - 1}{\sigma_k}$.

2. *The buyer's share of rents is:*

$$s_b^k = \frac{\rho_r^k}{\alpha_r^k \rho_k + \rho_r^k} \tag{14}$$

Proposition 1 shows that the buyer obtains a larger share when (1) it is easy to substitute across inputs ($\uparrow \zeta_r^k$), (2) the technological need for inputs is low ($\downarrow \alpha_r^k$), and (3) there is low competitive pressure to reduce costs ($\downarrow \sigma_k$). Moreover, note that because the revenue share of supplier ν depends on the value of $m_j(\nu)$, it has an incentive to pick a nonzero level of noncontractible specifications (see footnote 33).

4.4.2 Effective price

Given that suppliers know the share they will be able to extract from their buyer in advance (equation 13), they take this into account when choosing how much to invest in noncontractible specifications.[35] Likewise, given that the buyer also knows beforehand the revenue shares and the behavior of its suppliers, it takes these into account when designing the contract. Thus, we can obtain the equilibrium levels of contractible and noncontractible specifications by solving the buyer–supplier game backwards. The solution implies the following expression for the prices for relationship-specific inputs.

Proposition 2 (Price for relationship-specific inputs)
Let input ν be produced by a supplier in country j. The price that rationalizes trade flows from country j is:

$$p_{jh}^{rk}(\nu) = \frac{w_j \gamma_{jh} \tau_{j0}^k \Lambda_j^k}{a_j^{rk}(\nu)} \tag{15}$$

[35] Given the atomistic nature of suppliers, they take R and M_r as given when maximizing their payoff.

where:

$$\Lambda_j^k \equiv \frac{1}{s_b^{1-\mu_j}\left(\Xi_j^k\right)^{\mu_j}\left[\mu_j\Xi_j^k+(1-\mu_j)s_b\right]^{\frac{1-\rho_r^k}{\rho_r^k}}} \qquad \textit{and} \qquad \Xi_j^k \equiv \left[\frac{1-s_b(1-\mu_j)\rho_r^k}{1-(1-\mu_j)\rho_r^k}\right]$$

The following proposition characterizes the contracting frictions parameter Λ_j^k:

Proposition 3 (Properties of the contracting frictions parameter Λ_j^k)

1. *Λ_j^k behaves likes an iceberg variable cost, in the sense that it is bounded from below by one and it is unbounded from above:*

$$\Lambda_j^k \in [1, +\infty)$$

2. *Λ_j^k decreases (contracting frictions are lower) with (1) the strength of contract enforcement in the supplier's country (μ_j) and (2) the elasticity of substitution among relationship-specific inputs (ζ_r^k). On the other hand, it increases with (1) the buyer's technological need for these inputs (α_r^k) and (2) the elasticity of demand the buyer faces for its products (σ_k):*

$$\Lambda_j^k(\overbrace{\zeta_r^k}^{-},\overbrace{\alpha_r^k}^{+},\overbrace{\sigma_k}^{+},\overbrace{\mu_j}^{-})$$

3. *Λ_j^k converges to one (contracting frictions disappear) as (1) contract enforcement becomes perfect (μ_j), (2) the buyer's technological need for relationship-specific inputs goes to zero (α_r^k), or (3) the elasticity of demand the buyer faces for its products becomes unit elastic (σ_k). On the other hand, it diverges (trade collapses) as the elasticity of substitution among relationship-specific inputs becomes unit elastic (ζ_r^k).*

$$\lim_{\mu_j \nearrow 1}\Lambda_j^k = \lim_{\alpha_r^k \searrow 0}\Lambda_j^k = \lim_{\sigma_k \searrow 1}\Lambda_j^k = 1, \quad \textit{and} \quad \lim_{\zeta_r^k \searrow 1}\Lambda_j^k = +\infty$$

The closed-form expression for this price index is obtained following the same steps as the standardized inputs case: First, rewrite the integral in equation 7 in terms of prices and then use their cumulative density function to obtain:

$$P_h^{rk} = \left[\int_0^\infty p^{1-\zeta_r^k} d\Pr\left(p_h^{rk}(v) \leq p\right)\right]^{\frac{1}{1-\zeta_r^k}} = B_r^k\left[\sum_j T_j^{rk}(w_j\gamma_{jh}\tau_{j0}^k\Lambda_j^k)^{-\theta_r}\right]^{-\frac{1}{\theta_r}} \tag{16}$$

where $B_r^k \equiv \Gamma\left(\frac{\theta_r+1-\zeta_r^k}{\theta_r}\right)^{\frac{1}{1-\zeta_r^k}}$. The main difference between the two indices is that relationship-specific inputs are affected by the contracting frictions term Λ_j^k. Similarly, the closed-form expression for the share of expenditure in relationship-specific inputs from country j is equal to the probability that an input is sourced from that country (χ_{jh}^{rk}):

$$\frac{X_{jh}^{rk}}{\sum_{j'}X_{j'h}^{rk}} = \chi_{jh}^{rk} \equiv \frac{T_j^{rk}(w_j\gamma_{jh}\tau_{j0}^k\Lambda_j^k)^{-\theta_r}}{\sum_{j'}T_{j'}^{rk}(w_{j'}\gamma_{j'h}\tau_{j'0}^k\Lambda_{j'}^k)^{-\theta_r}} \tag{17}$$

4.5 Equilibrium

A **small open economy equilibrium** is defined as the vector of domestic wage (w_0), mass of domestic firms per industry ($\{N_0^k\}_k$), and consumer price indices per industry ($\{P_0^k\}_k$) such that (1) domestic consumers maximize utility, (2) all firms with plants in the SOE maximize profits, (3) all markets clear, and (4) the zero-profit condition is met. More detail can be found in appendix A.4.[36]

5. Counterfactual exercises

The empirical goal of this paper is to estimate the relative importance of the three potential channels that may advantage foreign multinationals over domestic firms in manufacturing. Each channel is represented in the theoretical model by a specific parameter. First, contracting frictions are represented by the parameter Λ_j^k. Because I allow international trade in inputs, firms are not constrained to local contract enforcement institutions since they can get around these by importing relationship-specific inputs.[37] Since this is a prerogative of both domestic and foreign firms, to allow for the possibility of a *contracting frictions channel*, I let the parameter vary by HQ country (Λ_{jh}^k), although this is not explicitly modeled. Second, international trade costs are represented by $\{\tau_{j0}^k, \tau_{0i}^k\}$ and $\{\gamma_{jh}, \gamma_{hi}\}$. Since the parameters in the first group (the traditional trade costs) are indexed by the place where production takes place (the SOE), foreign and domestic firms sourcing from the same country necessarily face the same wedge. In contrast, the communication parameters are indexed by the HQ country, which allows for differences between domestic and foreign firms. Hence, this is the set of parameters that embody the *geographical channel*, which can exist in sourcing (γ_{jh}) and in marketing (γ_{hi}). Finally, any other source of advantage unrelated to geography or contracting frictions (such as technology) is represented by the productivity parameter φ_h^k.

To isolate and compare the quantitative effects of these three channels, I follow the exact-hat algebra approach of Dekle et al. (2008) to estimating counterfactual equilibria. This method has the advantage of reducing the number of parameters for estimation/calibration by focusing only on those relevant to the exercise at hand. This approach has been extensively used in the literature to conduct counterfactual exercises in which the size of the shock (defined as a percentage change, e.g., "a 10 percent reduction in trade costs") is chosen by the researcher. One reason to frame counterfactual exercises this way is that the objects being shocked are unobservable. This paper is different because the size of a shock is implicitly defined as the percentage change that neutralizes a specific channel, which requires knowledge of its current size. For example, "turning off the productivity channel" means calculating an equilibrium in which φ_h^k becomes φ_0^k for all foreign firms. Whether this is a 10 percent or 50 percent reduction in φ_h^k hinges on the size of φ_0^k/φ_h^k. Given that none of the three parameters are directly observable, I devise a strategy to back out these relative

[36] In appendix A, I expand the model by adding two sectors (other nonmanufacturing tradables and nontradables) to map it to the data more accurately. I restate the equations in the main body whenever it is necessary to account for these additional sectors. I explain the aggregation of firms into the firm groups in appendix A.3 and define in detail the equilibrium in appendix A.4.

[37] This is true as long as the relevant contract enforcement institutions are the *exporter's*. The argument behind this is elaborated in section 3.5.2.

sizes using observable empirical moments.

In the next sections, I go over the details of each counterfactual exercise. I follow the convention in the literature of using $\hat{x}$ to represent $\frac{x'}{x}$, where x and x' are the initial and final values of variable x, respectively. I use colors to highlight the shocks to the contracting, geography, and productivity channels and the empirical moments.

5.1 Shocks

The three channels discussed above imply three different shocks to the equilibrium equations. I isolate each channel by shocking one parameter at a time and solving the model to obtain its corresponding counterfactual equilibrium. I can gauge the channels' relative importance by choosing an outcome (for example, the change in real GDP) and comparing its value with respect to that in the case when *all* parameters are shocked simultaneously. The shocks are:

1. *Contracting frictions:* if foreign affiliates faced the same contracting frictions as domestic firms when sourcing inputs:
$$\Lambda_{jh}^k \to \Lambda_{j0}^k, \quad \forall h \in \mathcal{J}_0^k$$

2. *Geography:* if foreign affiliates faced the same communication frictions as domestic firms when
   (a) sourcing inputs:
   $$\gamma_{jh} \to \gamma_{j0}, \quad \forall h \in \cup_k \mathcal{J}_0^k$$
   (b) marketing their products:
   $$\gamma_{hi} \to \gamma_{0i}, \quad \forall h \in \cup_k \mathcal{J}_0^k$$

3. *Productivity:* if foreign affiliates had the same productivity as domestic firms:
$$\varphi_h^k \to \varphi_0^k, \quad \forall h \in \mathcal{J}_0^k$$

where $\mathcal{J}_0^k$ is the subset of countries with plants in the SOE.

5.2 Identification

To compute the counterfactuals, I need to measure the empirical moments, estimate or calibrate the parameters, and identify the shocks from the equations in the last section. In this section, I explain how this is done.

5.2.1 Parameter calibration

The counterfactual exercises require the technology parameters (αs), trade elasticities for inputs (θs), and demand elasticities for final goods (σs). In addition, to calculate the counterfactual change in real GNP, I need to know the preferences parameters (βs). In this section, I explain how I calibrate their values. A table with the values can be found in appendix A.6.1.

Technology parameters. To calibrate the technology parameters, I rely on the Cobb–Douglas production function, which implies that, in equilibrium, inputs' shares in total cost are equal to their respective production function exponents. Hence, for each industry k, I calibrate the technology parameters according to the following formulas:[38]

$$\alpha_l^k = \frac{W_k}{W_k + IC_k} \qquad \alpha_s^k = \left(\frac{IC_g^{sk}}{IC_g^k}\right)\left(\frac{IC_g^k}{W_k + IC_k}\right)$$
$$\alpha_N^k = \frac{IC_N^k}{W_k + IC_k} \qquad \alpha_r^k = \left(\frac{IC_g^{rk}}{IC_g^k}\right)\left(\frac{IC_g^k}{W_k + IC_k}\right)$$

where W_k is the industry-level aggregate wage bill; IC_k is the industry-level intermediate consumption; IC_g^k and IC_N^k are the industry-level intermediate consumption of tradable and nontradable goods and services, respectively; and IC_g^{sk} and IC_g^{rk} are the industry-level intermediate consumption of tradable standardized and relationship-specific inputs, respectively. The first four are directly observable in the economic census. The last two are obtained by the strategy used to disaggregate total intermediate sales between input types.

Elasticities. I assume that trade in standardized and trade in relationship-specific inputs are governed by the same trade elasticity, which I make equal to $\theta = 4$, in line with the main result in Simonovska and Waugh (2014). For the demand elasticity σ_k, I rely on the model result that profits are proportional to total revenue (equation 30). This implies that the elasticity can be calibrated with the following formula:

$$\sigma_k = \frac{R_k}{R_k - (W_k + IC_k)}$$

where R_k is the industry-level aggregate revenue.[39]

Preference parameters. To calibrate the preference parameters, I rely on the Cobb–Douglas utility function, which implies that, in equilibrium, industries' shares in final expenditure are equal to their respective utility function exponents. Hence, for each industry k, I calibrate the preference parameters according to the following formulas:

$$\beta_k = \frac{E_0^k}{E_0} \qquad \beta_T = \frac{E_0^T}{E_0} \qquad \beta_N = \frac{E_0^N}{E_0}$$

[38] Since I assume that foreign affiliates and domestic firms use the same technology (allowing for different TFPs), I pool all firms in an industry when making these calculations.

[39] I use this same formula to calibrate the demand elasticity of the nonmanufacturing sector σ_T, in spite of its being a competitive sector with representative firms.

where E_0^k, E_0^T, and E_0^N are the industry-level aggregate final expenditures and E_0 is the total aggregated final expenditure, all of which are observed in the input–output tables.

5.2.2 Identification of shocks

In section 5.1, I showed that each counterfactual scenario involves a shock to a specific parameter and that, in all cases, the shock is defined in terms of the relative value between foreign and domestic firms. This means that I do not need to identify the parameter *levels* from the data. It is enough for me to back out, for each foreign HQ country $h \neq 0$, the relative size of the foreign firms' parameters with respect to their domestic counterparts'. The following proposition summarizes how this is done.

Proposition 4 (Identification of parameter shocks)
For any input source country j, destination market i, HQ country h, industry k, and variable x, let $\widehat{x}_h$ represent the relative value of x between domestic firms (country 0) and foreign affiliates from country h:

$$\widehat{x}_h \equiv \frac{x_0}{x_h}$$

The parameter shocks associated with each counterfactual scenario can be identified with the following formulas:

1. ***Contracting frictions:***

$$\left(\widehat{\Lambda}_{jh}^k\right)^{-\theta_r} = \left(\frac{\widehat{\chi}_{jh}^{rk}}{\widehat{\chi}_{j^*h}^{rk}}\right)\left(\frac{\widehat{\chi}_{j^*h}^{sk}}{\widehat{\chi}_{jh}^{sk}}\right)^{\frac{\theta_r}{\theta_s}} \tag{18}$$

   where j^* refers to any country with close to "perfect" contract enforcement institutions ($\mu_{j^*} \approx 1$).

2. ***Geography (input purchases):***

$$\left(\widehat{\gamma}_{jh}\right)^{-\theta_s} = \frac{\widehat{\chi}_{jh}^{sk}}{\widehat{\chi}_{0h}^{sk}} \tag{19}$$

   Alternatively:

$$\left(\widehat{\gamma}_{jh}\right)^{-\theta_r} = \left(\frac{\widehat{\chi}_{jh}^{rk}}{\widehat{\chi}_{0h}^{rk}}\right)\left(\frac{\widehat{\Lambda}_{0h}^k}{\widehat{\Lambda}_{jh}^k}\right)^{-\theta_r} \tag{20}$$

3. ***Geography (sales):***

$$\left(\widehat{\gamma}_{hi}\right)^{1-\sigma_k} = \frac{\widehat{R}_{hi}^{k,f}}{\widehat{R}_{h0}^{k,f}} \tag{21}$$

4. ***Productivity:***

$$\left(\widehat{\varphi}_h^k\right)^{\sigma_k - 1} = \left(\frac{\widehat{R}_{h0}^{k,f}}{\widehat{N}_h^k}\right)\left[\left(\frac{1}{\widehat{\chi}_{0h}^{sk}}\right)^{\frac{\alpha_s^k}{\theta_s}}\left(\frac{1}{\widehat{\chi}_{j^*h}^{rk}}\right)^{\frac{\alpha_r^k}{\theta_r}}\left(\frac{\widehat{\chi}_{j^*h}^{sk}}{\widehat{\chi}_{0h}^{sk}}\right)^{\frac{\alpha_r^k}{\theta_s}}\right]^{1-\sigma_k} \tag{22}$$

Alternatively:

$$\left(\widehat{\varphi}_h^k\right)^{\sigma_k-1} = \left(\frac{\widehat{R}_{h0}^{k,f}}{\widehat{N}_h^k}\right)\left\{\left[\sum_j \chi_{jh}^{sk}\left(\widehat{\gamma}_{jh}\right)^{-\theta_s}\right]^{\frac{\alpha_s^k}{\theta_s}}\left[\sum_j \chi_{jh}^{rk}\left(\widehat{\gamma}_{jh}\widehat{\Lambda}_{jh}^k\right)^{-\theta_r}\right]^{\frac{\alpha_r^k}{\theta_r}}\right\}^{1-\sigma_k} \tag{23}$$

5.3 Implied shocks

5.3.1 Contracting frictions

To back out contracting frictions using equation 18, I need to observe several coincidences: Firms from HQ country h *and* Mexico must source from j *and* j^* for *both* standardized and relationship-specific inputs. This is a requirement that many observations do not satisfy, mostly because firms source standardized and relationship-specific inputs from different countries. I work around this issue by making the following assumptions. First, instead of picking only one country as j^* (that is, one with contract enforcement institutions strong enough for contracting frictions not to be present for either the MN or Mexican firms), I use many from the top of the rule of law ranking.[40] Second, I exploit the assumption that the second term in parentheses in the formula does not vary by industry:

$$\left(\frac{\chi_{jh}^{sk}/\chi_{j^*h}^{sk}}{\chi_{j0}^{sk}/\chi_{j^*0}^{sk}}\right)^{\frac{\theta_r}{\theta_s}} = \left(\frac{\gamma_{jh}/\gamma_{j^*h}}{\gamma_{j0}/\gamma_{j^*0}}\right)^{\theta_r}$$

This means that I can also consider matches for which the ratios for relationship-specific and standardized inputs come from different industries (but same year and country pair). Third, I also consider matches between ratios from different years. However, when there is more than one measurement available, I prefer the one from the same year, reasoning that it is likelier that communication costs are the same across industries than they are constant in time. Finally, to increase my chance of finding a measurement based on ratios from the same year and industry, I pool all the measures from the three years of data and pick only one per industry and HQ–sourcing country pair. The result is a highly heterogeneous set of candidate measures, even for the same industry and country pair. In some cases, these measures reach very extreme values, which I interpret as most likely representing noise as opposed to real differences in effective contracting frictions.

Since there is not an obvious or established set of criteria to filter the most accurate measures, I devise one. I sequentially apply nine criteria to filter the competing measures. The first three are the most important, in the sense that they discard the largest number of candidates. First, I keep the measures with the best

[40]The sourcing locations for 2018 are, in rank order, Finland, Norway, Switzerland, Austria, New Zealand, Singapore, Sweden, Denmark, Luxembourg, Netherlands, Hong Kong, Canada, Greenland, Jersey, Iceland, Australia, Liechtenstein, Gibraltar, United Kingdom, Germany, Andorra, Japan, United States, Ireland, France, and Belgium. The lowest index value for an included location, that for Belgium, is percentile 88.5.

quality match.[41] Among the remaining candidates, I keep those "closest" (from either left or right) to one.[42] My goal is to avoid selecting extreme values without giving preference to the smallest measures. This second criterion is aimed at reducing the level of noise in measurement, although it is true that, by reducing variation, this method attenuates the quantitative effect of the contracting frictions channel. Third, among the remaining values, I prefer the one associated with the j^* country with the best position on the rule of law ranking, under the idea that the higher the rank, the closer the country is to the ideal of perfect contract enforcement institutions.

Table 5 shows summary statistics of the distribution of measured ratios for foreign firms disaggregated by industry. Overall, foreign firms enjoy advantages with some countries but disadvantages with others. The last row ("Total") shows a median ratio of one (neither advantage nor disadvantage). However, this is not one enjoyed across the board: The minimum ratio is 0.10 (domestic firms face only 10 percent of the contracting friction faced by foreign firms) and a maximum of 6.16 (some foreign firms face 16 percent of the contracting friction faced by domestic firms), with a mean value of 1.05.

[41] I assign the matches to three categories: (1) The "best" corresponds to instances in which both types of inputs were sourced from country j in the same year, (2) the "middle" category corresponds to cases in which there is a match of either the year (but different industry) or industry (but different year), and (3) the "worst" category corresponds to matches between different years and industries.

[42] I measure "distance" as the absolute value of the difference between the measure and one. For measures less than one (i.e., on which domestic firms have an advantage), I first invert their value to remove the bias that the zero lower bound creates. Before I apply this criterion, the ratios $\Lambda^k_{j0}/\Lambda^k_{jh}$ range from 0.01 to 70.91, with a mean of 1.60 and a median of one. After this step, the values range from 0.18 to 6.16, with a mean of 1.05 and a median of one.

Table 5. Model-implied values for $\Lambda^k_{j0}/\Lambda^k_{jh}$ for foreign firms in Mexico by industry, 2018

Industry	Min	p1	p5	p10	p25	median	p75	p90	p95	p99	Max	Mean	Obs
Food	0.27	0.27	0.86	0.92	1.00	1.00	1.03	1.17	1.41	1.66	1.66	1.02	35
Textile inputs	0.92	0.92	0.92	0.92	0.92	0.96	1.00	1.00	1.00	1.00	1.00	0.96	2
Nonapparel textiles	0.91	0.91	0.91	0.96	1.00	1.00	1.00	1.01	1.03	1.03	1.03	0.99	13
Apparel	0.84	0.84	0.88	0.93	0.98	1.00	1.00	1.11	1.16	1.27	1.27	1.00	30
Leather	0.59	0.59	0.59	0.98	0.99	1.00	1.00	1.01	1.01	1.01	1.01	0.97	14
Wood	0.95	0.95	0.95	1.00	1.00	1.00	1.01	1.02	3.94	3.94	3.94	1.27	11
Paper	0.18	0.18	0.89	0.91	0.99	1.00	1.00	1.02	1.03	1.18	1.18	0.97	27
Printing	0.89	0.89	0.98	1.00	1.00	1.00	1.02	1.07	1.17	1.20	1.20	1.02	25
Chemicals	0.71	0.72	0.91	0.95	1.00	1.00	1.00	1.16	1.27	1.96	2.53	1.04	120
Plastics & rubber	0.53	0.63	0.86	0.93	1.00	1.00	1.00	1.10	2.95	4.57	4.87	1.15	136
Nonmetal mineral goods	0.74	0.74	0.96	0.99	1.00	1.00	1.04	1.18	1.20	1.48	1.48	1.03	33
Basic metals	0.34	0.34	0.74	0.92	1.00	1.00	1.00	1.18	1.42	2.49	2.49	1.04	61
Metal goods	0.51	0.51	0.68	0.85	1.00	1.00	1.01	1.09	1.25	1.70	1.70	1.00	95
Machinery	0.43	0.74	0.92	0.93	1.00	1.00	1.01	1.11	1.29	2.80	5.99	1.09	102
Electronics	0.58	0.63	0.78	0.92	1.00	1.00	1.00	1.09	1.60	2.93	6.16	1.09	114
Electric equip.	0.43	0.43	0.82	0.89	1.00	1.00	1.01	1.06	1.14	1.98	1.98	1.00	87
Transport equip.	0.47	0.71	0.86	0.97	1.00	1.00	1.00	1.10	1.19	2.46	4.37	1.05	229
Furniture	0.10	0.10	0.40	0.76	0.99	1.00	1.00	1.41	1.84	2.12	2.12	1.03	33
Other	0.33	0.33	0.64	0.91	0.99	1.00	1.00	1.16	1.38	4.05	4.05	1.05	72
Total	0.10	0.53	0.85	0.93	1.00	1.00	1.00	1.10	1.27	2.95	6.16	1.05	1,239

Notes: The column "Obs" counts the number of ratios identified in each industry, not the total number of observations in the data. "Beverages and Tobacco" is omitted because there are no observations. "Petrochemicals" is omitted because it is an industry where restrictions on foreign direct investment remain in place.

5.3.2 Communication frictions

Given that *inbound* communication costs affect imports of both types of inputs, they can be identified by the expenditure shares of both types separately; that is, we no longer need firms to source both types from the same country j. Instead, the requirement is for a group of firms to source their standardized inputs domestically in addition to importing them from j and for domestic-owned firms to also buy from both locations (see equation 19). Unfortunately, given that firms source their standardized inputs from different countries than they do their relationship-specific ones, I need to also use equation 20 if I am to identify the ratio for a country from which standardized inputs are not imported. The downside of my using the second equation is that it requires me to control for contracting frictions differences, many of which are measured on the basis of the additional assumptions described previously. In contrast to my approach in measuring contracting frictions, the goal for the communication costs parameter is to have one measure per HQ–source country pair common to all industries in a given year. As was the case for the contracting

frictions, the measures are very heterogeneous and encompass a large range of values. Therefore, I apply the same strategy of choosing the value closest to one to filter out extreme values.[43]

Table 6 shows summary statistics of the distribution of ratios of inbound communication cost between domestic and foreign firms disaggregated by industry.[44] Overall, foreign firms enjoy advantages with some countries but disadvantages with others. The last row ("Total") shows a median ratio of one (neither advantage nor disadvantage) and extreme values that mirror each other: A minimum of 0.16 means domestic firms face only 16 percent of the communication costs faced by domestic firms, and a maximum of 6.17 indicates the exact same difference favoring foreign firms.

Table 6. Model-implied values for γ_{j0}/γ_{jh} for foreign firms in Mexico by industry, 2018.

Industry	Min	p1	p5	p10	p25	median	p75	p90	p95	p99	Max	Mean	Obs
Food	0.90	0.90	0.92	0.95	0.98	1.00	1.01	1.07	1.10	1.21	1.21	1.00	34
Textile inputs	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	1
Nonapparel textiles	0.90	0.90	0.90	0.97	0.98	0.99	1.01	1.01	1.08	1.08	1.08	0.99	15
Apparel	0.90	0.90	0.92	0.94	0.97	0.99	1.00	1.01	1.01	1.08	1.08	0.99	30
Leather	0.93	0.93	0.93	0.98	0.99	0.99	1.00	1.01	1.08	1.08	1.08	1.00	13
Wood	0.90	0.90	0.90	0.93	0.98	1.00	1.01	1.01	1.10	1.10	1.10	0.99	13
Paper	0.90	0.90	0.92	0.93	0.98	1.00	1.01	1.01	1.08	1.10	1.10	0.99	27
Printing	0.90	0.90	0.92	0.93	0.98	1.00	1.01	1.01	1.08	1.21	1.21	1.00	27
Chemicals	0.40	0.66	0.78	0.92	0.97	1.00	1.08	1.40	2.08	5.94	6.17	1.21	115
Plastics & rubber	0.38	0.49	0.66	0.86	0.95	1.00	1.05	1.18	1.36	2.08	2.19	1.01	129
Nonmetal mineral goods	0.90	0.90	0.92	0.95	0.97	0.99	1.01	1.07	1.10	1.21	1.21	1.00	37
Basic metals	0.38	0.38	0.76	0.90	0.97	1.00	1.01	1.10	1.23	2.89	2.89	1.03	63
Metal goods	0.57	0.57	0.83	0.93	0.97	1.00	1.05	1.12	1.28	2.08	2.08	1.02	94
Machinery	0.18	0.28	0.60	0.84	0.96	0.99	1.01	1.09	1.22	1.44	1.49	0.97	100
Electronics	0.16	0.43	0.61	0.78	0.95	1.00	1.03	1.16	1.32	1.83	2.53	1.00	106
Electric equip.	0.20	0.20	0.61	0.90	0.97	1.00	1.02	1.10	1.36	2.02	2.02	1.01	85
Transport equip.	0.18	0.29	0.43	0.60	0.90	1.00	1.07	1.37	1.49	3.41	3.97	1.02	221
Furniture	0.90	0.90	0.93	0.95	0.97	0.99	1.00	1.01	1.07	1.10	1.10	0.99	43
Other	0.77	0.77	0.92	0.94	0.97	1.00	1.04	1.08	1.11	1.30	1.30	1.01	69
Total	0.16	0.38	0.66	0.89	0.97	1.00	1.03	1.13	1.36	2.08	6.17	1.02	1,222

Notes: The column "Obs" counts the number of ratios identified in each industry, not the total number of observations in the data. "Beverages and Tobacco" is omitted because there are no observations. "Petrochemicals" is omitted because it is an industry where restrictions on foreign direct investment remain in place.

Outbound communication costs are more straightforward to identify; to do so, I require only observations

[43] I measure "distance" the same way I did for contracting frictions (see footnote 42). Before I apply this criterion, the ratios γ_{j0}/γ_{jh} range from 0.09 to 9.33, with a mean of 1.22 and a median of 1.02. After this step, the values range from 0.16 to 6.17, with a mean of 1.06 and a median of 0.99.

[44] I exclude domestic purchases given that they are normalized to one for all firms.

of firms that export and sell domestically and of domestic-owned firms that do so, as well, as in equation 21. Table 7 shows summary statistics of the distribution of ratios of outbound communication cost between domestic and foreign firms disaggregated by industry.[45] In this case, it is clearer that foreign firms may enjoy an advantage over domestic firms when selling to foreign markets. The last row ("Total") shows a median ratio of 1.47 and a mean ratio of 1.97, which means that Mexican firms face marketing costs almost 50 percent higher than those of foreign multinationals at the median market.

Table 7. Model-implied values for $\gamma_{0i}^k/\gamma_{hi}^k$ for foreign firms in Mexico by HQ country, 2018

Industry	Min	p1	p5	p10	p25	median	p75	p90	p95	p99	Max	Mean	Obs
Food	0.17	0.17	0.17	0.37	0.60	0.71	1.34	1.44	1.79	1.79	1.79	0.87	13
Textile inputs	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1
Nonapparel textiles	1.73	1.73	1.73	1.73	1.73	1.73	1.73	1.73	1.73	1.73	1.73	1.73	1
Apparel	0.80	0.80	0.80	0.98	1.31	1.71	1.95	2.77	3.52	3.52	3.52	1.76	10
Leather	2.51	2.51	2.51	2.51	2.51	2.51	2.51	2.51	2.51	2.51	2.51	2.51	1
Wood	3.62	3.62	3.62	3.62	3.62	3.62	3.62	3.62	3.62	3.62	3.62	3.62	1
Paper	0.77	0.77	0.77	0.77	1.45	2.29	4.30	6.12	6.12	6.12	6.12	2.87	4
Printing	1.98	1.98	1.98	1.98	1.98	1.98	1.98	1.98	1.98	1.98	1.98	1.98	1
Chemicals	0.30	0.42	0.63	0.75	1.22	1.72	2.67	3.90	5.91	6.57	10.70	2.16	101
Plastics & rubber	0.46	0.46	0.68	0.80	1.16	1.42	1.98	2.44	3.14	5.10	5.10	1.60	80
Nonmetal mineral goods	2.42	2.42	2.42	2.42	2.42	2.42	2.42	2.42	2.42	2.42	2.42	2.42	1
Basic metals	0.14	0.14	0.14	0.58	1.32	1.49	1.94	2.48	3.82	3.82	3.82	1.64	13
Metal goods	0.43	0.43	0.63	0.87	1.28	1.69	2.46	6.42	10.61	10.89	10.89	2.66	37
Machinery	0.34	0.34	0.39	0.44	0.50	0.77	1.29	1.86	2.10	2.83	2.83	0.98	45
Electronics	0.59	0.59	0.59	0.71	0.98	1.20	1.46	1.90	2.10	2.28	2.28	1.27	33
Electric equip.	0.35	0.35	0.41	0.52	0.80	1.24	1.64	2.74	3.38	5.50	5.50	1.51	32
Transport equip.	0.09	0.09	0.31	0.47	1.04	1.69	3.09	6.18	9.76	14.81	15.81	2.74	110
Furniture	2.94	2.94	2.94	2.94	2.94	2.94	2.94	2.94	2.94	2.94	2.94	2.94	1
Other	0.86	0.86	0.86	0.95	1.36	1.55	2.33	2.98	3.55	3.55	3.55	1.83	17
Total	0.09	0.27	0.45	0.62	0.96	1.47	2.14	3.67	5.35	10.70	15.81	1.97	502

Notes: The column "Obs" counts the number of ratios identified in each industry, not the total number of observations in the data. "Beverages and Tobacco" is omitted because there are no observations. "Petrochemicals" is omitted because it is an industry where restrictions on foreign direct investment remain in place.

5.3.3 Productivity differences

In proposition 4, I showed two ways to isolate productivity differences from the other two channels. In principle, it should not matter which equation I use given that they are equivalent, but the data limitations detailed in sections 5.3.1 and 5.3.2 mean that equation 23 is preferable in terms of increasing the number of usable observations. Table 8 shows summary statistics of the distribution of the ratio of inferred productivity,

[45] I exclude domestic sales given that they are normalized to one for all firms.

disaggregated by industry. With a median of 0.60 and mean of 0.62, **foreign firms have an advantage in productivity in most industries**. Another salient feature is that the industries with the largest presence of foreign firms (transportation equipment, electronics, and electric equipment) are *not* the ones with the largest productivity gaps. On the contrary, though foreign firms in these industries are between 9 percent and 13 percent more productive than their domestic peers at the median, in other industries, such as food and beverages and tobacco, the gap is around 1000 percent.

Table 8. Estimates of φ_0^k/φ_h^k for foreign firms in Mexico by industry, 2018

Industry	Min	p1	p5	p10	p25	median	p75	p90	p95	p99	Max	Mean	Obs
Food	0.04	0.04	0.04	0.07	0.08	0.11	0.22	0.27	1.11	1.11	1.11	0.21	11
Beverages & Tobacco	0.01	0.01	0.01	0.01	0.01	0.05	0.73	2.50	2.50	2.50	2.50	0.56	6
Textile inputs	0.25	0.25	0.25	0.25	0.28	0.37	0.45	0.46	0.46	0.46	0.46	0.36	6
Nonapparel textiles	0.22	0.22	0.22	0.22	0.22	0.23	0.25	0.25	0.25	0.25	0.25	0.23	2
Apparel	0.23	0.23	0.23	0.23	0.29	0.35	0.63	0.91	0.91	0.91	0.91	0.46	4
Leather	0.34	0.34	0.34	0.34	0.44	0.57	0.64	0.68	0.68	0.68	0.68	0.53	5
Wood	0.05	0.05	0.05	0.05	0.05	0.17	0.24	0.24	0.24	0.24	0.24	0.16	3
Paper	0.21	0.21	0.21	0.21	0.29	0.39	0.59	0.95	0.95	0.95	0.95	0.46	8
Printing	0.54	0.54	0.54	0.54	0.54	0.62	0.78	0.88	0.88	0.88	0.88	0.66	4
Chemicals	0.34	0.34	0.34	0.46	0.53	0.63	0.96	1.11	1.28	1.28	1.28	0.74	15
Plastics & rubber	0.47	0.47	0.47	0.56	0.59	0.65	0.71	0.99	1.13	1.13	1.13	0.70	18
Nonmetal mineral goods	0.10	0.10	0.10	0.10	0.15	0.18	0.25	0.26	0.26	0.26	0.26	0.19	6
Basic metals	0.33	0.33	0.33	0.65	0.66	0.81	1.08	1.13	2.04	2.04	2.04	0.90	11
Metal goods	0.16	0.16	0.16	0.24	0.26	0.29	0.35	0.42	0.60	0.60	0.60	0.32	15
Machinery	0.36	0.36	0.36	0.38	0.41	0.56	0.89	1.00	1.60	1.60	1.60	0.67	15
Electronics	0.75	0.75	0.75	0.80	0.81	0.88	0.95	1.13	1.24	1.24	1.24	0.92	13
Electric equip.	0.65	0.65	0.65	0.67	0.83	0.92	1.06	1.16	1.16	1.16	1.16	0.92	10
Transport equip.	0.39	0.39	0.39	0.61	0.73	0.88	1.11	1.51	1.56	1.56	1.56	0.94	17
Furniture	0.34	0.34	0.34	0.34	0.34	0.36	0.37	0.37	0.37	0.37	0.37	0.36	3
Other	0.33	0.33	0.33	0.33	0.38	0.47	0.50	1.02	1.02	1.02	1.02	0.51	9
Total	0.01	0.01	0.10	0.19	0.34	0.60	0.88	1.08	1.16	2.04	2.50	0.63	181

Note: The column "Obs" counts the number of ratios identified in each industry, not the total number of observations in the data. "Petrochemicals" is omitted because it is an industry where restrictions on foreign direct investment remain in place.

5.4 Results

In this section, I present the quantitative results of the counterfactual exercises. For each scenario, I evaluate three responses: the change in real GDP per capita, the change in real expenditure (GNP) per capita, the change in unit costs, and the reallocation of value added and employment across manufacturing industries. To conduct these exercises, I rewrite the system of equations that characterize the equilibrium in relative

changes. Details can be found in appendix A.5.

5.4.1 Benchmark scenario: Joint shocks

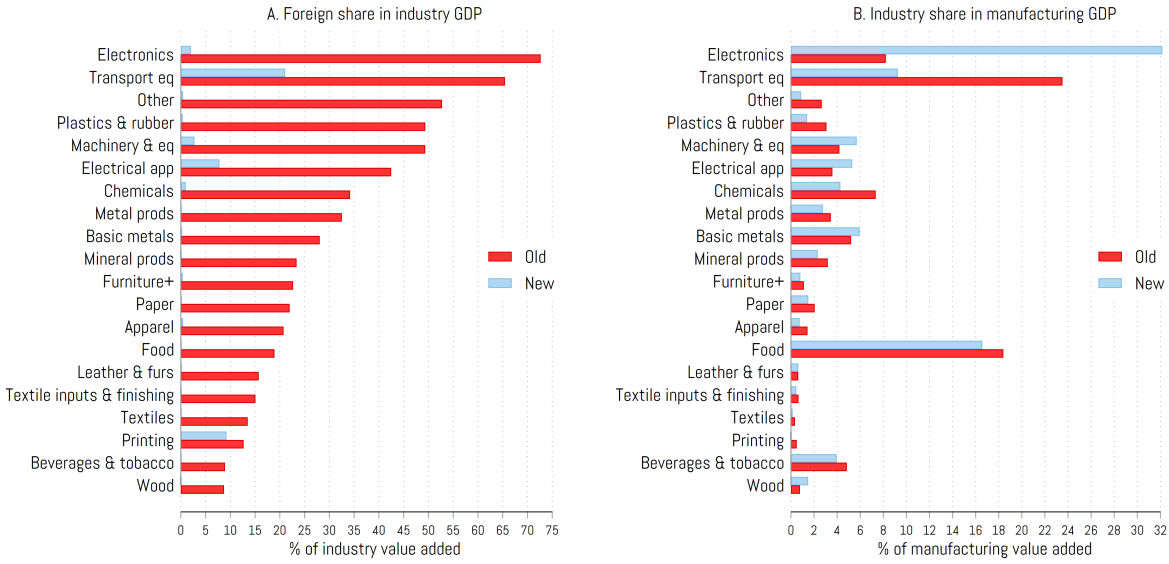
In the benchmark scenario, I ask: What if foreign affiliates were *identical* to domestic firms? To implement this scenario, I shock *all* parameters simultaneously. Its implementation is the least data-intensive since many of the expressions in proposition 4 cancel out to deliver simplified expressions for the counterfactual equations. For example, the counterfactual change in prices, which reflects all shocks except that to marketing communication costs, is:

$$\left(\widehat{c}_h^k\right)^{1-\sigma_k} = \left(\frac{R_{00}^{k,f}/N_0^k}{R_{h0}^{k,f}/N_h^k}\right)\left\{\left(\widehat{w}_0\right)^{\alpha_l^k+\alpha_N^k}\left[\chi_{00}^{sk}\left(\widehat{w}_0\right)^{-\theta_s}+1-\chi_{00}^{sk}\right]^{-\frac{\alpha_s^k}{\theta_s}}\left[\chi_{00}^{rk}\left(\widehat{w}_0\right)^{-\theta_r}+1-\chi_{00}^{rk}\right]^{-\frac{\alpha_r^k}{\theta_r}}\right\}^{1-\sigma_k}$$

This equation does not depend on the calibrated shocks since they cancel each other out. This means that these results do not rely on any of the assumptions I used when calibrating them. I find that the effect of the joint shocks is a **2.7 percent** reduction in real expenditure (welfare).

The role of foreign firms in the Mexican economy drops considerably, as part A in Figure 7 shows. This has an uneven effect on the industrial structure of Mexico's manufacturing sector, shown in part B.

Figure 7. All shocks: Industrial adjustment in the manufacturing sector



Three changes are salient. First, despite a large loss in foreign firms' productivity, their small initial share in the industry and a sizable entry of domestic firms (22 percent) leads to a modest reduction in the *food industry*'s share of manufacturing value added. Second, a modest increase in foreign firms' prices (10 and 13 percent at the median and mean, respectively) in *transportation equipment* manufacturing leads to

a reduction of this industry's share in manufacturing sector GDP of 14 percentage points. Third, another modest increase in foreign firms' prices (12 and 8 percent at the median and mean, respectively) in *electronics* manufacturing leads to an *increase* of this industry's share in manufacturing GDP of 24 percentage points. The reason behind the different effect of similar price increases in these two industries is their different demand elasticities (8.76 for electronics and 1.29 for transportation equipment). This implies that a similar increase in foreign prices leads to much higher profits for the incumbent domestic electronics firms. In fact, though the costs of *Mexican* firms drop by 8 percent in both industries in the new equilibrium, their effect on sales (conditional on market size) is 29 percent in the transportation equipment industry, but a staggering 97 percent in electronics. In the long run, this large increase in domestic competitiveness in the electronics industry results in massive entry of new domestic firms (1445 percent).

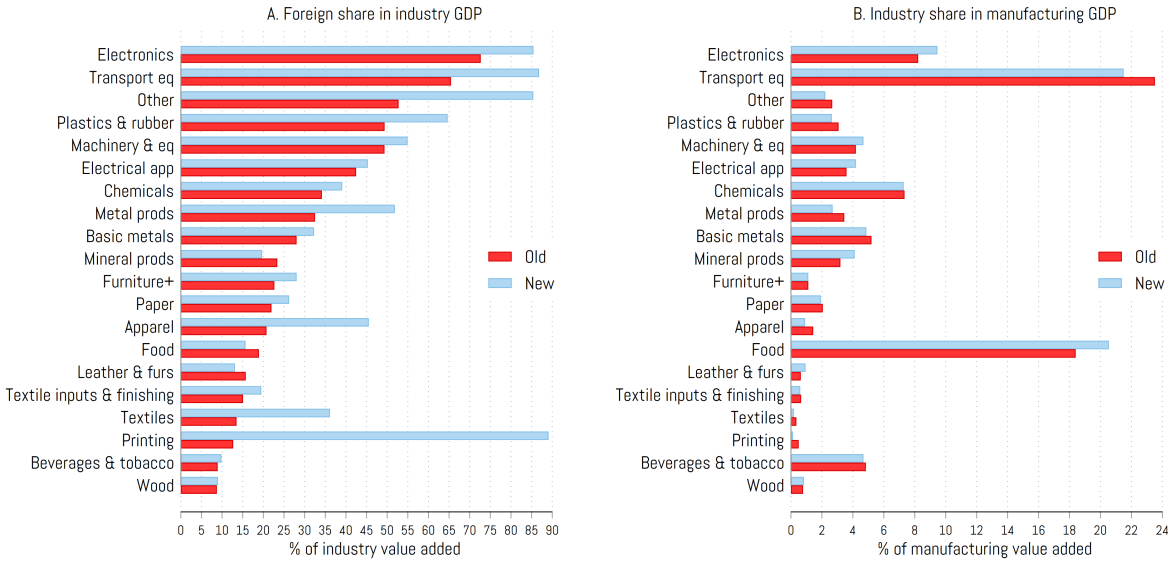
5.4.2 CF1: shock to contracting frictions

In general, matching foreign firms' contracting frictions to those of domestic firms actually *decreases* unit costs (at the mean and median). Given this, it is not surprising that when only **contracting frictions** are changed, real GDP increases by 9.7 percent. However, real expenditure is predicted to fall by **0.18 percent**. Why would the economy contract if foreign firms' prices *decrease*? This is a general equilibrium effect of the assumption that foreign firms send their all their profits to their home countries: 70 percent of foreign firms' value added is profits, which by assumption are sent abroad. Therefore, the increase in the foreign share in manufacturing leads to a *reduction* in labor demand.

In this scenario, foreign firms' share in their respective industries' GDP increases in almost every industry, in particular *printing*, in which it goes from 13 to 89 percent (panel A of Figure 8). This explosive increase in the foreign share is the result of (1) a large elasticity of demand (5.98), which takes a 6 percent reduction in prices and transforms it into a 36 percent increase in revenue (conditional on market size), and (2) important forward linkages, which imply that despite a 86 percent reduction in final sales, total sales rise by 27 percent.[46] Despite foreign firms' widespread expansion, in this scenario, the industrial structure of Mexico's manufacturing sector does not change much. This is because this increase is at the expense of domestic firms: Mexican firms exit all industries but five (food, leather products, nonmetal mineral products, machinery and equipment, and electric equipment). Hence, Mexican manufacturing becomes more dependent on foreign firms.

[46] A similar overcompensating input–output effect, although of lower magnitude, is found in the leather, food, electric, and wood industries.

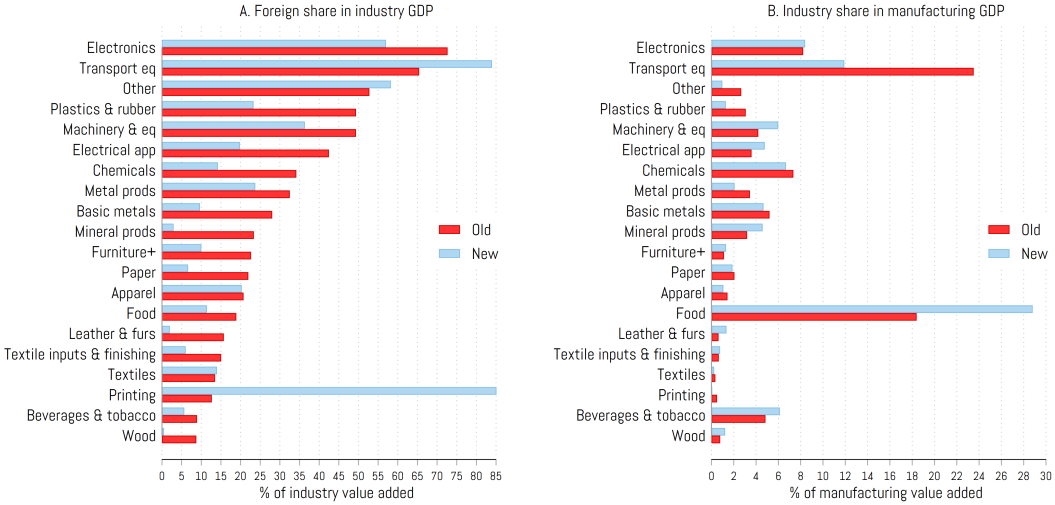
Figure 8. Contracting frictions shock: Industrial adjustment in the manufacturing sector



5.4.3 CF2: Shock to communication costs

In this counterfactual, similarly to the previous scenario, matching foreign firms' communication costs to those of domestic firms *decreases* unit costs (at the mean and median). Given this, when only **communication costs** are changed, real GDP increases by 4.9 percent. However, real expenditure is predicted to fall by **0.17 percent**. The difference is due to profits being sent abroad by foreign firms. Although crowding-out of domestic firms is not found across all industries as before, it is still present in the *printing* industry, for the same reasons as in scenario 0.

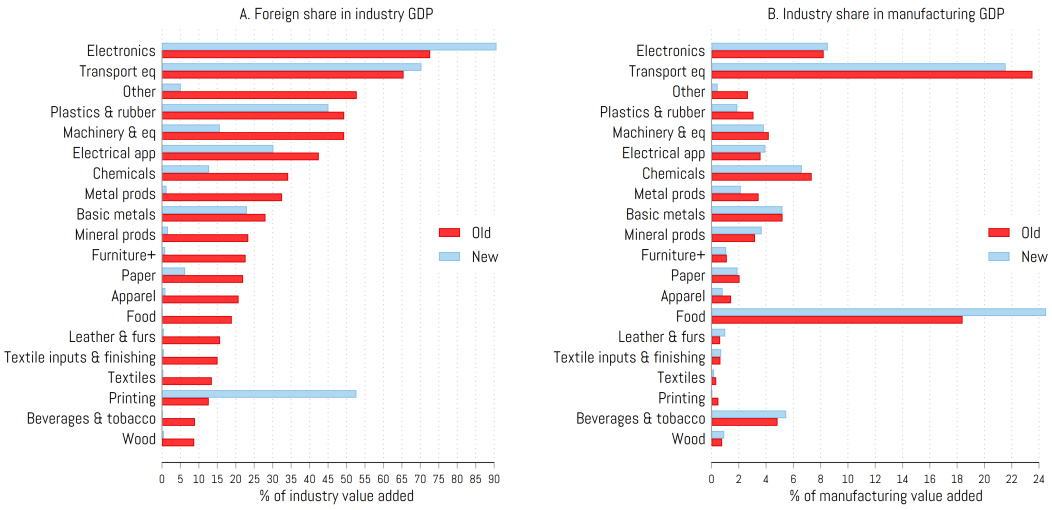
Figure 9. Communication costs shock (total): Industrial adjustment in the manufacturing sector



5.4.4 Shock to productivities

Given the small aggregate effect of the other two channels, it is expected that most of the effect found in the benchmark scenario is driven by differences in productivity. Indeed, in this counterfactual, real expenditure is predicted to fall by **2.2 percent**. Clearly, this accounts for most of the joint effect in scenario 0.

Figure 10. Productivity shock (total): Industrial adjustment in the manufacturing sector



6. Conclusion

How are foreign firms able to thrive in sectors where local firms cannot despite facing the same country fundamentals? The "off-the-shelf" answer given is always exogenous differences in productivity, but in this paper, I show that, at least in Mexico, other channels not related to productivity also benefit foreign firms. I conclude the paper by going over the areas that I think have potential for future research and commenting on policy implications.

The first candidate aspect for improvement is related to the data issues detailed in section **??**: The lack of a standardized product classification system forced me to rely on assumptions to back out some of the empirical moments of the data, which reduces the level of confidence in the results obtained with them. Applying this approach to a different country, or persuading INEGI to fix the issue, would constitute progress in this regard. Second, though the nature of the question asked in this paper required me to work with a multisector model, the theoretical results regarding the modeling of contracting friction do not rely on this approach. Therefore, they could be applied to more aggregate models of the global economy. Third, researchers could make progress by experimenting with alternative assumptions. For example, this model imposed the simplifying assumption of homogeneous firms within a country and a sector, but it would be interesting to see how the results are affected if one changes the model to account for heterogeneous firms,

as in Melitz (2003). Another example is the assumption that all firms in the same industry use the same technology, which could be relaxed to allow for multiple techniques per industry and their optimal choice, as in Boehm and Oberfield (2020). I leave these extensions for future research.

Finally, a comment about policy. Most papers in the recent literature on contracting frictions and development typically conclude by emphasizing how important judicial reform is for developing countries' development (for instance, see Boehm (2022)). However, improving a country's institutions not only could take years, if not decades (Haley (2006), Hammergren (2005), Messitte (2005)) but also has a circular flavor if one considers development synonymous with a country's having good institutions. Though they coincide on the importance of judicial system improvements in the long run, the results from this paper give some credibility to the idea that a policy of attracting foreign firms could help ameliorate the negative effects that weak contract enforcement has in an economy.

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A. Extended model

The economy is now divided into three main sectors: manufacturing, other tradables (T), and nontradables (N). The K industries of the main text now constitute the manufacturing sector.

A.1 Demand

The utility function's upper tier is expanded to include the two nonmanufacturing sectors:

$$U_i = \left(\frac{Q_i^T}{\beta_T}\right)^{\beta_T} \left(\frac{Q_i^N}{\beta_N}\right)^{\beta_N} \prod_{k=1}^{K} \left(\frac{Q_i^k}{\beta_k}\right)^{\beta_k}, \quad \text{with} \quad \beta_T + \beta_N + \sum_{k=1}^{K} \beta_k = 1$$

The lower tier for manufacturing industries is the same as in the body of the paper. The lower tier for other tradable goods is also a CES utility function with elasticity of substitution $\sigma_T \geq 1$. Unlike in manufacturing, in this sector, each country has a representative firm producing a unique variety (the Armington assumption):

$$Q_i^T = \left(\sum_{h=0}^{J} (q_{hi}^T)^{\frac{\sigma_T - 1}{\sigma_T}}\right)^{\frac{\sigma_T}{\sigma_T - 1}}$$

Consumer maximization implies the same demand function for manufacturing goods, except that expenditure is now $\beta_k E_i$ instead of simply E_i. The demand functions for other tradable goods and nontradable goods are, respectively:

$$\begin{aligned} q_{hi}^T &= (p_{hi}^T)^{-\sigma_T}(P_i^T)^{\sigma_T - 1}\beta_T E_i \\ Q_i^N &= \beta_N E_i / P_i^N \end{aligned}$$

where P_i^N is the price of domestic nontradable inputs and P_i^T is the CPI for other tradable goods:

$$P_i^T \equiv \left(\sum_{h=0}^{J} (p_{hi}^T)^{1-\sigma_T} \right)^{\frac{1}{1-\sigma_T}} \tag{24}$$

A.2 Producers of final goods

The description in the main body of the paper now applies only to manufacturing. In the other two sectors there is a representative firm producing a good priced at marginal cost. I assume that these sectors are closed off to foreign investment.

The production function in manufacturing is expanded to include nontradable inputs. I do this to be comprehensive when mapping the model to the data, in which plants spend nontrivial amounts on local nontradable inputs and services. Hence, the new production function in manufacturing will be:

$$q_{h0}^k(\omega) = \varphi_h^k A_0^k \left(\frac{L^k}{\alpha_l^k} \right)^{\alpha_l^k} \left(\frac{M_s^k}{\alpha_s^k} \right)^{\alpha_s^k} \left(\frac{M_r^k}{\alpha_r^k} \right)^{\alpha_r^k} \left(\frac{M_N^k}{\alpha_N^k} \right)^{\alpha_N^k} \quad \text{with } \alpha_l^k + \alpha_r^k + \alpha_s^k + \alpha_N^k = 1$$

where M_N^k is a composite of nontradable inputs with a share in revenue equal to $\alpha_N^k \in [0, 1]$. This implies that the marginal cost of a good ω now is:

$$c_h^k(\omega) = c_h^k \equiv \frac{(w_0)^{\alpha_l^k}(P_h^{sk})^{\alpha_s^k}(P_h^{rk})^{\alpha_r^k}(P_0^N)^{\alpha_N^k}}{\varphi_h^k A_0^k} \tag{25}$$

Nonmanufacturing goods are produced by representative firms with production functions linear in labor:

$$\begin{aligned} q_0^T &= A_0^T L_0^T \\ Q_0^N &= A_0^N L_0^N \end{aligned} \tag{26}$$

Since nonmanufactured goods are priced at marginal cost, their prices and total revenues are:

$$p_{0i}^T = \frac{w_0 \tau_{0i}^T \gamma_{0i}^T}{A_0^T} \quad \Rightarrow \quad R_0^T = \left(\frac{w_0}{A_0^T} \right)^{1-\sigma_T} \left[\beta_T E_0 (P_0^T)^{\sigma_T - 1} + FMA_0^T \right] \tag{27}$$

$$P_0^N = \frac{w_0}{A_0^N} \quad \Rightarrow \quad R_0^N = \beta_N E_0 \tag{28}$$

where FMA_0^T is defined similarly to FMA_h^k and is equal to:

$$FMA_0^T \equiv \sum_{i \neq 0} \beta_T E_i (P_i^T)^{\sigma_T - 1} (\tau_{0i}^T \gamma_{0i}^T)^{1-\sigma_T}$$

A.3 Aggregation

There are two main groups of firms: domestic, whose parent firm is Mexico ($h = 0$), and foreign, whose parent firm is headquartered abroad ($h \neq 0$). Since I am assuming away any fixed costs to export and import, all active firms trade globally. I do not model the foreign investment decisions of foreign firms, so I will take their presence as exogenously given. Given that empirically I cannot identify which domestic-owned firms are multinationals, I assume that Mexican firms cannot open plants abroad.

A.3.1 Revenue, profit, and disposable income

Given that all firms in the same group are identical by assumption, aggregate revenue and profits of domestic firms in industry k are equal to equations 9 and **??** multiplied by the mass of firms N_0^k:

$$R_0^k = N_0^k \left(\frac{\sigma_k}{\sigma_k - 1} \right)^{1-\sigma_k} (c_0^k)^{1-\sigma_k} \left[\sum_{i=1}^{J} \beta_k E_i (P_i^k)^{\sigma_k - 1} (\gamma_{0i} \tau_{0i}^k)^{1-\sigma_k} \right] \tag{29}$$

$$\Pi_0^k = \frac{R_0^k}{\sigma_k} - N_0^k w_0 f_e^k$$

Aggregate revenue for foreign firms is similar, but their profits are not. I assume that foreign firms paid the entry costs and the fixed cost related to multinational production in their home countries; thus, their aggregate profits are equal to:

$$\Pi_h^k = \frac{R_h^k}{\sigma_k} \tag{30}$$

Finally, as stated in section 3.2, consumers' disposable income (or aggregate nominal spending) is equal to the sum of the aggregate wage bill and aggregate profits from *domestic* firms since foreign firms are assumed to send their profits back to their home countries. In addition, since the nonmanufacturing sectors and the fringe of domestic manufacturing suppliers operate in perfectly competitive markets, they do not earn economic profits. Putting all together, we obtain the disposable income for domestic consumers:

$$E_0 = w_0 L_0 + \sum_k \Pi_0^k + D_0 \tag{31}$$

where D_0 is an exogenous variable to account for current account deficits in the data.

A.3.2 Consumer price index

The CPI was defined in equation 6. Given that all firms in the same group are identical, we can write the CPI in terms of their respective "group CPIs":

$$P_0^k = \left[\underbrace{N_0^k \left(p_{00}^k\right)^{1-\sigma_k}}_{\text{Domestic firms}} + \underbrace{\sum_{h\neq 0} N_h^k \left(p_{h0}^k\right)^{1-\sigma_k}}_{\text{Foreign affiliates}} + \underbrace{\sum_{n\neq 0} N_n^k \left(p_{n0}^k\right)^{1-\sigma_k}}_{\text{Imports}} \right]^{\frac{1}{1-\sigma_k}}$$

$$\Rightarrow P_0^k = \left[\left(P_{00}^k\right)^{1-\sigma_k} + \sum_{h\neq 0} \left(P_{h0}^k\right)^{1-\sigma_k} + CMA_0^k \right]^{\frac{1}{1-\sigma_k}} \tag{32}$$

where CMA_0^k, which stands for consumer market access, represents both the size and cost efficiency of foreign exporters as expressed in their prices. This object is exogenous to the SOE and will not change during the counterfactual exercises. The other two CPIs have the same interpretation (for domestic firms and foreign firms' affiliates producing and selling domestically) but they are endogenous and equal to:

$$\left(P_{h0}^k\right)^{1-\sigma_k} = N_h^k \left[\left(\frac{\sigma_k}{\sigma_k - 1} \right) c_h^k \right]^{1-\sigma_k} \tag{33}$$

where I normalize both the trade (τ) and communications (γ) costs of plants operating in the SOE to one (regardless of the home country).

A.4 Equilibrium

A **small open economy equilibrium** is defined as the vector of domestic wage (w_0), mass of domestic firms per industry ($\{N_0^k\}_k$), and CPIs per industry ($\{P_0^k\}_k$, P_0^T, P_0^N) such that (1) domestic consumers maximize utility, (2) all firms with plants in the SOE maximize profits, (3) all markets clear, and (4) the zero-profit condition is met.

A.4.1 Optimization

Consumer optimization is accounted for by the product demand and firm revenue functions (equations 5 and 9), which lead to the aggregate revenue function (equation 29). Firms' optimization is accounted for by their prices (equations 8, 27, and 28 for consumer goods and equations 10 and 15 for inputs), which lead to the aggregate revenue function and the price indices for consumer goods (equations 32 and 33) and intermediate inputs (equations 11 and 16 for standardized and relationship specific, respectively).

A.4.2 Market clearing of goods and inputs

Final goods markets always clear as long as consumers meet their budget constraint (equation 1) subject to disposable income being equal to equation 31. Intermediate input markets clearing is also implicit in the

market clearing for labor.

A.4.3 Market clearing of labor

The labor market clears if aggregate labor demand equals aggregate labor supply L_0, an exogenous parameter. Aggregate labor demand is equal to the sum of sector labor demands ($L_0 = \sum_k L_0^k + L_0^T + L_0^N$), which in turn are equal to the sum of the firms' labor demands. For ease of exposition, let us go over these components one by one before aggregating them.

Manufacturing industries. Manufacturing firms generate a direct and an indirect labor demand. The direct demand reflects the labor used in the assembly of manufacturing consumer goods. In addition, it includes the labor used to pay for the entry cost in the case of domestic firms:

$$L_h^k = \left(\frac{\alpha_l^k}{w_0}\right)\left(\frac{\sigma_k - 1}{\sigma_k}\right)R_h^k + \underbrace{N_h^k f_e^k}_{\text{only if } h=0}$$

where R_h^k represents the *aggregate* revenue of all firms in industry k from country h. The indirect demand is the result of firms' demand for intermediate inputs. The indirect demand for nontradable inputs is similar to that for assembly and is equal to:

$$L_h^{Nk} = \left(\frac{\alpha_N^k}{w_0}\right)\left(\frac{\sigma_k - 1}{\sigma_k}\right)R_h^k \tag{34}$$

The indirect labor demand for tradable inputs of type $x \in \{s, r\}$ can be found in four steps. First, note that the conditional demand for some input ν of type $x \in \{s, r\}$ by a firm ω is:

$$m_{jh}^{xk}(\omega, \nu) = (P_h^{xk})^{\zeta_x^k - 1} p_j^k(\nu)^{-\zeta_x^k} \alpha_x^k \left(\frac{\sigma_k - 1}{\sigma_k}\right) R_h^k(\omega)$$

Second, plug this into the linear production function for specifications (equation 3) to obtain the indirect labor demand for an input ν sourced from country j by a firm ω:

$$l_{jh}^{xk}(\omega, \nu) = \frac{\gamma_{jh}\tau_{j0}^k m_{jh}^{xk}(\nu)}{a_j^{xk}(\nu)} = \frac{(P_h^{xk})^{\zeta_x^k - 1} p_j^k(\nu)^{1-\zeta_x^k} \alpha_x^k \left(\frac{\sigma_k - 1}{\sigma_k}\right) R_h^k(\omega)}{w_j}$$

Third, obtain the indirect labor demand in country j generated by this firm by aggregating across all inputs ν that are, in equilibrium, purchased from this country:

$$\begin{aligned} l_{jh}^{xk}(\omega) &= \int_0^1 l_{jh}^{xk}(\omega, \nu) \mathbb{1}\Big(\nu \text{ is purchased from } j\Big) d\nu \\ &= \frac{(P_h^{xk})^{\zeta_x^k - 1} \alpha_x^k \left(\frac{\sigma_k - 1}{\sigma_k}\right) R_h^k(\omega)}{w_j} \left[\int_0^1 p_j^k(\nu)^{1-\zeta_x^k} \mathbb{1}\Big(\nu \text{ is purchased from } j\Big) d\nu\right] \end{aligned}$$

$$= \frac{(P_h^{xk})^{\zeta_x^k - 1} \alpha_x^k \left(\frac{\sigma_k - 1}{\sigma_k}\right) R_h^k(\omega)}{w_j} \chi_{jh}^{xk} \left[\int_0^\infty p^{1-\zeta_r^k} d\Pr\left(p_h^{xk}(\nu) \leq p\right) \right]$$

$$= \left(\frac{\alpha_x^k \chi_{jh}^{xk}}{w_j}\right) \left(\frac{\sigma_k - 1}{\sigma_k}\right) R_h^k(\omega)$$

Finally, summing across all firms in industry k from country h gives you the aggregate indirect demand for inputs from country j (which replaces only $R_h^k(\omega)$ in the previous result by R_h^k). Hence, the total (direct and indirect) *domestic* labor demand generated by firms with home country h in industry k is:

$$L_{0h}^k = \begin{cases} \left(\frac{\alpha_l^k + \alpha_s^k \chi_{0h}^{sk} + \alpha_r^k \chi_{0h}^{rk} + \alpha_N^k}{w_0}\right) \left(\frac{\sigma_k - 1}{\sigma_k}\right) R_h^k & \text{if } h \neq 0 \\ \left[\left(\frac{\alpha_l^k + \alpha_s^k \chi_{00}^{sk} + \alpha_r^k \chi_{00}^{rk} + \alpha_N^k}{w_0}\right) \left(\frac{\sigma_k - 1}{\sigma_k}\right) + \frac{1}{\sigma_k}\right] R_0^k & \text{if } h = 0 \end{cases} \tag{35}$$

where I used the zero-profit condition (equation 39) to obtain the demand stemming from domestic-owned firms.

Nonmanufacturing industries. The representative firm in the nonmanufacturing tradable sector sells only to final consumers (domestically and abroad), with sector-level revenue as in equation 27. Given that this sector is perfectly competitive, revenue equals expenditure, and since production uses only labor (equation 26), then revenue equals the wage bill, which implies the following labor demand:[47]

$$L_0^T = \frac{R_0^T}{w_0} \tag{36}$$

The representative firm in the nontradable sector sells only domestically (it does not export by definition), but to both the representative consumer and firms in manufacturing. The employment derived from its sales to other firms has already been accounted for by the indirect labor demand associated with nontradable inputs in equation 34, while that derived from the firm's sales to the final consumer is similar to that of the previous sector, with sales being an exogenous share of national disposable income, as in equation 28:

$$L_0^N = \frac{\beta_N E_0}{w_0} \tag{37}$$

Labor market clearing. Putting these pieces together, we have the labor market clearing condition:

$$L_0 = \sum_{k=1}^K \left\{ L_{00}^k + \sum_{h \neq 0} L_{0h}^k \right\} + L_0^T + L_0^N \tag{38}$$

[47] There is only direct labor demand since this sector does not use intermediate inputs.

A.4.4 Zero profit condition

Free entry and exit of firms in the manufacturing sectors imply that, despite domestic firms operating under monopolistic competition, their profits are zero in equilibrium (thus equation 31 becomes simply $E_0 = w_0 L_0 + D_0$), conditional on an exogenous number of foreign affiliates operating in the SOE:

$$\Pi_0^k = \frac{R_0^k}{\sigma_k} - N_0^k w_0 f_e^k = 0 \tag{39}$$

This condition implies that all variable profits of domestic firms go toward paying for the fixed cost (f_e^k). Given that this cost is measured in labor units, this constitutes an additional source of labor demand.

A.5 Equilibrium in relative changes

In this section, I show the equations that determine the equilibrium in relative changes (hats) when *all* shocks happen simultaneously. The exercises with single shocks are special cases in which the other shocks are set equal to one.

A.5.1 Changes in manufacturing unit costs

Unit costs for manufacturing goods are shown in equation 25. Their change is determined by the following four equations:

$$\widehat{c}_h^k = \frac{(\widehat{w}_0)^{\alpha_l^k}(\widehat{P}_h^{sk})^{\alpha_s^k}(\widehat{P}_h^{rk})^{\alpha_r^k}(\widehat{P}_0^N)^{\alpha_N^k}}{\widehat{\varphi}_h^k} \qquad \widehat{P}_h^{sk} = \left[\chi_{0h}^{sk}(\widehat{w}_0)^{-\theta_s} + \sum_{j\neq 0}\chi_{jh}^{sk}(\widehat{\gamma}_{jh})^{-\theta_s}\right]^{-\frac{1}{\theta_s}}$$

$$\widehat{P}_0^N = \widehat{w}_0 \qquad \widehat{P}_h^{rk} = \left[\chi_{0h}^{rk}(\widehat{w}_0\widehat{\Lambda}_{0h}^k)^{-\theta_r} + \sum_{j\neq 0}\chi_{jh}^{rk}(\widehat{\Lambda}_{jh}^k\widehat{\gamma}_{jh})^{-\theta_r}\right]^{-\frac{1}{\theta_r}}$$

These equations illustrate how the three channels affect different components of unit costs, which is what allows me to separately identify them (see section 5.2). The first difference is between productivity and the frictions: The former affects unit costs directly, while the other two do so via the input price indices, which in turn reflect firms' intensive-margin expenditure decisions, as shown in equations 12 and 17. This implies that though the expenditure shares have information on contracting and communication frictions, they are not informative about differences in productivity.[48] The second difference is between contracting and communication frictions: The former affect all sourcing countries, including the SOE, but are relevant only for relationship-specific inputs, but the latter affect both types of inputs, but only foreign sourcing

[48]The theoretical literature on multinational firms' global sourcing decisions has shown that differences in productivity are reflected in the set of countries firms source from (for example, see Antràs et al. (2017) and Antràs et al. (2022)). Nonetheless, given that all these models (and this one) rely on Fréchet distributional assumptions to obtain closed-form expressions for the expenditure shares, even if the set of countries a firm sources from depends on its productivity, the *relative* share between two sourcing countries does not, conditional on their being included in the sourcing set. This implies that ignoring the extensive margin should not be a source of bias for the measured shocks, given that all the expressions identifying these parameters are functions of these ratios (see proposition 4).

countries.

A.5.2 Changes in consumer price indices

The CPIs for manufacturing goods are shown in equations 32 and 33 and those for nonmanufacturing goods in equations 24 and 28. Their change is determined by the following two equations:

$$\widehat{P}_0^k = \left[\left(\frac{R_{00}^{k,f}}{E_0^k} \right) \widehat{N}_0^k \left(\widehat{c}_0^k \right)^{1-\sigma_k} + \sum_{h \neq 0} \left(\frac{R_{h0}^{k,f}}{E_0^k} \right) \left(\widehat{c}_h^k \right)^{1-\sigma_k} + \left(\frac{IMP_0^k}{E_0^k} \right) \right]^{\frac{1}{1-\sigma_k}}$$

$$\widehat{P}_0^T = \left[\left(\frac{R_{00}^{T,f}}{E_0^T} \right) \left(\widehat{w}_0 \right)^{1-\sigma_T} + \left(\frac{IMP_0^T}{E_0^T} \right) \right]^{\frac{1}{1-\sigma_T}}$$

where $R_{h0}^{k,f}$ and $R_{00}^{T,f}$ represent *domestic final* sales of manufacturing industry k (and HQ country h) and nonmanufacturing tradable sector T, respectively; IMP_0^k and IMP_0^T represent imports in the same sectors; and E_0^k and E_0^T represent aggregate final expenditure on these sectors.

A.5.3 Changes in the labor market equilibrium

The change in the labor market equilibrium can be obtained from equations 36-34. However, first I need to address the issue mentioned in footnote 34: The labor market clearing condition in relative changes requires that value added weights be assigned to the changes in labor demands if I am to follow standard national accounting practices. However, this would leave out the indirect labor demands since they are a part not of value added but of intermediate consumption. In a model with multiple sectors and *input–output linkages*, these indirect flows show up in the value added of upstream industries. Instead, this model features a competitive fringe of suppliers, which does not have an obvious mapping into actual industries. To solve this model–data mapping issue, I make the following assumptions:

Assumption 3. *Expenditures on standardized and relationship-specific inputs are distributed among all tradable industries on the basis of observable input–output linkages ($\lambda_{kk'}^{sh}$, and $\lambda_{kk'}^{rh}$).*

Assumption 4. *Foreign affiliates produce only final goods. Thus, all domestic intermediate inputs are produced by domestic firms.*

The first assumption allows me to split the total change in indirect labor demand of any industry k among its upstream industries by combining information from publicly available input–output tables (at the industry pair level) with that from plant-level data to construct separate input–output tables for standardized and relationship-specific inputs, respectively.[49] The second assumption is necessary because the available data do not have enough information for me to determine which portion of manufacturing

[49] Assumption 3 is an ad hoc change invoked to facilitate the mapping from model to data in the context of counterfactual exercises. It is not a proper modification of the model: Unit costs are still a function of just two input price indices.

intermediate inputs is sold by foreign affiliates. I circumvent this problem by simply assuming that only domestic firms sell to other industries. The following proposition summarizes the result:

Proposition 5 (Labor market clearing condition in relative changes)
The labor market clearing condition in relative changes is:

$$\widehat{GNP}_0 \equiv \widehat{w}_0 = \sum_{k=1}^{K} \left\{ \left(\frac{VA_0^k}{E_0} \right) \widehat{R}_0^k + \sum_{h \neq 0} \left(\frac{W_h^k}{E_0} \right) \widehat{R}_h^k \right\} + \left(\frac{VA_0^T}{E_0} \right) \widehat{R}_0^T + \left(\frac{VA_0^N}{E_0} \right) \widehat{R}_0^N$$

where VA_0^k, VA_0^T, and VA_0^N are the value added of domestic manufacturing industry k, nonmanufacturing tradable sector T and nontradable sector N, respectively; E_0 is the SOE's gross national product; and W_h^k is the wage bill of foreign manufacturing industry k. The relative change in revenue reflects the change in sales to final consumers and, in the case of domestic industries, also sales to downstream manufacturing industries:

$$\begin{aligned}
\widehat{R}_0^k &= \widehat{N}_0^k \left(\widehat{c}_0^k\right)^{1-\sigma_k} \left[\left(\frac{R_{00}^{k,f}}{R_0^k} \right) \widehat{E}_0 \left(\widehat{P}_0^k\right)^{\sigma_k - 1} + \left(\frac{EXP_0^k}{R_0^k} \right) \right] + \sum_{k'} \sum_{h} \left[\left(\frac{R_{00,h}^{k,sk'}}{R_0^k} \right) \widehat{\chi}_{0h}^{sk'} + \left(\frac{R_{00,h}^{k,rk'}}{R_0^k} \right) \widehat{\chi}_{0h}^{rk'} \right] \widehat{R}_h^{k'} \\
\widehat{R}_h^k &= \left(\widehat{c}_h^k\right)^{1-\sigma_k} \left[\left(\frac{R_{h0}^{k,f}}{R_h^k} \right) \widehat{E}_0 \left(\widehat{P}_0^k\right)^{\sigma_k - 1} + \sum_{i \neq 0} \left(\frac{R_{hi}^{k,f}}{R_h^k} \right) \left(\widehat{\gamma}_{hi}\right)^{1-\sigma_k} \right], \quad \text{for} \quad h \neq 0 \\
\widehat{R}_0^T &= \left(\widehat{w}_0\right)^{1-\sigma_T} \left[\left(\frac{R_{00}^{T,f}}{R_0^T} \right) \widehat{E}_0 \left(\widehat{P}_0^T\right)^{\sigma_T - 1} + \left(\frac{EXP_0^T}{R_0^T} \right) \right] + \sum_{k'} \sum_{h} \left[\left(\frac{R_{00,h}^{T,sk'}}{R_0^T} \right) \widehat{\chi}_{0h}^{sk'} + \left(\frac{R_{00,h}^{T,rk'}}{R_0^T} \right) \widehat{\chi}_{0h}^{rk'} \right] \widehat{R}_h^{k'} \\
\widehat{R}_0^N &= \left(\frac{R_{00}^{N,f}}{R_0^N} \right) \widehat{E}_0 + \sum_{k'} \sum_{h} \left(\frac{R_{0h}^{N,k'}}{R_0^N} \right) \widehat{R}_h^{k'}
\end{aligned} \tag{40}$$

where $R_{hi}^{k,f}$ and R_h^k are the sales of final goods to market i and total sales of firms in industry k and HQ in country h; EXP_0^k is total exports of domestic firms; $R_{00,h}^{k,sk'}$ and $R_{00,h}^{k,sk'}$ are the domestic sales of standardized and relationship-specific inputs from industry k to firms in industry k' and HQ in country h; $R_{00}^{T,f}$, EXP_0^T, $R_{00,h}^{T,sk'}$, and $R_{00,h}^{T,rk'}$ represent domestic sales of final goods, total exports, domestic sales of standardized inputs, and domestic sales of relationship-specific inputs of nonmanufacturing tradable sector goods to firms in industry k' with HQ in country h; $R_{00}^{N,f}$ and R_0^N are the domestic and total sales of nontradable services to final consumers; and $R_{0h}^{N,k'}$ is the domestic sales of nontradable services to firms in industry k' with HQ in country h. The relative change in expenditure shares for standardized and relationship-specific inputs, respectively, are:

$$\widehat{\chi}_{0h}^{sk} = \left(\frac{\widehat{w}_0}{\widehat{P}_h^{sk}} \right)^{-\theta_s} \qquad \widehat{\chi}_{0h}^{rk} = \left(\frac{\widehat{w}_0 \widehat{\Lambda}_{0h}^k}{\widehat{P}_h^{rk}} \right)^{-\theta_r}$$

Finally, the changes in aggregate expenditure and in the number of domestic firms are:

$$\widehat{E}_0 = \left(\frac{GNP_0}{E_0} \right) \widehat{w}_0 + 1 - \frac{GNP_0}{E_0} \qquad \text{and} \qquad \widehat{N}_0^k = \frac{\widehat{R}_0^k}{\widehat{w}_0}$$

Proof. First, I show how they affect the revenue equations in levels and then in changes. After that, I circle back to the labor market equilibrium.

Revenues in levels. Aggregate revenue for manufacturing firms is shown in equation 29, while that of the other two sectors is shown in equations 27 and 28. Assumptions 3 and 4 imply that total revenue must also include sales to other sectors. In addition, assumption 4 implies that foreign affiliates' revenue function does not change. The new aggregate revenue for the domestic k industry is:

$$R_0^k = \overbrace{N_0^k\left(\frac{\sigma_k}{\sigma_k-1}\right)^{1-\sigma_k}(c_0^k)^{1-\sigma_k}\left[\beta_k w_0 L_0 (P_0^k)^{\sigma_k-1} + FMA_0^k\right]}^{\text{sales to final consumers}} + \overbrace{\sum_{k'}\sum_h \lambda_{kk'}^{sh}\alpha_s^{k'}\left(\frac{\sigma_{k'}-1}{\sigma_{k'}}\right)\chi_{0h}^{sk'}R_h^{k'}}^{\text{domestic sales of std. inputs}}$$
$$+ \underbrace{\sum_{k'}\sum_h \lambda_{kk'}^{rh}\alpha_r^{k'}\left(\frac{\sigma_{k'}-1}{\sigma_{k'}}\right)\chi_{0h}^{rk'}R_h^{k'}}_{\text{domestic sales of RS inputs}}$$

Similarly, the nonmanufacturing sectors also sell inputs to the manufacturing industries (I do not allow intrasector linkages).[50] Hence, their new revenue functions are:

$$R_0^T = \overbrace{\left(\frac{w_0}{A_0^T}\right)^{1-\sigma_T}\left[\beta_T w_0 L_0 (P_0^T)^{\sigma_T-1} + FMA_0^T\right]}^{\text{sales to final consumers}} + \overbrace{\sum_{k'}\sum_h \lambda_{Tk'}^{sh}\alpha_s^{k'}\left(\frac{\sigma_{k'}-1}{\sigma_{k'}}\right)\chi_{0h}^{sk'}R_h^{k'}}^{\text{domestic sales of std. inputs}} + \overbrace{\sum_{k'}\sum_h \lambda_{Tk'}^{rh}\alpha_r^{k'}\left(\frac{\sigma_{k'}-1}{\sigma_{k'}}\right)\chi_{0h}^{rk'}R_h^{k'}}^{\text{domestic sales of RS inputs}}$$

$$R_0^N = \underbrace{\beta_N w_0 L_0}_{\substack{\text{sales to final}\\ \text{consumers}}} + \underbrace{\sum_{k'}\sum_h \alpha_N^{k'}\left(\frac{\sigma_{k'}-1}{\sigma_{k'}}\right)R_h^{k'}}_{\text{domestic sales of inputs}}$$

Revenues in relative changes. The change in foreign multinationals' revenue is:

$$\widehat{R}_h^k = \left(\widehat{c}_h^k\right)^{1-\sigma_k}\left[\left(\frac{R_{h0}^k}{R_h^k}\right)\widehat{w}_0\left(\widehat{P}_0^k\right)^{\sigma_k-1} + \sum_{i\neq 0}\left(\frac{R_{hi}^k}{R_h^k}\right)\left(\widehat{\gamma}_{hi}\right)^{1-\sigma_k}\right]$$

where R_{hi}^k and R_h^k represent sales to market i and total sales, respectively. Next, we can write the change in revenue for domestic manufacturing firms using matrix algebra:

$$\widehat{\boldsymbol{R}}_{\boldsymbol{0}} = \left(\boldsymbol{I} - \widehat{\boldsymbol{A}}_{\boldsymbol{00}}\right)^{-1}\left[\widehat{\boldsymbol{F}}_{\boldsymbol{0}} + \sum_{h\neq 0}\widehat{\boldsymbol{A}}_{\boldsymbol{0h}}\widehat{\boldsymbol{R}}_{\boldsymbol{h}}\right]$$

[50] This assumption is innocuous for the nonmanufacturing tradable sector (which groups agriculture, mining, and oil) since most of its domestic intermediate sales go to the manufacturing sector (87 percent in 2018). However, this is not the case for the nontradable sector: 62 percent of its sales were intrasector in 2018. Nonetheless, ignoring this feedback is expected to ameliorate the quantitative results, not exacerbate them. Thus, we can take them as relatively conservative estimates.

where $\widehat{\boldsymbol{R}}_{\mathbf{0}}$, $\widehat{\boldsymbol{F}}_{\mathbf{0}}$, and $\widehat{\boldsymbol{R}}_{\boldsymbol{h}}$ are the $K \times 1$ vectors of changes in total revenue and final revenue for domestic firms, respectively, and total revenue for foreign affiliates from HQ country h. The $K \times K$ matrices $\widehat{\boldsymbol{A}}_{\mathbf{00}}$ and $\widehat{\boldsymbol{A}}_{\mathbf{0}\boldsymbol{h}}$ account for input–output linkages between industries, for domestic and foreign downstream industries, respectively. They are written in hats to emphasize that they depend on the shares of domestic purchases $\chi_{0h'}^{xk}$, which are also changing. A representative element for these matrices is:

$$\widehat{a}_{0h}^{kk'} = \left(\frac{R_{00,h}^{k,sk'}}{R_0^k}\right)\widehat{\chi}_{0h}^{sk'} + \left(\frac{R_{00,h}^{k,rk'}}{R_0^k}\right)\widehat{\chi}_{0h}^{rk'}$$

where $R_{00,h}^{k,sk'}$ and $R_{00,h}^{k,rk'}$ are the domestic sales to firms from HQ country h in industry k' of standardized and relationship-specific inputs, respectively. $\widehat{\boldsymbol{F}}_{\mathbf{0}}^{\boldsymbol{k}}$ is the vector with the changes in sales to final consumers:

$$\widehat{F}_0^k = \widehat{N}_0^k \left(\widehat{c}_0^k\right)^{1-\sigma_k} \left[\left(\frac{R_{00}^{k,f}}{R_0^k}\right)\widehat{w}_0 \left(\widehat{P}_0^k\right)^{\sigma_k - 1} + \left(\frac{EXP_0^k}{R_0^k}\right)\right]$$

where $R_{00}^{k,f}$ and EXP_0^k represent domestic sales and total exports, respectively. Finally, the revenue changes for the nonmanufacturing sectors are:

$$\widehat{R}_0^T = \left(\widehat{w}_0\right)^{1-\sigma_T} \left[\left(\frac{R_{00}^{T,f}}{R_0^T}\right)\widehat{w}_0 \left(\widehat{P}_0^T\right)^{\sigma_T - 1} + \left(\frac{EXP_0^T}{R_0^T}\right)\right] + \sum_{k'}\sum_{h}\left[\left(\frac{R_{00,h}^{T,sk'}}{R_0^T}\right)\widehat{\chi}_{0h}^{sk'} + \left(\frac{R_{00,h}^{T,rk'}}{R_0^T}\right)\widehat{\chi}_{0h}^{rk'}\right]\widehat{R}_h^{k'}$$

$$\widehat{R}_0^N = \left(\frac{R_0^{N,f}}{R_0^N}\right)\widehat{w}_0 + \sum_{k'}\sum_{h}\left(\frac{R_{0h}^{N,k'}}{R_0^N}\right)\widehat{R}_h^{k'}$$

where $R_{00}^{T,f}$ and EXP_0^T represent domestic sales and exports, respectively; $R_{00,h}^{T,sk'}$ and $R_{00,h}^{T,rk'}$ the domestic sales to firms from HQ country h in industry k' of standardized and relationship-specific inputs, respectively; and $R_0^{N,f}$ and $R_{0h}^{N,k'}$ the domestic sales of nontradable services to final consumers and to firms from country h in industry k', respectively.

New labor market clearing condition. By distributing the indirect labor demands of manufacturing firms among *observable* industries, the previous assumptions simplify the accounting of the labor market clearing condition in relative changes. Specifically, since the indirect effects now show up as interindustry sales, I can redefine the components of equation 38 as reflecting only direct labor demands. In addition, if I multiply both sides of the equation by w_0, the labor market condition is equivalent to the national accounting definition of GNP, denoted by E_0:

$$\underbrace{w_0 L_0}_{GNP_0} = \sum_{k=1}^{K}\left\{\underbrace{w_0 L_{00}^k}_{VA_0^k} + \sum_{h\neq 0}\underbrace{w_0 L_{0h}^k}_{W_h^k}\right\} + \underbrace{w_0 L_0^T}_{VA_0^T} + \underbrace{w_0 L_0^N}_{VA_0^N}$$

where VA stands for value added and W for the wage bill. When we take into account this new labor market clearing condition, together with equations 34, 36, and 37, the equation in relative changes is:

$$\widehat{w}_0 = \sum_{k=1}^{K} \left\{ \left(\frac{VA_0^k}{E_0} \right) \widehat{R}_0^k + \sum_{h \neq 0} \left(\frac{W_h^k}{E_0} \right) \widehat{R}_h^k \right\} + \left(\frac{VA_0^T}{E_0} \right) \widehat{R}_0^T + \left(\frac{VA_0^N}{E_0} \right) \widehat{R}_0^N$$

Finally, the relative change in expenditure shares can be written as follows:

$$\widehat{\chi}_{0h}^{sk} = \frac{\left(\widehat{w}_0\right)^{-\theta_s}}{\chi_{0h}^{sk}\left(\widehat{w}_0\right)^{-\theta_s} + \sum\limits_{j \neq 0} \chi_{jh}^{sk}\left(\widehat{\gamma}_{jh}\right)^{-\theta_s}} = \frac{\left(\widehat{w}_0\right)^{-\theta_s}}{\left(\widehat{P}_h^{sk}\right)^{-\theta_s}}$$

$$\widehat{\chi}_{0h}^{rk} = \frac{\left(\widehat{w}_0 \widehat{\Lambda}_{0h}^k\right)^{-\theta_r}}{\chi_{0h}^{rk}\left(\widehat{w}_0 \widehat{\Lambda}_{0h}^k\right)^{-\theta_r} + \sum\limits_{j \neq 0} \chi_{jh}^{rk}\left(\widehat{\gamma}_{jh} \widehat{\Lambda}_{jh}^k\right)^{-\theta_r}} = \frac{\left(\widehat{w}_0 \widehat{\Lambda}_{0h}^k\right)^{-\theta_r}}{\left(\widehat{P}_h^{rk}\right)^{-\theta_r}}$$

□

I use GNP instead of GDP because foreign affiliates' profits (which are part of value added and thus of GDP) are repatriated by assumption. In contrast, since all domestic firms' profits are used to pay entry costs according to the zero-profit condition (equation 39) and entry costs require hiring labor, the full value added of domestic firms ends up in workers' pockets.

A.5.4 Welfare

I will use the change in real expenditure as the yardstick to compare the different counterfactual scenarios. From consumer optimization, this is equal to:

$$\widehat{rE}_0 \equiv \frac{\widehat{E}_0}{\left(\widehat{P}_0^T\right)^{\beta_T} \left(\widehat{P}_0^N\right)^{\beta_N} \Pi_k \left(\widehat{P}_0^k\right)^{\beta_k}}$$

A.6 Identification in more detail

A.6.1 Parameter calibrated values

The following table reports the values for the model parameters for each industry:

Table 9. Calibrated parameter values by industry

Industry	α_l^k	α_s^k	α_r^k	α_N^k	β_k	σ_k
Food	0.06	0.37	0.36	0.20	0.08	2.93
Beverages & tobacco	0.09	0.36	0.36	0.19	0.01	2.12
Textile inputs	0.14	0.36	0.29	0.21	<0.01	4.51
Nonapparel textiles	0.17	0.31	0.31	0.21	<0.01	4.24
Apparel	0.23	0.30	0.31	0.17	0.01	4.13
Leather	0.21	0.32	0.32	0.15	<0.01	4.82
Wood	0.17	0.35	0.31	0.17	<0.01	3.14
Paper	0.10	0.33	0.32	0.25	<0.01	3.53
Printing	0.15	0.31	0.31	0.23	<0.01	5.98
Petrochemicals	0.02	0.42	0.42	0.14	0.02	4.52
Chemicals	0.07	0.37	0.35	0.21	0.02	3.65
Plastics & rubber	0.10	0.34	0.34	0.22	0.01	5.09
Nonmetal mineral goods	0.10	0.34	0.34	0.23	<0.01	2.97
Basic metals	0.04	0.37	0.37	0.22	0.01	3.75
Metal goods	0.14	0.34	0.34	0.19	0.01	4.32
Machinery	0.13	0.34	0.34	0.18	0.03	3.99
Electronics	0.11	0.39	0.43	0.06	0.03	8.76
Electric equip.	0.12	0.38	0.38	0.12	0.01	5.08
Transport equip.	0.06	0.36	0.39	0.19	0.06	4.06
Furniture	0.21	0.31	0.31	0.17	<0.01	4.09
Other	0.14	0.40	0.39	0.07	0.01	5.56
Non-mnf tradables					0.02	1.74
Nontradables					0.68	1.99

A.6.2 Empirical moments

The counterfactual exercises require three types of moments: expenditure shares (for inputs and consumption goods), revenue shares, and aggregate shares in GNP.

Expenditure shares. To compute the change in input price indices, I need to measure the country expenditure shares for standardized and relationship-specific inputs, for both domestic and foreign-owned firms (χ_{jh}^{xk}). For each HQ country and input type type, I observe the share of expenditures that comes from the domestic market in the economic census data and the share of imports that come from each country in the

customs data. After rescaling the latter using the former, I obtain the complete vectors of shares.

To compute the change in CPIs of manufacturing industries, I need the share in aggregate final expenditure of domestic firms' ($R_{00}^{k,f}$), foreign affiliates' ($R_{h0}^{k,f}$), and imported goods (IMP_0^k). Domestic purchases and imports by industry are directly observable in official input–output tables. I split the former between domestic and foreign-owned firms by combining information from the input–output table with that of the economic census, together with assumption 4.[51] For the nonmanufacturing tradable sector, domestic purchases by final consumers ($R_{00}^{T,f}$) and imports (IMP_0^T) by industry are directly observable in official input–output tables.

Revenue shares. To compute the change in revenues for manufacturing industries, I need to observe two sets of revenue shares: sales to final consumers and sales of intermediate inputs. For the first set, I need to disaggregate final sales by destination market and HQ country ($R_{hi}^{k,f}$). For each HQ country, I observe total domestic sales in the economic census and disaggregated exports by market in the customs data. I isolate the part of total domestic sales that corresponds to *final consumer goods* following the same strategy from footnote 51. I use the exports share in revenue from the census to rescale the shares from customs to obtain the complete vector of shares of final goods in total revenue. For nonmanufacturing sectors, domestic final sales ($R_{00}^{T,f}$ and $R_{00}^{N,f}$) and total exports ($\sum_{i\neq 0} R_{0i}^{T,f}$) are directly observable in the input–output tables.

For the second set, I need to disaggregate the total intermediate sales among downstream industries k' and between input types (the latter only for tradable sectors). To do so, I use the auxiliary census dataset with information on input purchases at the product level for a subset of plants, described in section 2.1.1.[52] This rich information allowed me to construct a separate input–output matrix for standardized and relationship-specific inputs.[53]

GNP shares. Industry-level value added and wage bill are available in the input–output tables. To disaggregate them by HQ country, I use the HQ country shares in the industry totals (for instance, $VA_0^k / \sum_h VA_h^k$), which can be calculated with data from the economic census. For the denominator of these shares, I use aggregate expenditure (E_0) from the input–output tables.

[51] Let the domestic sales of intermediate and final goods by domestic firms be x and y, respectively, and the sales by foreign firms (which by assumption sell only to final consumers) be a. I need to distinguish between x and y, but in the economic census, I can observe only their sum, $b = x + y$. Since in the input–output table I observe the industry's (domestic and foreign firms together) intermediate, C, and final sales, D, I can back out domestic firms' intermediate and final sales using $x = \left(\frac{C}{C+D}\right)(a+b)$, which implies $\frac{R_{00}^{k,f}}{E_0^k} = \left[\frac{b}{a+b} - \left(\frac{a}{a+b}\right)\left(\frac{C}{D}\right)\right]\frac{R_0^{k,f}}{E_0^k}$, where the second ratio is observable in the input–output table. This strategy works as long as $y > 0$. If $y < 0$, I assume that all sales of domestic firms are intermediate inputs ($x = b$), thus $\frac{R_{00}^{k,f}}{E_0^k} = 0$.

[52] The economic census does not use any international standard system to classify goods. I was able to exploit the information in the auxiliary dataset only because INEGI shared with me the internal correspondence tables (one for each census year) that it uses to construct the official input–output tables. These tables assign a six-digit SCIAN industry of origin to each input (including nonmanufacturing ones), which allows me construct industry-to-industry transactions. However, to differentiate between standardized and relationship-specific transactions, I also need to group inputs according to the Rauch (1999) classification. For this, only the 2009 table is useful, since it is the only one that also assigns a CPC Rev.2 code to each input. Using publicly available correspondence tables to go from CPC to SITC classification (the one used by Rauch), I can classify inputs as standardized or relationship specific.

[53] I could also obtain these matrices for each HQ country with firms in the downstream industry. I do not do this because of the large number of missing observations, which implies that these tables are less reliable or representative the more disaggregated they are.

A. Proofs for Section 4 (Solving the model)

Proposition 1 (Revenue shares)
Let input v be produced by a supplier in country j.

1. *The supplier's share of rents is:*

$$s_j^k(v) = \left(\frac{\alpha_r^k \rho_k}{\alpha_r^k \rho_k + \rho_r^k}\right)\left(\frac{m_j^k(v)}{M_r^k}\right)^{\rho_r^k} \tag{13}$$

where $\rho_r^k \equiv \frac{\zeta_r^k - 1}{\zeta_r^k}$ and $\rho_k \equiv \frac{\sigma_k - 1}{\sigma_k}$.

2. *The buyer's share of rents is:*

$$s_b^k = \frac{\rho_r^k}{\alpha_r^k \rho_k + \rho_r^k} \tag{14}$$

Proof. Suppose the number of intermediate suppliers is finite and equal to $\mathcal{I}$ and that each one of them controls a range $\varepsilon = 1/\mathcal{I}$ of the continuum of inputs. The Shapley value for a supplier v is equal to its "average" marginal contribution across all possible coalitions of agents. The marginal contribution to a coalition of $n \leq \mathcal{I}$ suppliers plus the buyer is the increase in revenue from adding the $n+1$ supplier (supplier v).

Revenue with n suppliers. The presence of iceberg frictions implies that for every unit sold in market i, plants need to manufacture and ship $\tau_{0i}^k \gamma_{hi} \geq 1$ units. This and equation 5 allow us to write the *global* demand function faced by a plant as:

$$\overbrace{q_h^k}^{\text{global demand}} = \sum_i \underbrace{\overbrace{q_{hi}^k}^{\text{sold}} (\tau_{0i}^k \gamma_{hi})}_{\text{produced}} = \left[\left(\frac{\sigma_k}{\sigma_k - 1}\right) c_h^k\right]^{-\sigma_k} \left[\sum_{i=0}^{J} \beta_k E_i (P_i^k)^{\sigma_k - 1} (\gamma_{jh} \tau_{j0}^k)^{1-\sigma_k}\right]$$

Defining the plant's free-on-board (FOB) price as $p_h^k \equiv \left(\frac{\sigma_k}{\sigma_k - 1}\right) c_h^k$, we can write the global inverse demand function as:

$$p_h^k = (q_h^k)^{-\frac{1}{\sigma_k}} \left[\sum_{i=0}^{J} \beta_k E_i (P_i^k)^{\sigma_k - 1} (\gamma_{jh} \tau_{j0}^k)^{1-\sigma_k}\right]^{\frac{1}{\sigma_k}}$$

We can multiply by q_h^k to express the plant's total revenue in terms of its relationship-specific input bundle:

$$\begin{aligned} R_h^k &= (q_h^k)^{\frac{\sigma_k - 1}{\sigma_k}} \left[\sum_{i=0}^{J} \beta_k E_i (P_i^k)^{\sigma_k - 1} (\gamma_{jh} \tau_{j0}^k)^{1-\sigma_k}\right]^{\frac{1}{\sigma_k}} \\ &= \underbrace{\left[\varphi_h^k A_0^k \left(\frac{L^k}{\alpha_l^k}\right)^{\alpha_l^k} \left(\frac{M_s^k}{\alpha_s^k}\right)^{\alpha_s^k} \left(\frac{1}{\alpha_r^k}\right)^{\alpha_r^k}\right]^{\frac{\sigma_k - 1}{\sigma_k}} \left[\sum_{i=0}^{J} \beta_k E_i (P_i^k)^{\sigma_k - 1} (\gamma_{jh} \tau_{j0}^k)^{1-\sigma_k}\right]^{\frac{1}{\sigma_k}}}_{A_h^k} (M_r^k)^{\alpha_r^k \left(\frac{\sigma_k - 1}{\sigma_k}\right)} \end{aligned}$$

Finally, omitting the plant's industry and home country indices, the total revenue of a coalition between the buyer and n suppliers is defined as:

$$R(n) = A\left(\sum_{i=1}^{n} m_i^{\rho_r^k}\right)^{\frac{\alpha_r^k \rho_k}{\rho_r^k}} \tag{41}$$

where $\rho_r^k \equiv \frac{\zeta_r^k - 1}{\zeta_r^k}$ and $\rho_k \equiv \frac{\sigma_k - 1}{\sigma_k}$.

Marginal contribution. If the coalition includes the buyer and $n-1$ other suppliers, the marginal contribution of supplier v is:

$$\Delta R_v(n;\varepsilon) = A\left[\left(\sum_{i=1}^{n-1} \varepsilon \cdot {m_i}^{\rho_r^k} + \varepsilon \cdot {m_v}^{\rho_r^k}\right)^{\frac{\alpha_r^k \rho_k}{\rho_r^k}} - \left(\sum_{i=1}^{n-1} \varepsilon \cdot {m_i}^{\rho_r^k}\right)^{\frac{\alpha_r^k \rho_k}{\rho_r^k}}\right]$$

If the coalition does not includes the buyer, then there is no production and no revenue. In this case, the value of a coalition would have simply been the sum of the outside value of the inputs and, therefore, the marginal contribution of supplier v would have been its outside value. However, assumption 1 makes this outside value zero, so the marginal contribution will be positive only if the buyer is in the coalition.

The Shapley value. As was defined in section 3.7.2, the Shapley value has an intuitive interpretation: It is the average marginal contribution of each player across all coalitions. Since for any coalition size, the marginal contribution of a player may depend on the "order" in which the player is added, this average must take into account all possible orders.

In our model, there are $\mathcal{I}$ suppliers and one buyer ("player 0"), numbered $0, 1, 2, \cdots, \mathcal{I}$. Let g represent one particular sequence of these $\mathcal{I}+1$ players. g can be thought of a function that assigns a position n to each player i:

$$\begin{aligned} g : \{0, 1, \cdots, \mathcal{I}\} &\to \{0, 1, \cdots, \mathcal{I}\} \\ g(i) &= n \end{aligned}$$

Denote by z_g^v the set of players in sequence g who were added before player v. Under this notation, $v(z_g^v)$ is the value of the coalition before player v joined, and $v\left(z_g^v \cup v\right)$ is the value after v joined. Given that there are $(\mathcal{I}+1)!$ possible sequences (permutations) of players, the Shapley value for player v is:

$$S_v = \frac{1}{(\mathcal{I}+1)!} \sum_{g \in G} \left[v\left(z_g^v \cup v\right) - v(z_g^v)\right]$$

This expression is directly connected to the definition, but it is more useful to rewrite it in terms of all possible *positions* of player v. To do so, first note that if we fix player v in some position n, there will be $\mathcal{I}$ positions to fill (n positions to fill before and $\mathcal{I}-n$ positions after) with $\mathcal{I}$ players. Hence, the total number of possible sequences where player v is in position n is $\mathcal{I}!$

Second, note that player v's marginal contribution depends on whether the buyer is in the coalition. Hence, we need to calculate the probability that player 0 is already in the coalition, conditional on player v's position in the coalition (n). If we fix player 0 in some position $n' \neq n$, there are $(\mathcal{I}-1)!$ possible sequences where players v and 0 are in positions n and n', respectively. Given that there are n positions in the coalition before v's is added (positions $0, 1, \cdots, n-1$), the probability that the buyer was already in the coalition is $\frac{\sum_{n'=0}^{n-1}(\mathcal{I}-1)!}{\mathcal{I}!} = \frac{n(\mathcal{I}-1)!}{\mathcal{I}!} = \frac{n}{\mathcal{I}}$.

Putting all these pieces together, we can rewrite the Shapley value as:[54]

$$\begin{aligned} S_v &= \frac{1}{(\mathcal{I}+1)!} \sum_{n=1}^{\mathcal{I}} (\mathcal{I}!)\left(\frac{n}{\mathcal{I}}\right) \Delta R_v(n;\varepsilon) \\ &= \frac{1}{(\mathcal{I}+1)\mathcal{I}} \sum_{n=1}^{\mathcal{I}} n \cdot \Delta R_v(n;\varepsilon) \end{aligned}$$

[54]Note that the summation need start only at $n = 1$ since when player v is the first in the coalition (position 0), its marginal contribution must be zero since necessarily the buyer has not been added yet: $v\left(z_g^v \cup v\right) = v(z_g^v) = 0$.

$$= \left(\frac{1}{1+\varepsilon}\right)\sum_{n=1}^{1/\varepsilon}(n\varepsilon)\cdot A\left[\left(\sum_{i=1}^{n-1}\varepsilon\cdot {m_i}^{\rho_r^k}+\varepsilon\cdot {m_v}^{\rho_r^k}\right)^{\frac{\alpha_r^k\rho_k}{\rho_r^k}}-\left(\sum_{i=1}^{n-1}\varepsilon\cdot {m_i}^{\rho_r^k}\right)^{\frac{\alpha_r^k\rho_k}{\rho_r^k}}\right]\cdot\varepsilon \tag{42}$$

where in the last step, $\mathcal{I}$ was replaced by $1/\varepsilon$.

Approximation to the Shapley value. I will approximate equation 42 using a linear MacLaurin expansion, where S_v is defined as a function of variable ${m_v}^{\rho_r^k}$:

$$S_v \approx \left[\left(\frac{\alpha_r^k\rho_k}{\rho_r^k}\right)A{m_v}^{\rho_r^k}\cdot\varepsilon\right]\left(\frac{1}{1+\varepsilon}\right)\sum_{n=1}^{1/\varepsilon}(n\varepsilon)\left(\sum_{i=1}^{n-1}\varepsilon\cdot {m_i}^{\rho_r^k}\right)^{\frac{\alpha_r^k\rho_k}{\rho_r^k}-1}\cdot\varepsilon$$

Asymptotic Shapley value. The final step is to divide both sides by ε and take limits as the number of suppliers $\mathcal{I}\to\infty$ (which implies that $\varepsilon\to 0$). The outside summation with respect to n, going from 1 to $1/\varepsilon$, converges to an integral with respect to $x\equiv n\varepsilon$, going from 0 to 1, with $dx\equiv\varepsilon$.[55] The inside summation from $\varepsilon\cdot m_1^{\rho_r^k}$ to $\varepsilon\cdot m_{n-1}^{\rho_r^k}$ converges to an integral of $m(v)^{\rho_r^k}$, where v goes from 0 to x[56]

$$\begin{aligned}
S(v)\equiv\lim_{\varepsilon\to 0}\left(\frac{S_v}{\varepsilon}\right) &= \left[\left(\frac{\alpha_r^k\rho_k}{\rho_r^k}\right)Am(v)^{\rho_r^k}\right]\left[\int_0^1 x\left(\int_0^x m(v)^{\rho_r^k}dv\right)^{\frac{\alpha_r^k\rho_k}{\rho_r^k}-1}dx\right]\\
&= \left[\left(\frac{\alpha_r^k\rho_k}{\rho_r^k}\right)Am(v)^{\rho_r^k}\right]\left[\int_0^1 x\left(x\int_0^1 m(v)^{\rho_r^k}dv\right)^{\frac{\alpha_r^k\rho_k}{\rho_r^k}-1}dx\right]\\
&= \left[\left(\frac{\alpha_r^k\rho_k}{\rho_r^k}\right)Am(v)^{\rho_r^k}\right]\underbrace{\left(\int_0^1 x^{\frac{\alpha_r^k\rho_k}{\rho_r^k}}dx\right)}_{\frac{\rho_r^k}{\alpha_r^k\rho_k+\rho_r}}\underbrace{\left(\int_0^1 m(v)^{\rho_r^k}dv\right)^{\frac{\alpha_r^k\rho_k}{\rho_r^k}-1}}_{M_r^{\alpha_r^k\rho_k}M_r^{-\rho_r^k}}\\
&\underbrace{=}_{\text{eq. 41}}\left(\frac{\alpha_r^k\rho_k}{\alpha_r^k\rho_k+\rho_r}\right)\cdot\left(\frac{m(v)}{M_r}\right)^{\rho_r^k}\cdot R
\end{aligned}$$

The buyer's asymptotic Shapley value is the residual from subtracting the sum of suppliers' Shapley values from total revenue:

$$\begin{aligned}
S_b &= R-\int_0^1 S(v)dv = R-\left(\frac{\alpha_r^k\rho_k}{\alpha_r^k\rho_k+\rho_r}\right)\cdot R\\
&= \left(\frac{\rho_r^k}{\alpha_r^k\rho_k+\rho_r}\right)\cdot R
\end{aligned}$$

□

Proposition 2 (Price for relationship-specific inputs)
Let input v be produced by a supplier in country j. The price that rationalizes trade flows from country j is:

$$p_{jh}^{rk}(v)=\frac{w_j\gamma_{jh}\tau_{j0}^k\Lambda_j^k}{a_j^{rk}(v)} \tag{15}$$

[55] At $n=1$, $x=\varepsilon\to 0$, and at $n=\frac{1}{\varepsilon}$, $x=\frac{\varepsilon}{\varepsilon}=1$.
[56] $(n-1)$ in terms of x is $(n-1)\varepsilon=(n\varepsilon)-\varepsilon\to x$.

where:

$$\Lambda_j^k \equiv \frac{1}{s_b^{1-\mu_j}\left(\Xi_j^k\right)^{\mu_j}\left[\mu_j\Xi_j^k+(1-\mu_j)s_b\right]^{\frac{1-\rho_r^k}{\rho_r^k}}} \qquad \textit{and} \qquad \Xi_j^k \equiv \left[\frac{1-s_b(1-\mu_j)\rho_r^k}{1-(1-\mu_j)\rho_r^k}\right]$$

Proof. Let $c_j^k(\nu)$ be the marginal cost plus trade costs of input ν. For example, if the input is sourced from country j, then $c_j^k(\nu)$ is:

$$c_j^k(\nu) = \frac{w_j\gamma_{jh}\tau_{j0}^k}{a_j^{rk}(\nu)}$$

All agents can predict the result of the bargaining; thus, when suppliers have to choose how much to invest in noncontractible specifications, they will do so taking into account their post-bargaining payoff.

Noncontractible specifications. Let input ν be sourced from a supplier in country j. The supplier's optimization problem is:[57]

$$\begin{aligned}\max_{m_{n,j}^k(\nu)} \pi_j^k(\nu) &= S_j^k(\nu) - c_j^k(\nu)(1-\mu_j)m_{n,j}^k(\nu) - c_j^k(\nu)\mu_j m_{c,j}^k(\nu)\\ &= \left(\frac{\alpha_r^k\rho_k}{\alpha_r^k\rho_k+\rho_r^k}\right)R_h^k\left[\frac{m_j^k(\nu)}{M_r^k}\right]^{\rho_r^k} - c_j^k(\nu)(1-\mu_j)m_{n,j}^k(\nu) - c_j^k(\nu)\mu_j m_{c,j}^k(\nu)\\ &= \left(\frac{\alpha_r^k\rho_k}{\alpha_r^k\rho_k+\rho_r^k}\right)R_h^k\left(M_r^k\right)^{-\rho_r^k} m_{c,j}^k(\nu)^{\mu_j\rho_r^k} m_{n,j}^k(\nu)^{(1-\mu_j)\rho_r^k} - c_j^k(\nu)(1-\mu_j)m_{n,j}^k(\nu) - c_j^k(\nu)\mu_j m_{c,j}^k(\nu)\end{aligned}$$

Solving the first-order condition (FOC), we obtain:

$$m_{n,j}^k(\nu) = \left[\rho_r^k\left(\frac{\alpha_r^k\rho_k}{\alpha_r^k\rho_k+\rho_r^k}\right)\frac{R_h^k\left(M_r^k\right)^{-\rho_r^k}}{c_j^k(\nu)}\right]^{\frac{1}{1-(1-\mu_j)\rho_r^k}} m_{c,j}^k(\nu)^{\frac{\mu_j\rho_r^k}{1-(1-\mu_j)\rho_r^k}} \tag{43}$$

This result translates into the following expression for $m_j^k(\nu)$:

$$m_j^k(\nu) \equiv m_{n,j}^k(\nu)^{1-\mu_j} m_{c,j}^k(\nu)^{\mu_j} \Rightarrow m_j^k(\nu) = \left[\rho_r^k\left(\frac{\alpha_r^k\rho_k}{\alpha_r^k\rho_k+\rho_r^k}\right)\frac{R_h^k\left(M_r^k\right)^{-\rho_r^k}}{c_j^k(\nu)}\right]^{\frac{1-\mu_j}{1-(1-\mu_j)\rho_r^k}} m_{c,j}^k(\nu)^{\frac{\mu_j}{1-(1-\mu_j)\rho_r^k}} \tag{44}$$

Contractible specifications. Given the previous result, the buyer's problem is to pick $m_c^k(\nu)$ and $f^k(\nu)$ to maximize its payoff:[58]

$$\max_{\{m_c^k(\nu),f^k(\nu)\}} \pi_b^k = S_b^k - \int_0^1 f^k(\nu)d\nu \quad \text{s.t.} \quad \pi^k(\nu) + f^k(\nu) \geq 0, \forall\nu \in [0,1]$$

Given that the buyer is able to offer a take-it-or-leave-it contract, it will choose the fee $f^k(\nu)$ to make the supplier just indifferent between accepting or rejecting the contract; i.e., the constraint will bind. The buyer's problem becomes:

$$\max_{\{m_c^k(\nu)\}} \pi_b^k = S_b^k + \int_0^1 \pi^k(\nu)d\nu$$

[57] $S_j^k(\nu)$ depends on $m_j^k(\nu)$, which should be interpreted as the amount of input that *effectively* contributes to revenue. Since there are trade costs, for every $m_j^k(\nu)$ that makes it into production, the supplier must ship $\gamma_{jh}\tau_{j0}^k m_j^k(\nu)$. Thus, $c_j^k(\nu)$ should be interpreted as the marginal cost of production *inclusive* of the trade costs.

[58] Note that variables indexed by ν implicitly have a country of origin $j = j(\nu)$ associated with them. I omit it to avoid notational clutter, except for the contractibility parameter μ.

$$\Rightarrow \max_{\{m_c^k(\nu)\}} \pi_b^k = S_b^k + \int_0^1 \left\{ S^k(\nu) - c^k(\nu)\left[1-\mu_{j(\nu)}\right] m_n^k(\nu) - c^k(\nu)\mu_{j(\nu)} m_c^k(\nu) \right\} d\nu$$

$$\Rightarrow \max_{\{m_c^k(\nu)\}} \pi_b^k = R_h^k - \int_0^1 \left\{ c^k(\nu)\left[1-\mu_{j(\nu)}\right] m_n^k(\nu) + c^k(\nu)\mu_{j(\nu)} m_c^k(\nu) \right\} d\nu$$

$$\text{s.t.} \quad R_h^k \equiv A_h^k \left(\int_0^1 m^k(\nu)^{\rho_r^k} d\nu \right)^{\frac{\alpha_r^k \rho_k}{\rho_r^k}}$$

and equations 43 and 44.

Using the expression for the buyer's share from equation 14, the optimal contractible specifications bundle for an input sourced from country j is:

$$m_{c,j}^k(\nu) = \left[\frac{\alpha_r^k \rho_k R_h^k \left(M_r^k\right)^{-\rho_r^k}}{c_j^k(\nu)} \right]^{\frac{1}{1-\rho_r^k}} \underbrace{\left\{ \left(s_b^k\right)^{(1-\mu_j)\rho_r^k} \left[\frac{1 - s_b^k(1-\mu_j)\rho_r^k}{1-(1-\mu_j)\rho_r^k} \right]^{1-(1-\mu_j)\rho_r^k} \right\}^{\frac{1}{1-\rho_r^k}}}_{\Gamma_{c,j}^k} \tag{45}$$

We can plug this result back into equation 43 to obtain:

$$m_{n,j}^k(\nu) = \left[\frac{\alpha_r^k \rho_k R_h^k \left(M_r^k\right)^{-\rho_r^k}}{c_j^k(\nu)} \right]^{\frac{1}{1-\rho_r^k}} \underbrace{\left\{ \left(s_b^k\right)^{1-\mu_j\rho_r^k} \left[\frac{1 - s_b^k(1-\mu_j)\rho_r^k}{1-(1-\mu_j)\rho_r^k} \right]^{\mu_j\rho_r^k} \right\}^{\frac{1}{1-\rho_r^k}}}_{\Gamma_{n,j}^k} \tag{46}$$

Finally, we can plug the result back into equation 44 to obtain:

$$m_j^k(\nu) = \left[\frac{\alpha_r^k \rho_k R_h^k \left(M_r^k\right)^{-\rho_r^k}}{c_j^k(\nu)} \right]^{\frac{1}{1-\rho_r^k}} \underbrace{\left\{ \left(s_b^k\right)^{1-\mu_j} \left[\frac{1 - s_b^k(1-\mu_j)\rho_r^k}{1-(1-\mu_j)\rho_r^k} \right]^{\mu_j} \right\}^{\frac{1}{1-\rho_r^k}}}_{\Gamma_j^k} \tag{47}$$

The expressions in square brackets on the right-hand side (RHS) of equations 47, 45, and 46 are the benchmark results when there is perfect contractibility. In that case, the distinction between contractible and noncontractible specifications is not meaningful since $m_j^k(\nu) = m_{c,j}^k(\nu) = m_{n,j}^k(\nu)$.

Implicit prices. The first step is to determine the *net* payment to suppliers. On one hand, a supplier from country j receives $s_j^k(\nu) R_h^k$ as a result of bargaining. On the other hand, to win the contract in the first place, it had to pay the buyer $f_j^k(\nu)$. Therefore, its net payment is:

$$X_j^k(\nu) = c_j^k(\nu) \left[\mu_j m_{c,j}^k(\nu) + (1-\mu_j) m_{n,j}^k(\nu) \right]$$

Using equations 47 and 45, this expression becomes:[59]

$$\begin{aligned} X_j^k(\nu) &= c_j^k(\nu)^{1-\zeta_r^k} \left[\alpha_r^k \rho_k R_h^k \left(M_r^k\right)^{-\rho_r^k} \right]^{\zeta_r^k} \left[\mu_j (\Gamma_{c,j}^k)^{\zeta_r^k} + (1-\mu_j)(\Gamma_{n,j}^k)^{\zeta_r^k} \right] \\ &= c_j^k(\nu)^{1-\zeta_r^k} \left[\alpha_r^k \rho_k R_h^k \left(M_r^k\right)^{-\rho_r^k} \right]^{\zeta_r^k} (\Gamma_j^k)^{\zeta_r^k - 1} \left\{ \mu_j \Xi_j^k + (1-\mu_j) s_b^k \right\} \end{aligned} \tag{48}$$

[59] I have replaced $\frac{1}{1-\rho_r^k}$ by its equivalent ζ_r^k.

One intuitive way to obtain the price is to divide this expression by the quality-adjusted quantity of $m_j^k(\nu)$ found in equation 46. This gives us the implicit price per unit of quality of input ν:

$$p_j^k(\nu) \equiv \frac{X_j^k(\nu)}{m_j^k(\nu)} = c_j^k(\nu)\left\{\frac{\mu_j \Xi_j^k + (1-\mu_j)s_b^k}{\Gamma_j^k}\right\}$$

This approach has two disadvantages. First, the expression in brackets is not a monotonic function of either μ_j or ζ_r^k, which leads to less intuitive and clear insights.[60] Second, although it is still possible to obtain closed-forms expressions for the price index and the share of *inputs* purchased from a country j, the latter is not equal to the share of *expenditure* on inputs from country j, as in Eaton and Kortum (2002), which unnecessarily complicates the analysis.

An alternative is to define prices as those that *rationalize* the expenditure flows $X_j^k(\nu)$ "as if" buyers were able to optimally choose $m_j^k(\nu)$ and contracting frictions affected these quantities only *indirectly* via their prices. To implement this formulation, we need only to rewrite equation 48 as:

$$X_j^k(\nu) = \left[\underbrace{c_j^k(\nu)\Lambda_j^k}_{p_j^k(\nu)}\right]^{1-\zeta_r^k}\left[\alpha_r^k \rho_k R_h^k \left(M_r^k\right)^{-\rho_r^k}\right]^{\frac{\zeta_r^k}{}}$$

where:

$$\Lambda_j^k \equiv \frac{1}{\left(s_b^k\right)^{1-\mu_j}\left[\frac{1-s_b^k(1-\mu_j)\rho_r^k}{1-(1-\mu_j)\rho_r^k}\right]^{\mu_j}\left[\mu_j\left[\frac{1-s_b^k(1-\mu_j)\rho_r^k}{1-(1-\mu_j)\rho_r^k}\right]+(1-\mu_j)s_b^k\right]^{\frac{1-\rho_r^k}{\rho_r^k}}}$$

□

Proposition 3 (Properties of the contracting frictions parameter Λ_j^k)

1. *Λ_j^k behaves likes an iceberg variable cost, in the sense that it is bounded from below by one and it is unbounded from above:*

$$\Lambda_j^k \in [1, +\infty)$$

2. *Λ_j^k decreases (contracting frictions are lower) with (1) the strength of contract enforcement in the supplier's country (μ_j) and (2) the elasticity of substitution among relationship-specific inputs (ζ_r^k). On the other hand, it increases with (1) the buyer's technological need for these inputs (α_r^k) and (2) the elasticity of demand the buyer faces for its products (σ_k):*

$$\Lambda_j^k(\overbrace{\ \zeta_r^k\ }^{-}, \overbrace{\ \alpha_r^k\ }^{+}, \overbrace{\ \sigma_k\ }^{+}, \overbrace{\ \mu_j\ }^{-})$$

3. *Λ_j^k converges to one (contracting frictions disappear) as (1) contract enforcement becomes perfect (μ_j), (2) the buyer's technological need for relationship-specific inputs goes to zero (α_r^k), or (3) the elasticity of demand the buyer faces for its products becomes unit elastic (σ_k). On the other hand, it diverges (trade collapses) as the elasticity of substitution among relationship-specific inputs becomes unit elastic (ζ_r^k).*

$$\lim_{\mu_j \nearrow 1} \Lambda_j^k = \lim_{\alpha_r^k \searrow 0} \Lambda_j^k = \lim_{\sigma_k \searrow 1} \Lambda_j^k = 1, \quad \textit{and} \quad \lim_{\zeta_r^k \searrow 1} \Lambda_j^k = +\infty$$

Proof. Let us start by establishing Λ_j^k's lower and upper bounds.

[60] For example, $p_j(\nu)$ is increasing in contract enforcement quality (μ_j) at low levels of μ_j (i.e., among countries with weak institutions, those with relatively stronger ones are *less* attractive sources of inputs, while among countries with a minimum quality of contract enforcement, the relationship reverses), and increasing in input elasticity of substitution (ζ_r^k) at high levels of ζ_r^k and low levels of μ_j (i.e., in countries with weak contract enforcement institutions, industries for which it is "too" easy to substitute inputs would face higher frictions than industries for which it is less easy to do so). These patterns can be a consequence of the fact that the contracting frictions parameters for $X_j(\nu)$ and $m_j(\nu)$ have the same qualitative relationship with the fundamental parameters (increasing in μ_j, concave in ζ_r^k, and decreasing in α_r^k and σ_k).

Lower and upper bounds of Λ_j^k.- Given that $s_b^k \in [0,1]$ and $\mu_j \in [0,1]$, then:

$$\Xi_j^k \equiv \frac{1-s_b^k(1-\mu_j)\rho_r^k}{1-(1-\mu_j)\rho_r^k} = 1 + \frac{(1-s_b^k)(1-\mu_j)\rho_r^k}{1-(1-\mu_j)\rho_r^k} \geq 1$$

Define $\Gamma_j^k \equiv \left(s_b^k\right)^{1-\mu_j}\left(\Xi_j^k\right)^{\mu_j}$. Since $s_b^k \geq 0$ and $\Xi_j^k \geq 1$, then $\Gamma_j^k \geq 0$. Its upper bound can be determined by first noting that Γ_j^k is always increasing in ζ_r^k:

$$\frac{d\Gamma_j^k}{d\zeta_r^k} = \frac{(1-\mu_j)\Gamma_j^k(1-s_b^k)}{(\zeta_r^k)^2}\left\{\frac{1}{\rho_r^k} + \frac{\mu_j[(1-s_b^k)+s_b^k(1-\mu_j)\rho_r^k]}{[1-s_b^k(1-\mu_j)\rho_r^k][1-(1-\mu_j)\rho_r^k]}\right\} \geq 0 \tag{49}$$

Hence, Γ_j^k is bounded from above by its limit when ζ_r^k becomes infinitely large:

$$\Gamma_j^k \leq \lim_{\zeta_r^k \to +\infty} \Gamma_j^k = \lim_{\rho_r^k \to 1} \Gamma_j^k = \left(\frac{1}{1+\alpha_r^k\rho_k}\right)\left(1+\frac{\alpha_r^k\rho_k}{\mu_j}\right)^{\mu_j} \in [0.5,1] \leq 1$$

Now, note that the second bracket in the denominator of Λ_j^k is between zero and one:

$$\mu_j\Xi_j^k + (1-\mu_j)s_b^k = 1 - \frac{(1-\mu_j)(1-s_b^k)(1-\rho_r^k)}{1-(1-\mu_j)\rho_r^k} \in [0,1] \tag{50}$$

Given that $\rho_r^k \in [0,1]$, then $\frac{1-\rho_r^k}{\rho_r^k} \geq 0$. Therefore, $\left[\mu_j\Xi_j^k + (1-\mu_j)s_b^k\right]^{\frac{1-\rho_r^k}{\rho_r^k}} \in [0,1]$, too. Putting all together:

$$\Lambda_j^k \equiv \left\{ \underbrace{\Gamma_j^k}_{\in[0,1]} \underbrace{\left[\mu_j\Xi_j^k + (1-\mu_j)s_b^k\right]^{\frac{1-\rho_r^k}{\rho_r^k}}}_{\in[0,1]} \right\}^{-1} \in [1,+\infty)$$

First derivatives and limits of Λ_j^k with respect to fundamental parameters.

- *Elasticity of substitution among inputs ζ_r^k*
  The results in equations 49 and 50 imply:

$$\frac{d\Lambda_j^k}{d\zeta_r^k} = -\Lambda_j^k\left\{ \underbrace{\frac{1}{\Gamma_j^k}\left(\frac{d\Gamma_j^k}{d\zeta_r^k}\right)}_{\geq 0} + \underbrace{\left(\frac{1-\rho_r^k}{\rho_r^k}\right)\left(\frac{1}{\mu_j\Xi_j^k + (1-\mu_j)s_b^k}\right)\left[\mu_j\frac{d\Xi_j^k}{d\zeta_r^k} + (1-\mu_j)\frac{ds_b^k}{d\zeta_r^k}\right]}_{\geq 0} \underbrace{-\frac{\log\left(\mu_j\Xi_j^k + (1-\mu_j)s_b^k\right)}{(\zeta_r^k-1)^2}}_{\geq 0} \right\} \leq 0$$

  The limits of Λ_j^k with respect to the bounds of ζ_r^k are:

$$\lim_{\zeta_r^k \to 1} \Lambda_j^k = \lim_{\rho_r^k \to 0} \Lambda_j^k = +\infty$$

$$\lim_{\zeta_r^k \to +\infty} \Lambda_j^k = \lim_{\rho_r^k \to 1} \Lambda_j^k = \left(\lim_{\zeta_r^k \to +\infty} \Gamma_j^k\right)^{-1} \in [1,2]$$

- *Technological intensity α_r^k or product demand elasticity σ_k*
  Λ_j^k is an increasing function of α_r^k and ρ_k:

$$\frac{d\Lambda_j^k}{d(\alpha_r^k\rho_k)} = -\Lambda_j^k \Bigg\{ \underbrace{\frac{1}{\Gamma_j^k}\frac{d\Gamma_j^k}{d(\alpha_r^k\rho_k)}}_{\leq 0} \underbrace{- \left(\frac{1-\rho_r^k}{\rho_r^k}\right)\left(\frac{1}{\mu_j\Xi_j^k + (1-\mu_j)s_b^k}\right)\left[\frac{(1-\mu_j)(s_b^k)^2(1-\rho_r^k)}{[1-(1-\mu_j)\rho_r^k]\rho_r^k}\right]}_{\leq 0} \Bigg\} \geq 0$$

Given that ρ_r^k is an increasing function of σ_k, the derivative of Λ_j^k with respect to σ_k is also positive. The limits of Λ_j^k with respect to the bounds of α_r^k or σ_k are:

$$\lim_{\alpha_r^k\to 0}\Lambda_j^k = \lim_{\sigma_k\to 1}\Lambda_j^k = \lim_{\rho_k\to 0}\Lambda_j^k = 1$$

$$\lim_{\alpha_r^k\to 1}\Lambda_j^k = \lim_{\sigma_k\to +\infty}\Lambda_j^k = \lim_{\rho_k\to 1}\Lambda_j^k$$

$$= \frac{1+\rho_r^k}{\rho_r^k}\left\{\frac{\rho_r^k\left[1-(1-\mu_j)\rho_r^k\right]}{1+\rho_r^k\left[1-(1-\mu_j)\rho_r^k\right]}\right\}^{\mu_j}\left\{\frac{(1+\rho_r^k)\left[1-(1-\mu_j)\rho_r^k\right]}{(1+\rho_r^k)\left[1-(1-\mu_j)\rho_r^k\right]-(1-\mu_j)(1-\rho_r^k)}\right\}^{\frac{1}{\rho_r^k}-1} \geq 1$$

- *Contractibility* μ_j^k

  Λ_j^k is a decreasing function of μ_j as:

$$\frac{d\Lambda_j^k}{d\mu_j} = -\Lambda_j^k\left\{\log\Xi_j^k - \log s_b^k - \frac{\mu_j(1-s_b^k)}{\left[1-s_b^k(1-\mu_j)\rho_r^k\right]\left[\mu_j+(1-\mu_j)s_b^k(1-\rho_r^k)\right]}\right\} \leq 0$$

To justify the last inequality, we need to follow three steps. First, since both s_b^k and Ξ_j^k are nonnegative, the properties regarding bounds of logarithmic functions imply the following lower bound for $\log\Xi_j^k - \log s_b^k$:[61]

$$\log\Xi_j^k - \log s_b^k \geq \frac{(1-s_b^k)[1+(1-s_b^k)(1-\mu_j)\rho_r^k]}{1-s_b^k(1-\mu_j)\rho_r^k}$$

Second, given that $\rho_r^k \leq 1$, then:

$$\frac{\mu_j\rho_r^k}{1-(1-\mu_j)\rho_r^k} = \frac{\mu_j\rho_r^k}{(1-\rho_r^k)+\mu_j\rho_r^k} \in [0,1]$$

Putting these pieces together, we obtain:

$$\log\Xi_j^k - \log s_b^k \geq \left[\frac{1-s_b^k}{1-s_b^k(1-\mu_j)\rho_r^k}\right]\underbrace{\left[1+(1-s_b^k)(1-\mu_j)\rho_r^k\right]}_{\geq 1} \geq \left[\frac{1-s_b^k}{1-s_b^k(1-\mu_j)\rho_r^k}\right]\underbrace{\left[\frac{\mu_j}{\mu_j+(1-\mu_j)s_b^k(1-\rho_r^k)}\right]}_{\leq 1}$$

$$\Rightarrow \log\Xi_j^k - \log s_b^k - \frac{\mu_j(1-s_b^k)}{\left[1-s_b^k(1-\mu_j)\rho_r^k\right]\left[\mu_j+(1-\mu_j)s_b^k(1-\rho_r^k)\right]} \geq 0$$

The limits of Λ_j^k with respect to the bounds of μ_j are:

$$\lim_{\mu_j\to 0}\Lambda_j^k = \left(\frac{\alpha_r^k\rho_k+\rho_r^k}{\rho_r^k}\right)^{\frac{1}{\rho_r^k}} \geq 1 \qquad \text{and} \qquad \lim_{\mu_j\to 1}\Lambda_j^k = 1$$

□

[61] I am using the following property of logarithmic functions: $1-\frac{1}{x} \leq \log x \leq x-1$.

B. Proofs for Section 5 (Counterfactual exercises)

Proposition 4 (Identification of parameter shocks)
For any input source country j, destination market i, HQ country h, industry k, and variable x, let $\widehat{x}_h$ represent the relative value of x between domestic firms (country 0) and foreign affiliates from country h:

$$\widehat{x}_h \equiv \frac{x_0}{x_h}$$

The parameter shocks associated with each counterfactual scenario can be identified with the following formulas:

1. ***Contracting frictions:***

$$\left(\widehat{\Lambda}_{jh}^{k}\right)^{-\theta_r} = \left(\frac{\widehat{\chi}_{jh}^{rk}}{\widehat{\chi}_{j^*h}^{rk}}\right)\left(\frac{\widehat{\chi}_{j^*h}^{sk}}{\widehat{\chi}_{jh}^{sk}}\right)^{\frac{\theta_r}{\theta_s}} \tag{18}$$

   where j^* refers to any country with close to "perfect" contract enforcement institutions ($\mu_{j^*} \approx 1$).

2. ***Geography (input purchases):***

$$\left(\widehat{\gamma}_{jh}\right)^{-\theta_s} = \frac{\widehat{\chi}_{jh}^{sk}}{\widehat{\chi}_{0h}^{sk}} \tag{19}$$

   Alternatively:

$$\left(\widehat{\gamma}_{jh}\right)^{-\theta_r} = \left(\frac{\widehat{\chi}_{jh}^{rk}}{\widehat{\chi}_{0h}^{rk}}\right)\left(\frac{\widehat{\Lambda}_{0h}^{k}}{\widehat{\Lambda}_{jh}^{k}}\right)^{-\theta_r} \tag{20}$$

3. ***Geography (sales):***

$$\left(\widehat{\gamma}_{hi}\right)^{1-\sigma_k} = \frac{\widehat{R}_{hi}^{k,f}}{\widehat{R}_{h0}^{k,f}} \tag{21}$$

4. ***Productivity:***

$$\left(\widehat{\varphi}_{h}^{k}\right)^{\sigma_k-1} = \left(\frac{\widehat{R}_{h0}^{k,f}}{\widehat{N}_{h}^{k}}\right)\left[\left(\frac{1}{\widehat{\chi}_{0h}^{sk}}\right)^{\frac{\alpha_s^k}{\theta_s}}\left(\frac{1}{\widehat{\chi}_{j^*h}^{rk}}\right)^{\frac{\alpha_r^k}{\theta_r}}\left(\frac{\widehat{\chi}_{j^*h}^{sk}}{\widehat{\chi}_{0h}^{sk}}\right)^{\frac{\alpha_r^k}{\theta_s}}\right]^{1-\sigma_k} \tag{22}$$

   Alternatively:

$$\left(\widehat{\varphi}_{h}^{k}\right)^{\sigma_k-1} = \left(\frac{\widehat{R}_{h0}^{k,f}}{\widehat{N}_{h}^{k}}\right)\left\{\left[\sum_j \chi_{jh}^{sk}\left(\widehat{\gamma}_{jh}\right)^{-\theta_s}\right]^{\frac{\alpha_s^k}{\theta_s}}\left[\sum_j \chi_{jh}^{rk}\left(\widehat{\gamma}_{jh}\widehat{\Lambda}_{jh}^{k}\right)^{-\theta_r}\right]^{\frac{\alpha_r^k}{\theta_r}}\right\}^{1-\sigma_k} \tag{23}$$

Proof. The identification of parameter shocks is based on different sets of assumptions. Let us delve into the details of each one.

Contracting frictions.- The identification of the contracting friction shock is based on my exploiting two features of the model. The first is that the relative expenditure shares between two countries, regardless on whether inputs are standardized or relationship specific, has the same functional form and depends on almost the same variables: relative productivities (T_j^{xk}/T_0^{xk}), labor costs (w_j/w_0), and bilateral trade and communication frictions ($\tau_{j0}^k\gamma_{jh}/\tau_{00}^k\gamma_{0h}$). The exception is that contracting frictions do not affect standardized inputs (assumption 2). This implies that the double difference (domestic vs. foreign firms and

standardized vs. relationship-specific inputs) in expenditure shares from some country j reveals information about the relative contracting frictions faced by domestic firms and foreign affiliates when they source from that country (assuming firms in both groups source from this place):

$$\frac{\chi_{j0}^{rk}/\chi_{jh}^{rk}}{\left(\chi_{j0}^{sk}/\chi_{jh}^{sk}\right)^{\frac{\theta_r}{\theta_s}}} = \left(\frac{\Lambda_{j0}^{k}}{\Lambda_{jh}^{k}}\right)^{-\theta_r}\left[\frac{\Theta_{h}^{rk}/\Theta_{0}^{rk}}{\left(\Theta_{h}^{sk}/\Theta_{0}^{sk}\right)^{\frac{\theta_r}{\theta_s}}}\right] \tag{51}$$

This double ratio would be sufficient if the terms Θ_h^{xk} and Θ_0^{xk}, known in the literature as *sourcing capabilities*,[62] were identical. Unfortunately, even if I assume that foreign firms source from the same set of countries as domestic firms (that is, if I shut down extensive-margin differences), these objects will be different as long as domestic and foreign firms face different contracting and communication frictions:

$$\frac{\Theta_{h}^{rk}}{\Theta_{0}^{rk}} = \frac{\sum_{j'} T_{j'}^{rk}(w_{j'}\gamma_{j'h}\tau_{j'0}^{k}\Lambda_{j'h}^{k})^{-\theta_r}}{\sum_{j'} T_{j'}^{rk}(w_{j'}\gamma_{j'0}\tau_{j'0}^{k}\Lambda_{j'0}^{k})^{-\theta_r}}$$

The second model feature is introduced to solve this issue. The third numeral in proposition 3 states that the contracting friction parameter disappears ($\Lambda_j^k \to 1$) as contract enforcement institutions approach perfection ($\mu_j \to 1$). Let j^* denote a country with perfect contract enforcement institutions. For such a country, equation 51 becomes:

$$\frac{\chi_{j^*0}^{rk}/\chi_{j^*h}^{rk}}{\left(\chi_{j^*0}^{sk}/\chi_{j^*h}^{sk}\right)^{\frac{\theta_r}{\theta_s}}} = \frac{\Theta_{h}^{rk}/\Theta_{0}^{rk}}{\left(\Theta_{h}^{sk}/\Theta_{0}^{sk}\right)^{\frac{\theta_r}{\theta_s}}}$$

This means that I can use the same double difference in expenditure shares from some country j^* to control for differences in sourcing capabilities between domestic and foreign firms as long as this is a country with near-perfect contract enforcement institutions, which can be inferred from any of the publicly available rankings of institutional quality:

$$\left(\frac{\chi_{j0}^{rk}/\chi_{j^*0}^{rk}}{\chi_{jh}^{rk}/\chi_{j^*h}^{rk}}\right)\left(\frac{\chi_{jh}^{sk}/\chi_{j^*h}^{sk}}{\chi_{j0}^{sk}/\chi_{j^*0}^{sk}}\right)^{\frac{\theta_r}{\theta_s}} = \left(\frac{\Lambda_{j0}^{k}}{\Lambda_{jh}^{k}}\right)^{-\theta_r} \equiv \left(\widehat{\Lambda}_{jh}^{k}\right)^{-\theta_r}$$

Geography (input purchases).- The identification of the *inbound* communication costs shock is based on the assumption that communications costs are equal between domestic and foreign firms when they source domestically ($\gamma_{0h} = \gamma_{00} = 1$). This implies that the difference in their relative expenditure shares (for standardized inputs) between some country j and the SOE reveals information about the different inbound communication costs faced by these two groups of firms:

$$\begin{aligned}\frac{\chi_{jh}^{sk}}{\chi_{0h}^{sk}} &= \frac{T_j^{sk}\left(w_j\tau_{j0}^{k}\right)^{-\theta_s}}{T_0^{sk}\left(w_0\right)^{-\theta_s}}(\gamma_{jh})^{-\theta_s} \\ \frac{\chi_{j0}^{sk}}{\chi_{00}^{sk}} &= \frac{T_j^{sk}\left(w_j\tau_{j0}^{k}\right)^{-\theta_s}}{T_0^{sk}\left(w_0\right)^{-\theta_s}}(\gamma_{j0})^{-\theta_s}\end{aligned} \Rightarrow \left(\frac{\gamma_{j0}}{\gamma_{jh}}\right)^{-\theta_s} = \frac{\chi_{j0}^{sk}/\chi_{00}^{sk}}{\chi_{jh}^{sk}/\chi_{0h}^{sk}}$$

The alternative specification is theoretically equivalent to this one, but its implementation involves an extra step given that it requires that the contracting friction shocks be previously calibrated. A reason to take into account is that the set of sourcing countries for standardized inputs and that for relationship-specific inputs overlap but differ considerably. Having two theoretically equivalent ways of backing out the same object increases the number of parameters that can be estimated in practice.

[62] See Antràs et al. (2017).

Geography (sales).- The identification of the *outbound* communication costs shock is based on the fact that firm characteristics interact with destination market features to determine sales in a multiplicative manner (see equation 29). This means that the component associated with firm characteristics is the same for markets, so dividing exports to market i by domestic sales isolates the market access differences. Therefore, the difference in their relative sales between some market i and the SOE reveals information of the different outbound communication costs faced by these two groups of firms:

$$\left.\begin{aligned}\frac{R_{hi}^k}{R_{h0}^k} &= \frac{E_i(P_i^k)^{\sigma_k-1}\left(\tau_{0i}^k\gamma_{hi}\right)^{1-\sigma_k}}{E_0(P_0^k)^{\sigma_k-1}}\\ \frac{R_{0i}^k}{R_{00}^k} &= \frac{E_i(P_i^k)^{\sigma_k-1}\left(\tau_{0i}^k\gamma_{0i}\right)^{1-\sigma_k}}{E_0(P_0^k)^{\sigma_k-1}}\end{aligned}\right\} \Rightarrow \frac{R_{h0}^{k,f}/R_{00}^{k,f}}{R_{hi}^k/R_{0i}^k} = \left(\frac{\gamma_{0i}}{\gamma_{hi}}\right)^{1-\sigma_k} \equiv \left(\widehat{\gamma}_{hi}\right)^{1-\sigma_k}$$

Productivity.- The identification of productivity differences is based on my putting together some of the approaches used for the previous shocks. First, it exploits the multiplicative nature of the revenue function by using the relative sales between foreign and domestic firms to isolate the underlying factors behind cost differences between them. Second, it focuses on *domestic* sales because all firms face the same communication costs when selling to the SOE, which thus lets me isolate marginal cost differences from trade cost differences. Third, the model predicts that these cost differences are the outcome of differences in productivity, contracting frictions, and inbound communication costs, so to isolate the first one, we need to write the ratio in terms of the other shocks:

$$\begin{aligned}\frac{R_{h0}^{k,f}}{R_{00}^{k,f}} &= \frac{N_h^k(c_h^k)^{1-\sigma_k}}{N_0^k(c_0^k)^{1-\sigma_k}} \underbrace{=}_{\text{eq. 25}} \frac{N_h^k}{N_0^k}\left[\left(\frac{P_h^{sk}}{P_0^{sk}}\right)^{\alpha_s^k}\left(\frac{P_h^{rk}}{P_0^{rk}}\right)^{\alpha_r^k}\frac{\varphi_0^k}{\varphi_h^k}\right]^{1-\sigma_k}\\ &\underbrace{=}_{\substack{\text{eq. 11}\\ \text{eq. 16}}} \frac{N_h^k}{N_0^k}\left\{\left[\frac{\sum_j T_j^{sk}(w_j\gamma_{j0}\tau_{j0}^k)^{-\theta_s}}{\sum_j T_j^{sk}(w_j\gamma_{jh}\tau_{j0}^k)^{-\theta_s}}\right]^{\frac{\alpha_s^k}{\theta_s}}\left[\frac{\sum_j T_j^{rk}(w_j\gamma_{j0}\tau_{j0}^k\Lambda_{j0}^k)^{-\theta_r}}{\sum_j T_j^{rk}(w_j\gamma_{jh}\tau_{j0}^k\Lambda_{jh}^k)^{-\theta_r}}\right]^{\frac{\alpha_r^k}{\theta_r}}\right\}^{1-\sigma_k}\left(\widehat{\varphi}_h^k\right)^{1-\sigma_k}\\ &= \frac{N_h^k}{N_0^k}\left\{\left[\sum_j \chi_{jh}^{sk}\left(\widehat{\gamma}_{jh}\right)^{-\theta_s}\right]^{\frac{\alpha_s^k}{\theta_s}}\left[\sum_j \chi_{jh}^{rk}\left(\widehat{\gamma}_{jh}\widehat{\Lambda}_{jh}^k\right)^{-\theta_r}\right]^{\frac{\alpha_r^k}{\theta_r}}\right\}^{1-\sigma_k}\left(\widehat{\varphi}_h^k\right)^{1-\sigma_k}\end{aligned}$$

If we replace the parameter shocks in this equation by their formulas and simplify, we obtain:

$$\begin{aligned}\frac{R_{h0}^{k,f}}{R_{00}^{k,f}} &= \frac{N_h^k}{N_0^k}\left\{\left[\left(\frac{\chi_{0h}^{sk}}{\chi_{00}^{sk}}\right)\sum_j \chi_{j0}^{sk}\right]^{\frac{\alpha_s^k}{\theta_s}}\left[\left(\frac{\chi_{0h}^{rk}}{\chi_{00}^{rk}}\right)\left(\widehat{\Lambda}_{0h}^k\right)^{-\theta_r}\sum_j \chi_{j0}^{rk}\right]^{\frac{\alpha_r^k}{\theta_r}}\right\}^{1-\sigma_k}\left(\widehat{\varphi}_h^k\right)^{1-\sigma_k}\\ &= \frac{N_h^k}{N_0^k}\left\{\left(\frac{\chi_{0h}^{sk}}{\chi_{00}^{sk}}\right)^{\frac{\alpha_s^k}{\theta_s}}\left[\left(\frac{\chi_{j^*h}^{rk}}{\chi_{j^*0}^{rk}}\right)\left(\frac{\chi_{0h}^{sk}}{\chi_{00}^{sk}}\right)^{\frac{\theta_r}{\theta_s}}\left(\frac{\chi_{j^*0}^{sk}}{\chi_{j^*h}^{sk}}\right)^{\frac{\theta_r}{\theta_s}}\right]^{\frac{\alpha_r^k}{\theta_r}}\right\}^{1-\sigma_k}\left(\widehat{\varphi}_h^k\right)^{1-\sigma_k}\\ &= \frac{N_h^k}{N_0^k}\left[\left(\frac{\chi_{0h}^{sk}}{\chi_{00}^{sk}}\right)^{\frac{\alpha_s^k+\alpha_r^k}{\theta_s}}\left(\frac{\chi_{j^*h}^{rk}}{\chi_{j^*0}^{rk}}\right)^{\frac{\alpha_r^k}{\theta_r}}\left(\frac{\chi_{j^*0}^{sk}}{\chi_{j^*h}^{sk}}\right)^{\frac{\alpha_r^k}{\theta_s}}\right]^{1-\sigma_k}\left(\widehat{\varphi}_h^k\right)^{1-\sigma_k}\end{aligned}$$

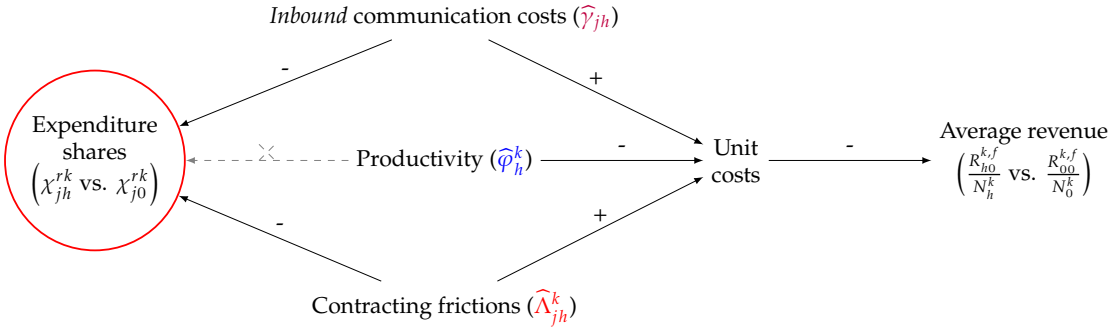
Finally, rearranging terms to solve for $\left(\widehat{\varphi}_h^k\right)^{\sigma_k-1}$, we obtain:

$$\left(\widehat{\varphi}_h^k\right)^{\sigma_k-1} = \frac{R_{00}^{k,f}/N_0^k}{R_{h0}^{k,f}/N_h^k}\left[\left(\frac{\chi_{0h}^{sk}}{\chi_{00}^{sk}}\right)^{\frac{\alpha_s^k+\alpha_r^k}{\theta_s}}\left(\frac{\chi_{j^*h}^{rk}}{\chi_{j^*0}^{rk}}\right)^{\frac{\alpha_r^k}{\theta_r}}\left(\frac{\chi_{j^*0}^{sk}}{\chi_{j^*h}^{sk}}\right)^{\frac{\alpha_r^k}{\theta_s}}\right]^{1-\sigma_k}$$

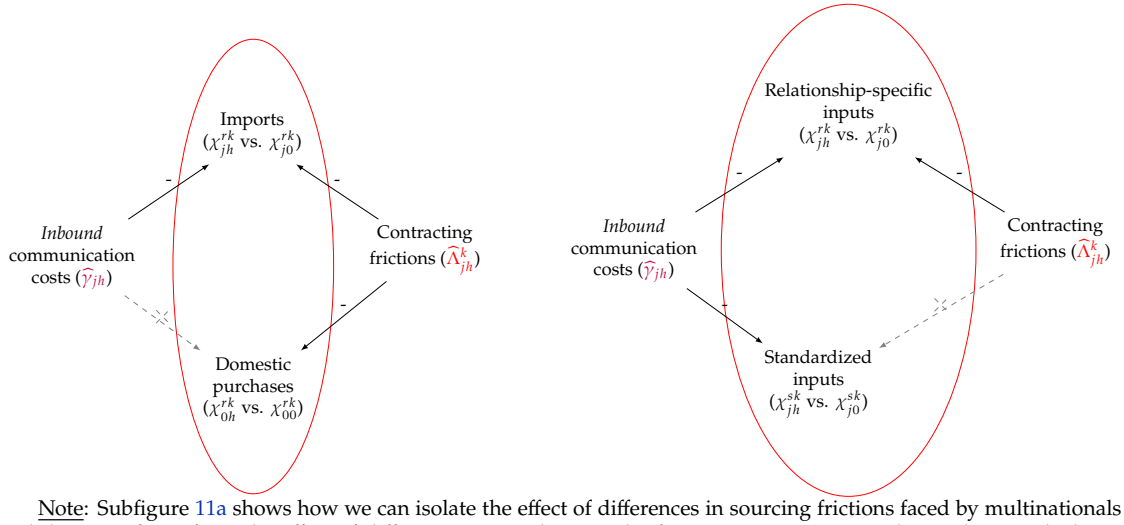
The following figure summarizes the identification strategy.

Figure 11. Identification strategy for counterfactual shocks

(a) Disentangling differences in *productivity* from differences in *sourcing frictions*



(b) Disentangling differences in *trade costs* from differences in *contracting frictions*



Note: Subfigure 11a shows how we can isolate the effect of differences in sourcing frictions faced by multinationals and domestic firms from the effect of differences in productivity by focusing on input expenditure shares, which are not affected by the latter. Subfigure 11b shows how we can distinguish between the effect of differences in inbound communication costs from the effect of differences in contracting frictions by exploiting the fact that the former does not affect domestic input purchases (but the latter does) and the fact that the latter does not affect imports of standardized inputs (but the former does).

□